

Results — revised submission — enhanced style

1 Datasets

#	Name	N	P	K	MeanIR	CVIR
1	GpositivePseAAC	519	440	4	3.86	1.15
2	CHD_49	555	49	6	5.77	1.60
3	Emotions	593	72	6	1.48	0.16
4	Scene	2407	294	6	1.25	0.11
5	PlantPseAAC	978	440	12	6.69	0.68
6	Water-quality	1060	16	14	1.77	0.29
7	Yeast	2417	103	14	7.20	1.82
8	HumanPseAAC	3106	440	14	15.29	1.05
9	enron	1702	1001	53	10.45	0.80

Table 1: (Table 3) Datasets used in the revised experiments. N , P , K are the number of instances, features, labels. MeanIR and CVIR report per-label imbalance (higher = more imbalanced).

Pending follow-ups (to fill before final submission)

- **LGBM run on enron.** Sequentially training (2–3h per task due to $K=53$ pairwise classifiers); will be merged into `full_enron_split_lgbm_summary/` when complete and the LGBM aggregate panels regenerated.
- **Extended PartialAbstention metrics.** The evaluator has been extended with 11 new PA metrics (`jaccard_pa`, `{example,macro,micro}-{precision,recall,f1}_pa`, `mfrd_pa`, `afrd_pa`). Existing runs were summarised with the previous metric set; the new columns will populate automatically on the next end-to-end training+evaluation pass.

2 Method encoding

For every panel that follows, the x-axis encodes the classifier as one of 12 methods, grouped into three blocks:

Block A — Order-based predictors (ours, indices 1–8). Each name has the form `<order>-<target>-<height>`; $order \in \{\text{PA} = \text{partial order}, \text{PR} = \text{pre-order}\}$; $target \in \{\text{H} = \text{Hamming}, \text{S} = \text{Subset } 0/1\}$; $height \in \{2, \infty \text{ (omitted in the label)}\}$.

- **1 = PA-H-2:** partial order, Hamming target, height ≤ 2 .

- 2 = **PA-H**: partial order, Hamming target, unrestricted height.
- 3 = **PA-S-2**: partial order, Subset 0/1 target, height ≤ 2 .
- 4 = **PA-S**: partial order, Subset 0/1, unrestricted.
- 5 = **PR-H-2**: pre-order, Hamming, height ≤ 2 .
- 6 = **PR-H**: pre-order, Hamming, unrestricted.
- 7 = **PR-S-2**: pre-order, Subset 0/1, height ≤ 2 .
- 8 = **PR-S**: pre-order, Subset 0/1, unrestricted.

Block B — Standard MLC baselines from the original paper (indices 9–11).

- 9 = **BR** (Binary Relevance): K independent binary classifiers, one per label.
- 10 = **CC** (Classifier Chain): sequential chain that uses previous labels as features.
- 11 = **CLR** (Calibrated Label Ranking): pairwise label ranking with a calibration label that separates relevant from irrelevant.

Block C — Extended baseline added for the revision (index 12).

- 12 = **ECC** (Ensemble of Classifier Chains): averages over multiple CCs with random chain orders.

Reading the panels. Each panel plots one dotted line with markers per noise level. **Noise is encoded by warm-hue intensity** (ColorBrewer YlOrRd): yellow = clean ($\alpha = 0.0$), dark red = noisy ($\alpha = 0.3$): ■ $\alpha = 0.0$, ■ $\alpha = 0.1$, ■ $\alpha = 0.2$, ■ $\alpha = 0.3$. Below the numeric x-axis (1–12), a second tier prints the short method name so the reader can identify each classifier without referring back to this section. *PartialAbstention* and *ScoreVector* panels show only indices 1–8. A vertical dashed line at $x = 8.5$ separates the order-based block (1–8) from the standard MLC baselines (9–12). The y-axis is zoomed to the actual data range to keep close-clustered curves visually separable. Missing methods show as gaps in the line (with a small grey “x” on the axis when the method has no data at all for that panel).

3 Paper results (RF base learner)

Reproduces the layout of the original paper’s main experiment tables (Tables 5–7): 3 datasets across, 4 metrics down, one table per prediction type (binary vector, partial abstention, score vector). Per the revision, $f_{\text{MLC}}^{\text{ham}}$ and $f_{\text{MLC}}^{\text{sub}}$ are dropped in favour of $f_{\text{MLC}}^{\text{Jacc}}$.

#	Name	N	P	K	MeanIR	CVIR
1	GpositivePseAAC	519	440	4	3.86	1.15
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Table 2: (Table 3) Datasets used in the revised experiments. N , P , K are the number of instances, features, labels. MeanIR and CVIR report per-label imbalance (higher = more imbalanced).

3.1 Standard MLC predictions (BinaryVector)

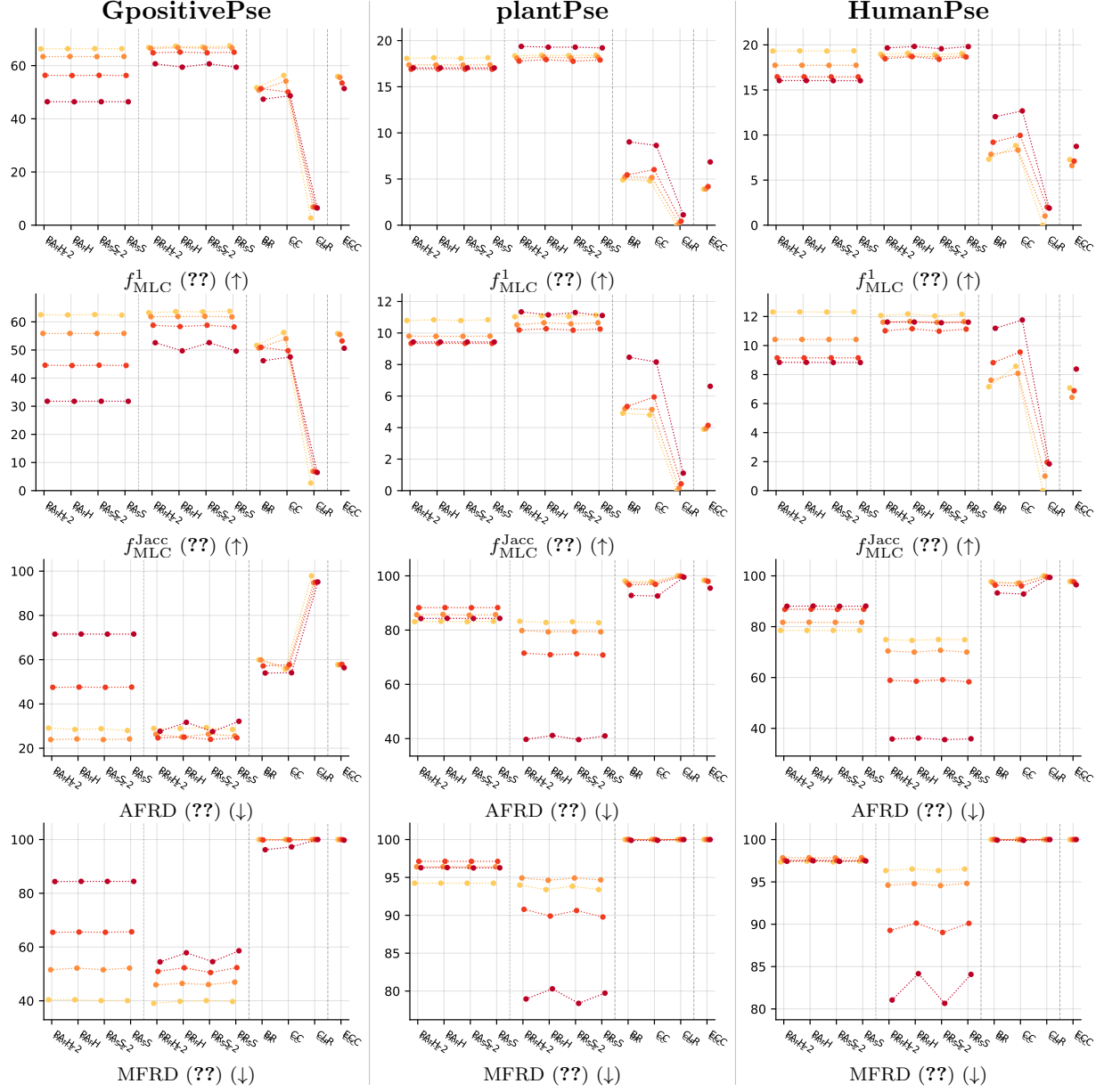


Table 3: (Table 4) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the classifiers (x -axis): partial-order predictors (PA-H-2, PA-H, PA-S-2, PA-S), preorder predictors (PR-H-2, PR-H, PR-S-2, PR-S), and standard MLC baselines (BR, CC, CLR). Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

3.2 Partial abstention (PartialAbstention)

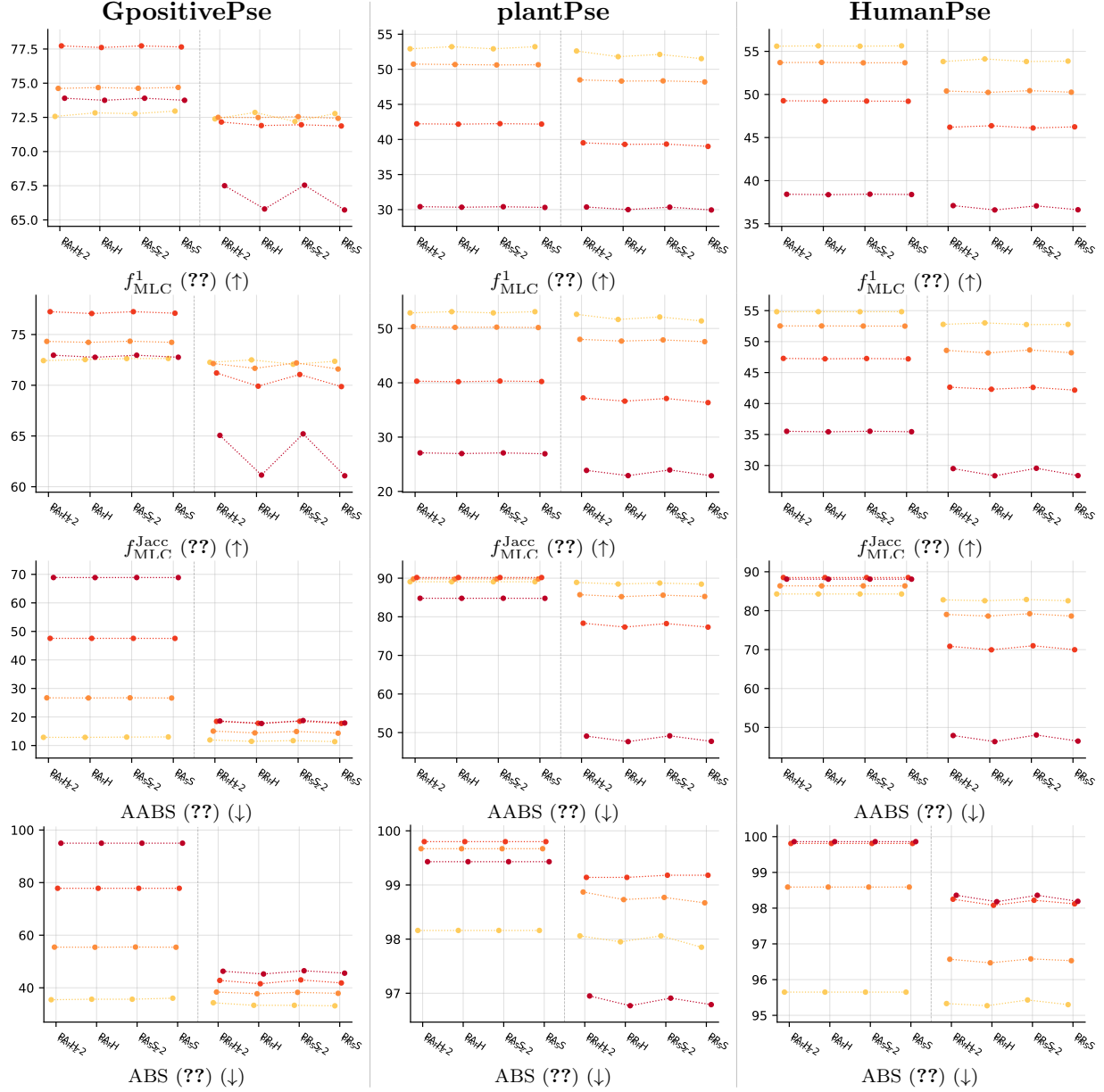


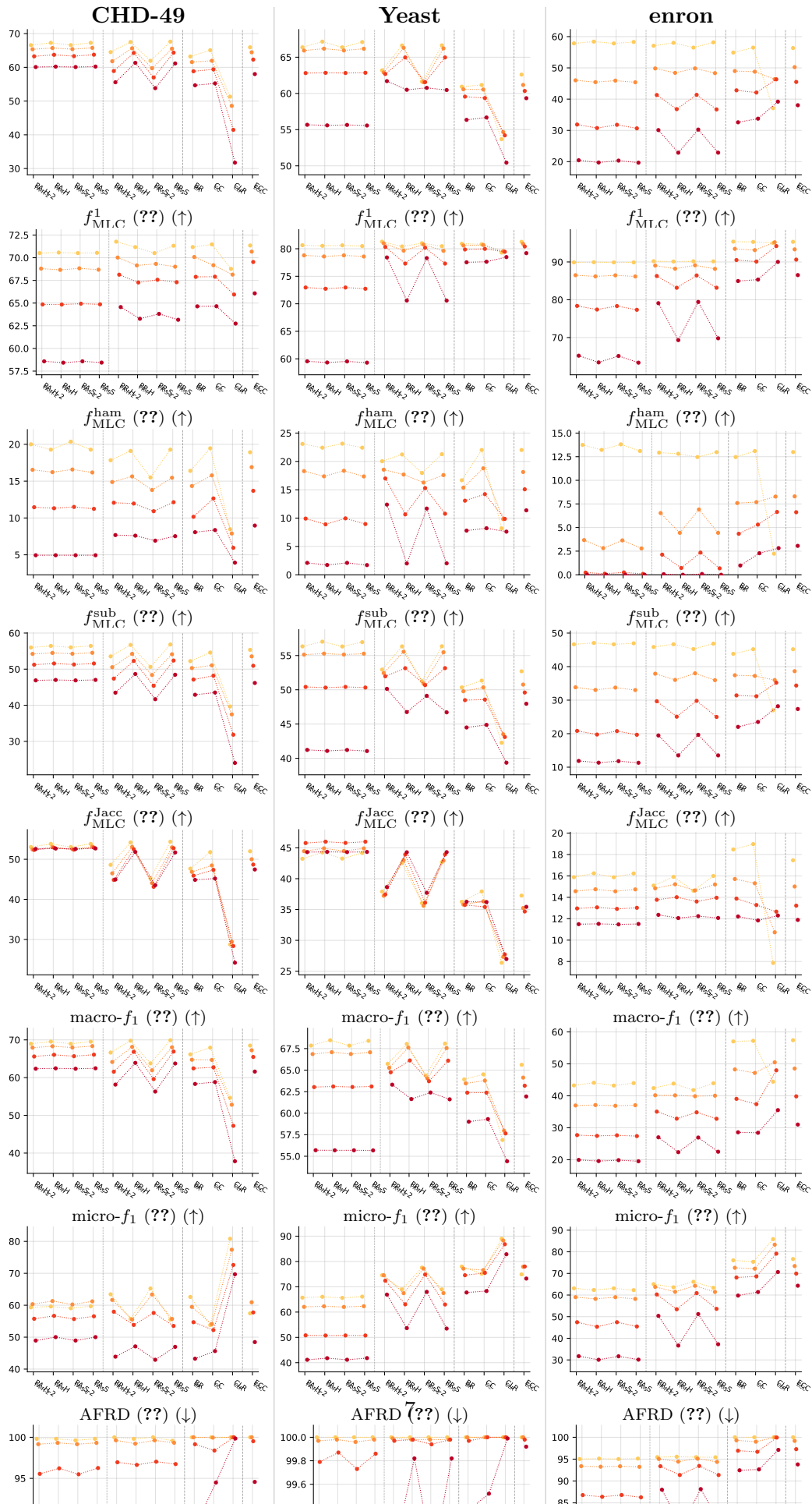
Table 4: (Table 5) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the classifiers (x -axis): partial-order predictors (PA-H-2, PA-H, PA-S-2, PA-S) and preorder predictors (PR-H-2, PR-H, PR-S-2, PR-S). Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

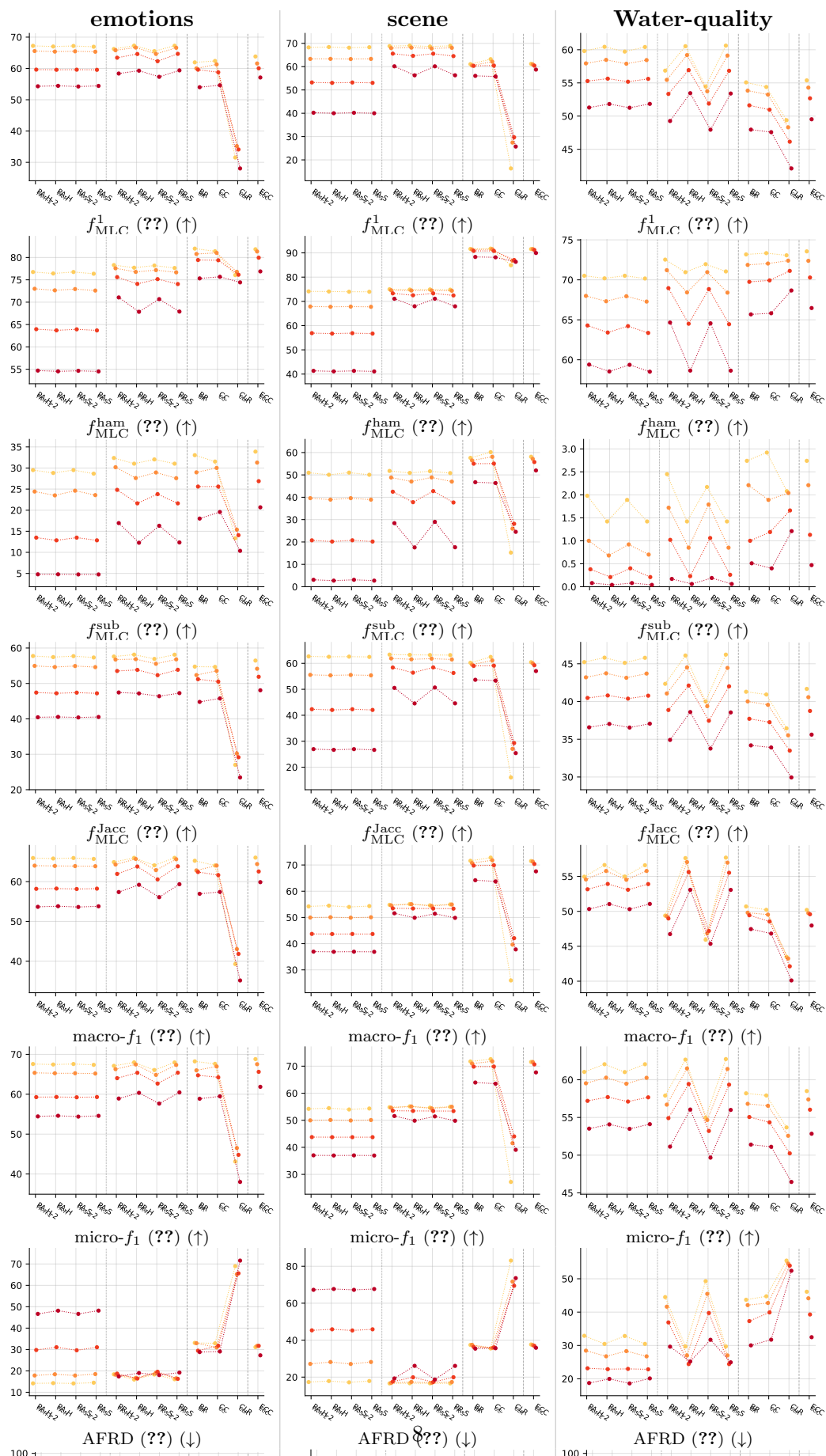
3.3 Experimental results — appendix tables (RF base learner)

Appendix counterpart of the main Paper results section. Tables All tables share the same 8-metric set (original 5 + 3 revision additions: f_{MLC}^{Jacc} , macro- f_1 , micro- f_1). G2/G4 = imbalanced datasets, G3/G5 = balanced datasets (split follows the original paper’s appendix). G6/G7 = main-paper

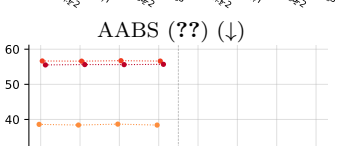
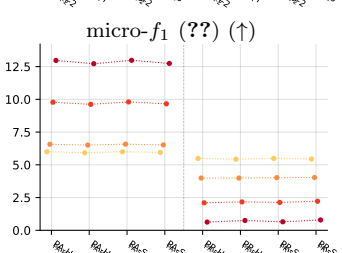
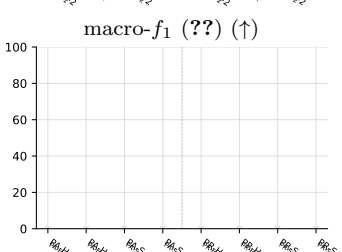
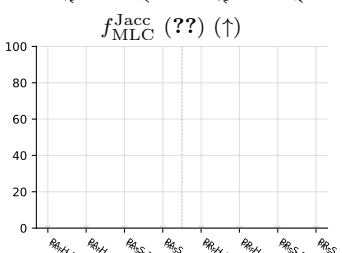
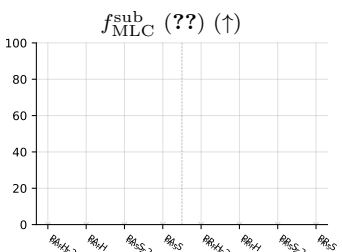
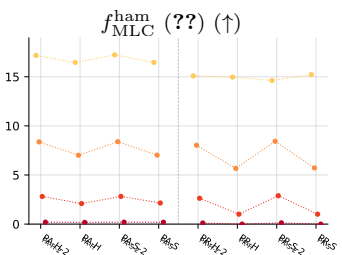
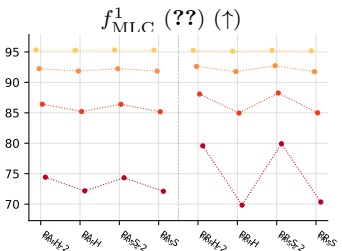
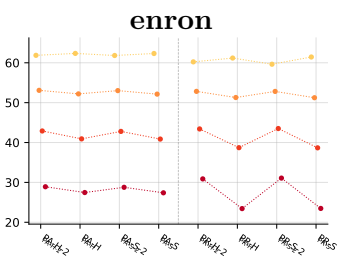
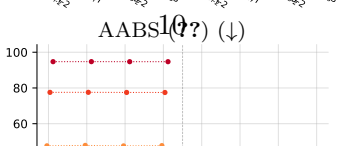
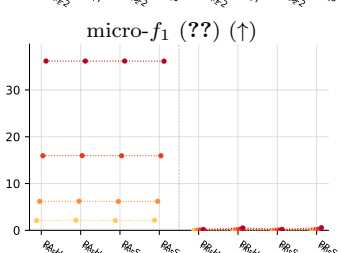
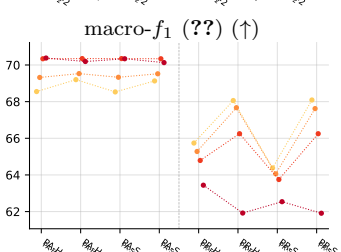
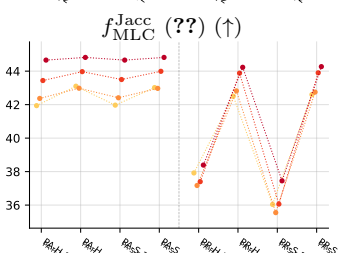
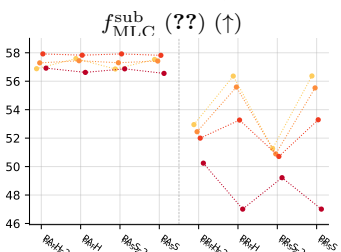
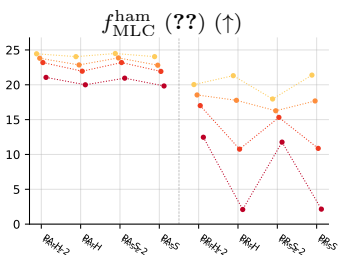
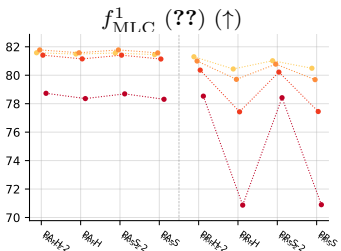
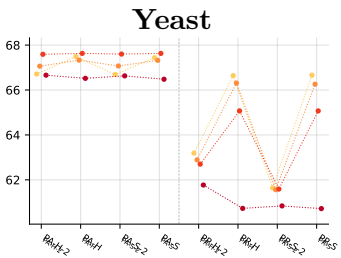
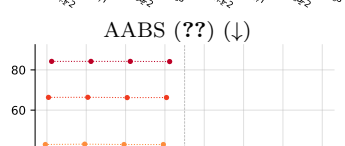
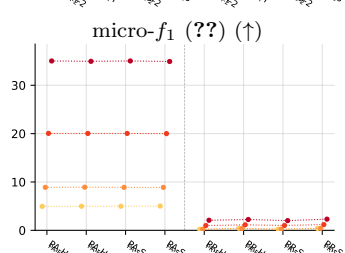
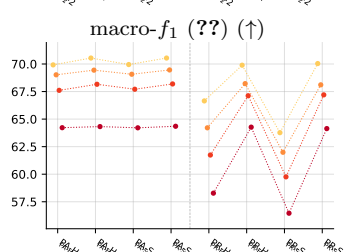
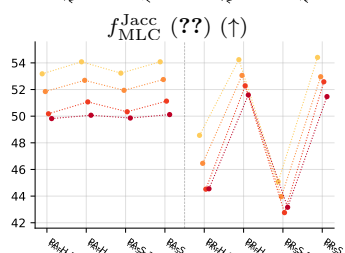
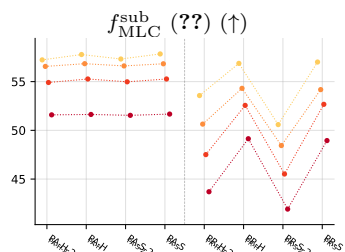
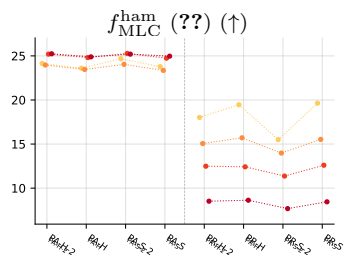
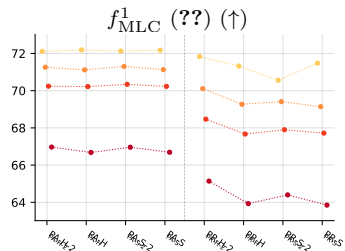
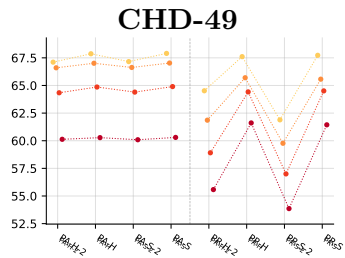
datasets (GpositivePse, PlantPse, HumanPse) shown with the full metric set; the main body only carries the revised compact Tables 4/5.

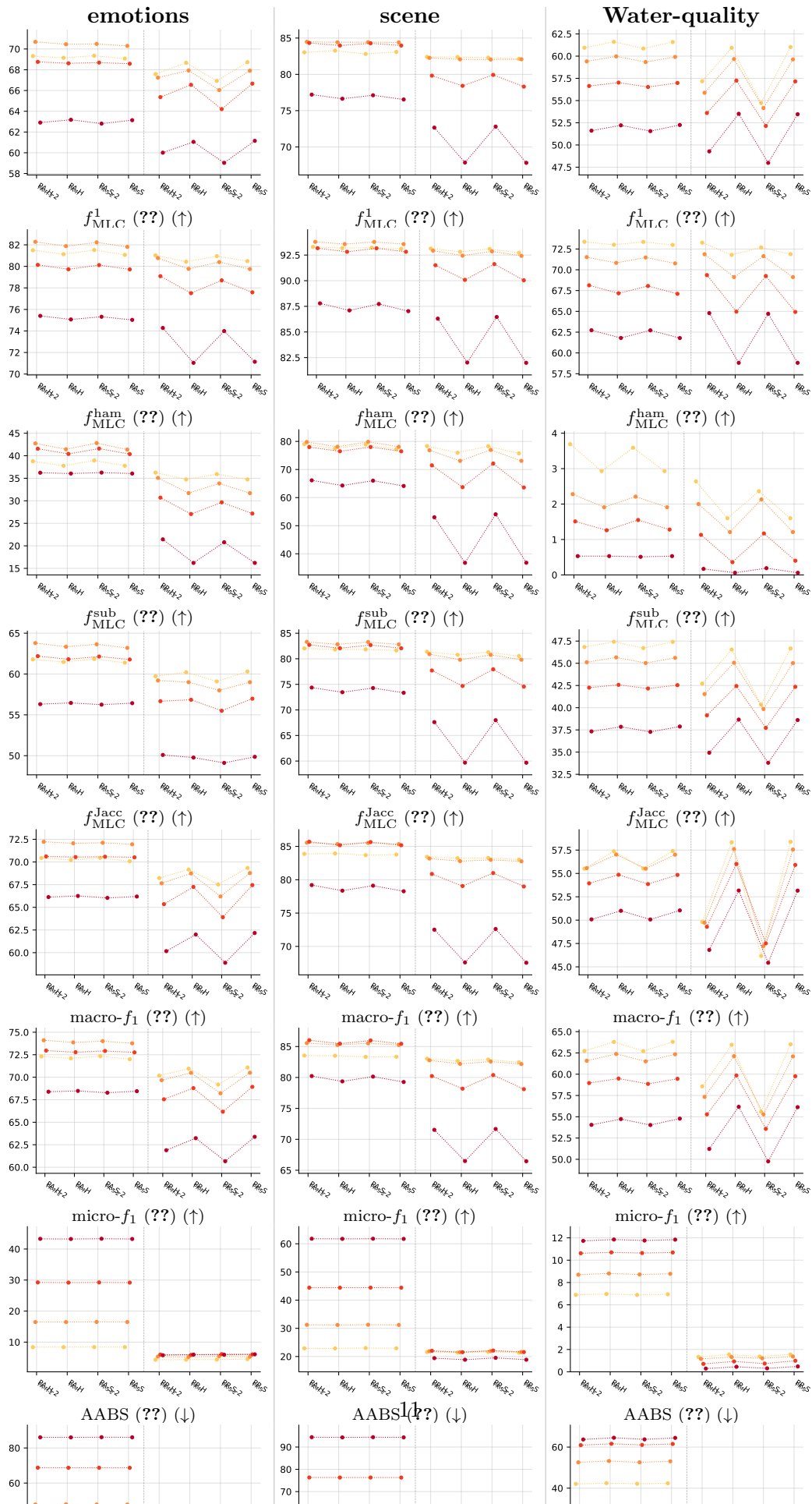
3.3.1 Standard MLC predictions (BinaryVector)





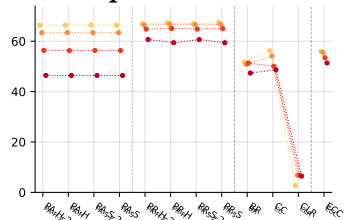
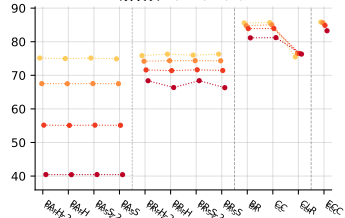
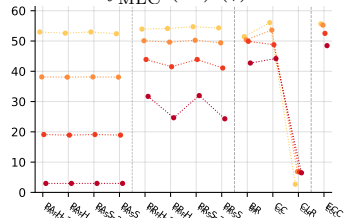
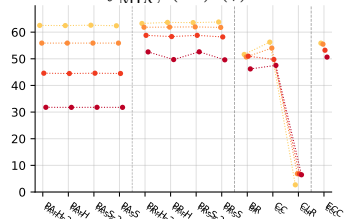
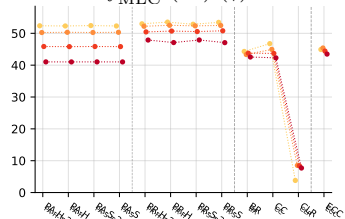
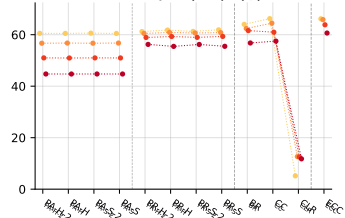
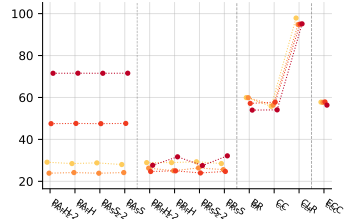
3.3.2 Partial abstention (PartialAbstention)



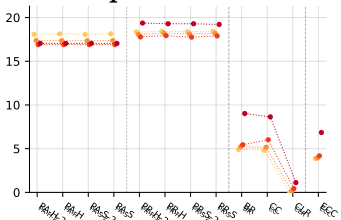
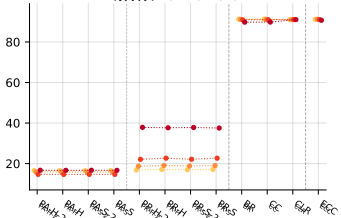
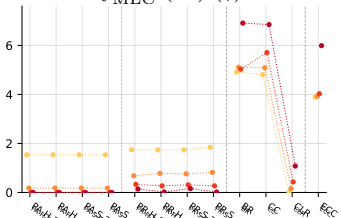
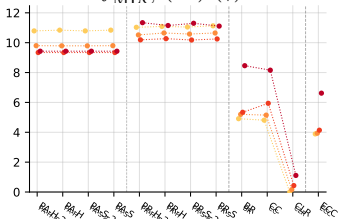
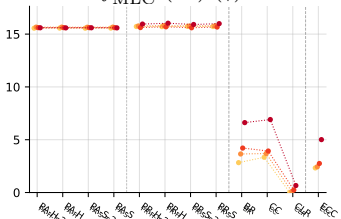
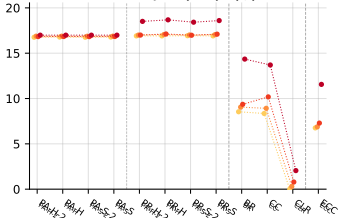
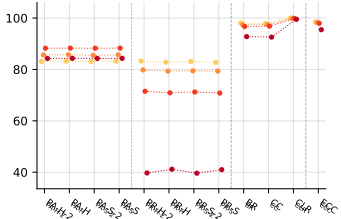
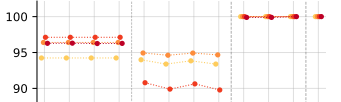


3.3.3 Standard MLC predictions (BinaryVector)

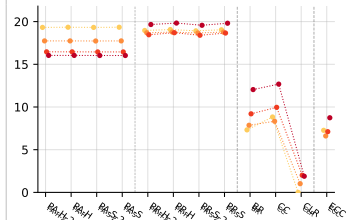
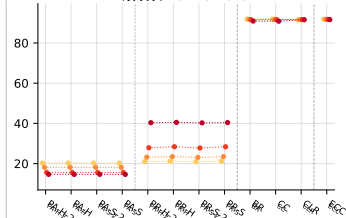
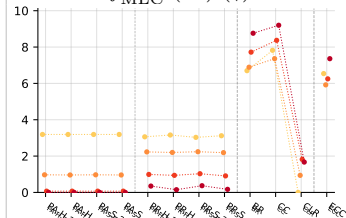
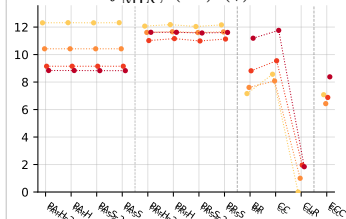
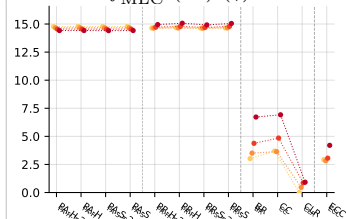
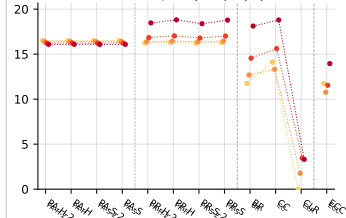
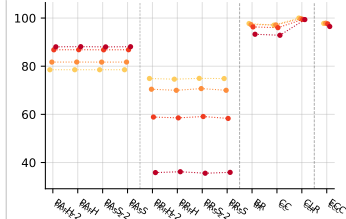
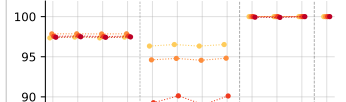
GpositivePse

 f_{MLC}^1 (??) ($\uparrow$) f_{MLC}^{ham} (??) ($\uparrow$) f_{MLC}^{sub} (??) ($\uparrow$) f_{MLC}^{Jacc} (??) ($\uparrow$)macro- f_1 (??) ($\uparrow$)micro- f_1 (??) ($\uparrow$)AFRD (??) ($\downarrow$)

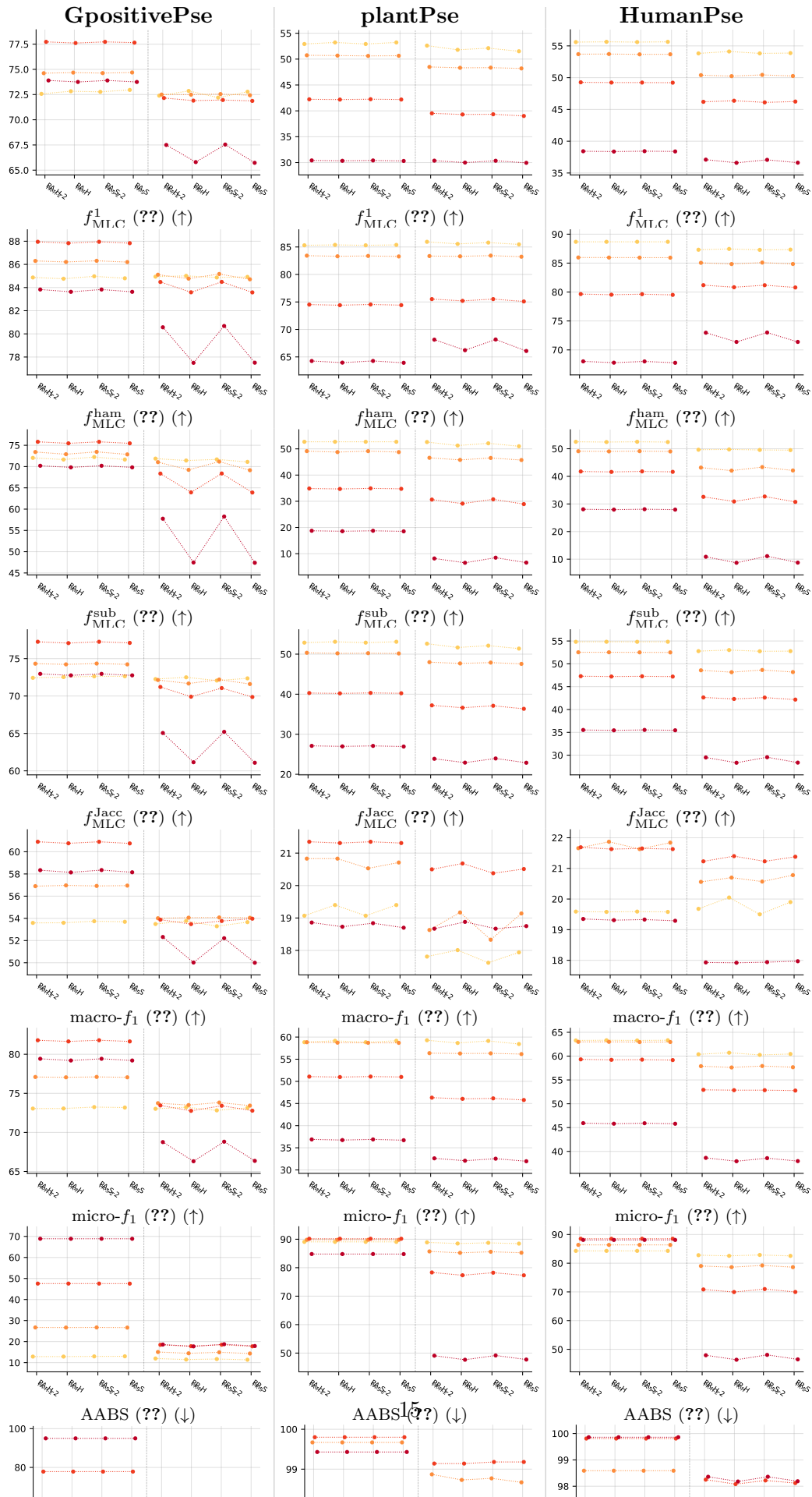
plantPse

 f_{MLC}^1 (??) ($\uparrow$) f_{MLC}^{ham} (??) ($\uparrow$) f_{MLC}^{sub} (??) ($\uparrow$) f_{MLC}^{Jacc} (??) ($\uparrow$)macro- f_1 (??) ($\uparrow$)micro- f_1 (??) ($\uparrow$)AFRD (??) ($\downarrow$)

HumanPse

 f_{MLC}^1 (??) ($\uparrow$) f_{MLC}^{ham} (??) ($\uparrow$) f_{MLC}^{sub} (??) ($\uparrow$) f_{MLC}^{Jacc} (??) ($\uparrow$)macro- f_1 (??) ($\uparrow$)micro- f_1 (??) ($\uparrow$)AFRD (??) ($\downarrow$)

3.3.4 Partial abstention (PartialAbstention)



3.4 Abstention overview: aggregate trends

Purpose. This compact section shows the headline abstention-vs-standard-MLC trend across noise levels, averaged over all datasets, for both base learners (RF, LGBM) and both retained quality metrics (f_1 , Jaccard). Use it as a quick visual summary before diving into the detailed per-method / per-dataset figures in Section 5.

How to read. In each panel: blue bar = standard MLC (no abstention) on its native quality metric; coloured bar = with-abstention quality on retained labels. Green “+ $x.y$ ” label = score-point gain of abstention over the standard-MLC bar of the same method. The four sub-panels per figure correspond to $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$.

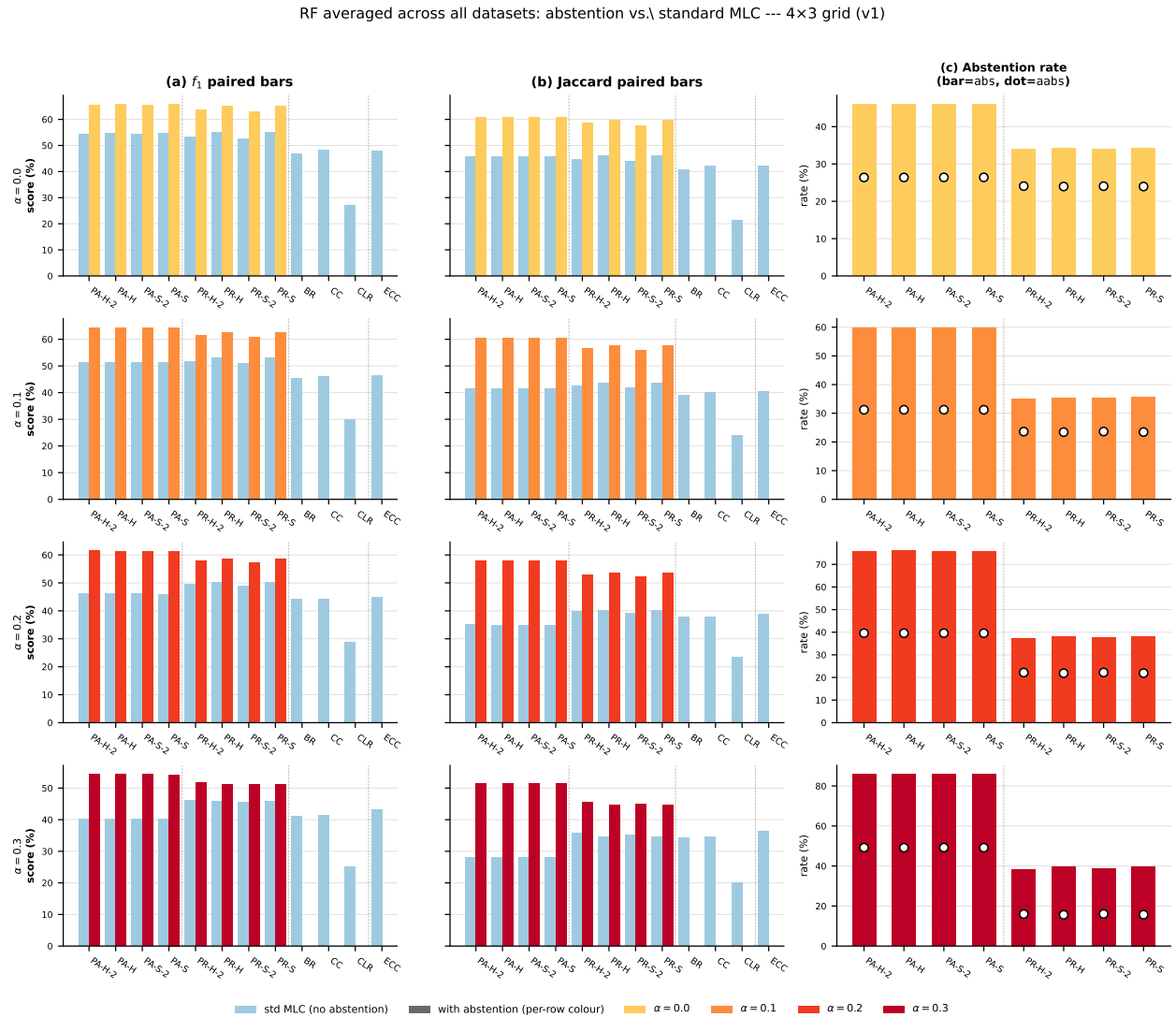
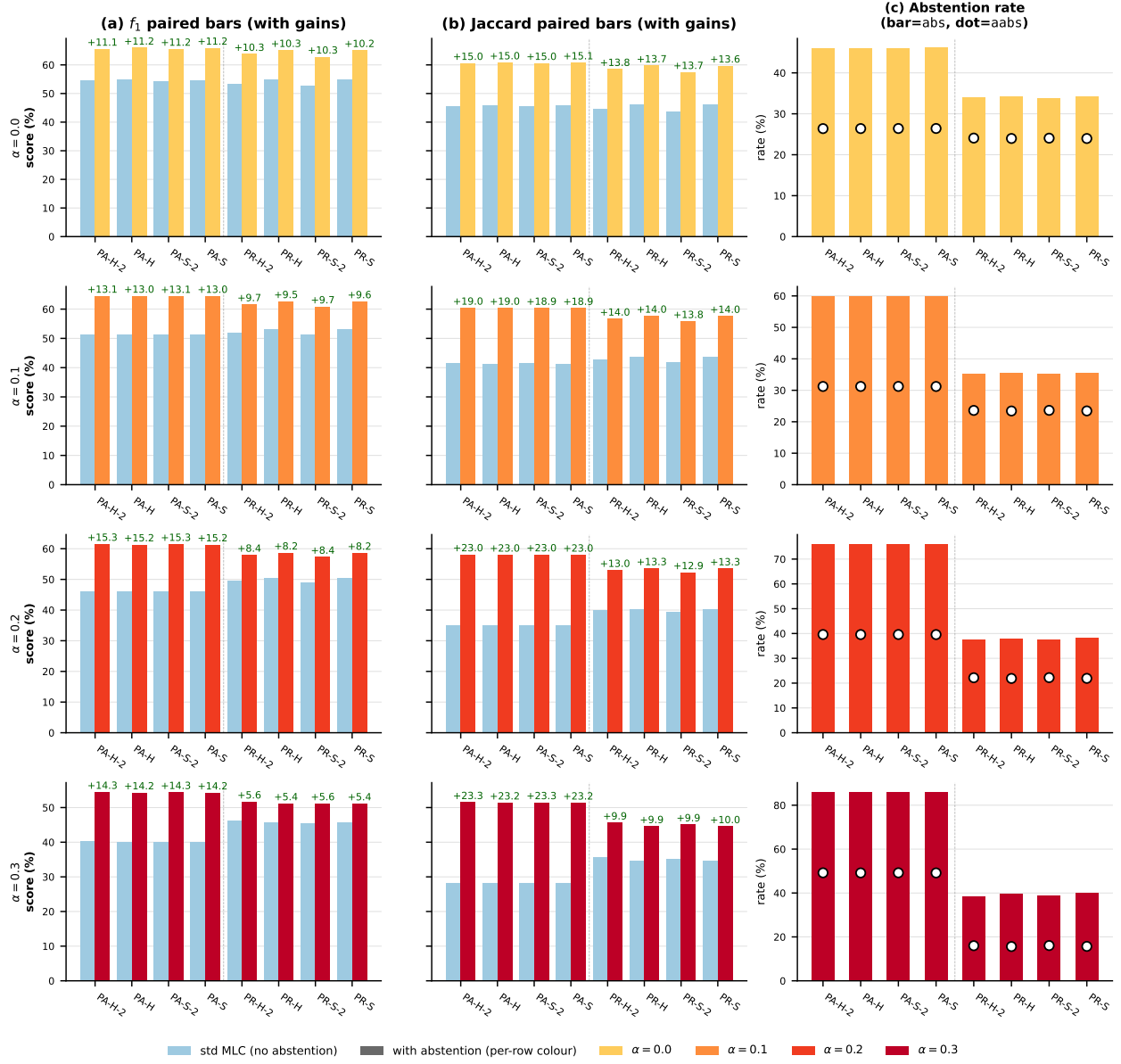


Figure 1: RF averaged across all datasets, 4×3 grid (rows = α , cols = f_1 / Jaccard / abstention rate) — (version 1) all 12 methods on the x -axis (PA + PR + baselines BR/CC/CLR/ECC); baselines plot only the std-MLC bar since they cannot abstain. Light blue = standard MLC, α -coloured = with-abstention; in (c), bar = **abs** (instance-level), dot = **aabs** (per-cell). Color = $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$.

RF averaged across all datasets: abstention vs. standard MLC --- 4×3 grid (v2, 8 methods + gains)



LGBM averaged across all datasets: abstention vs. standard MLC --- 4×3 grid (v1)

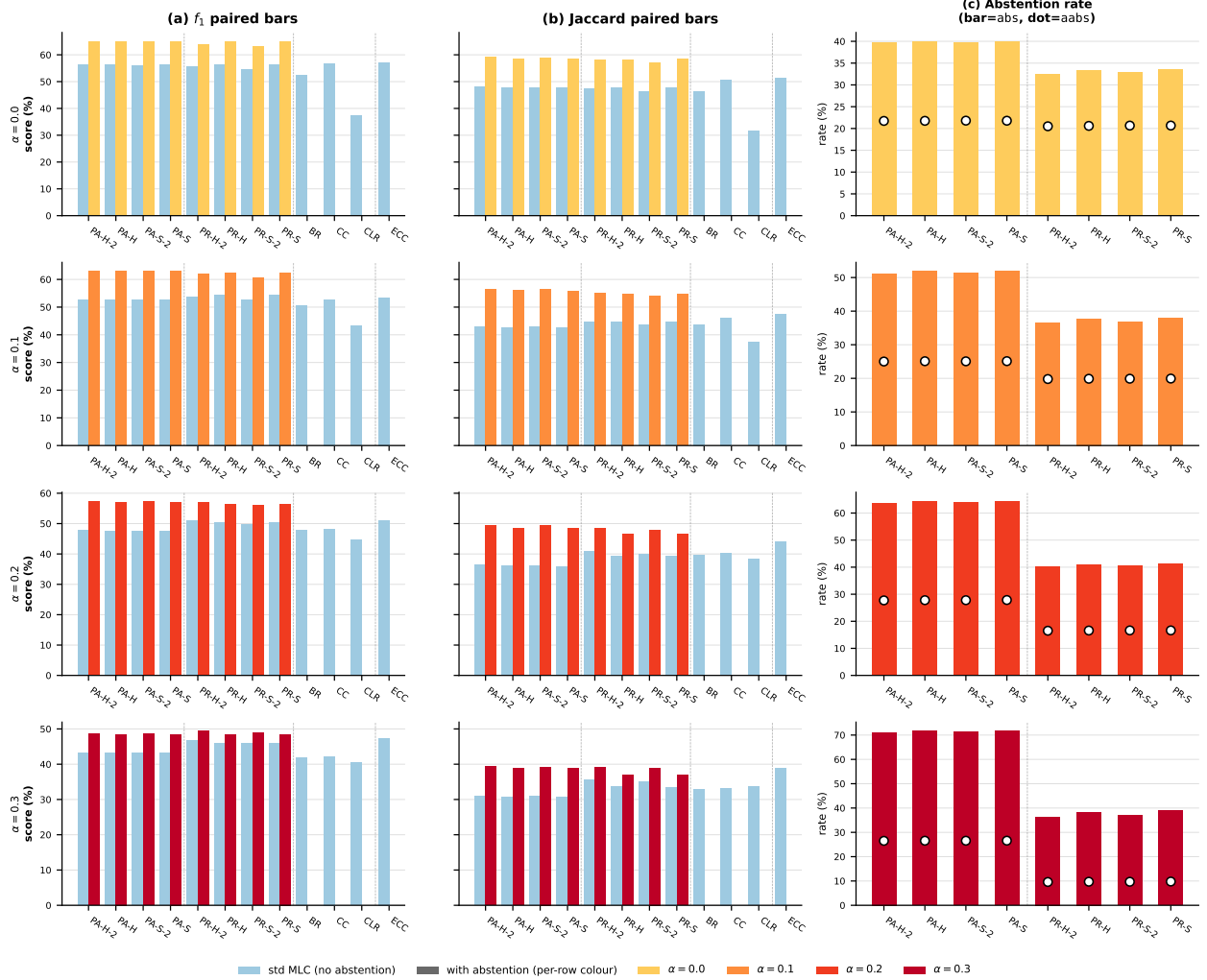


Figure 3: LGBM averaged across all datasets, 4×3 grid (rows = α , cols = f_1 / Jaccard / abstention rate) — (version 1) all 12 methods on the x -axis (PA + PR + baselines BR/CC/CLR/ECC); baselines plot only the std-MLC bar since they cannot abstain. Light blue = standard MLC, α -coloured = with-abstention; in (c), bar = **abs** (instance-level), dot = **aabs** (per-cell). Color = $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$.

LGBM averaged across all datasets: abstention vs. standard MLC --- 4×3 grid (v2, 8 methods + gains)

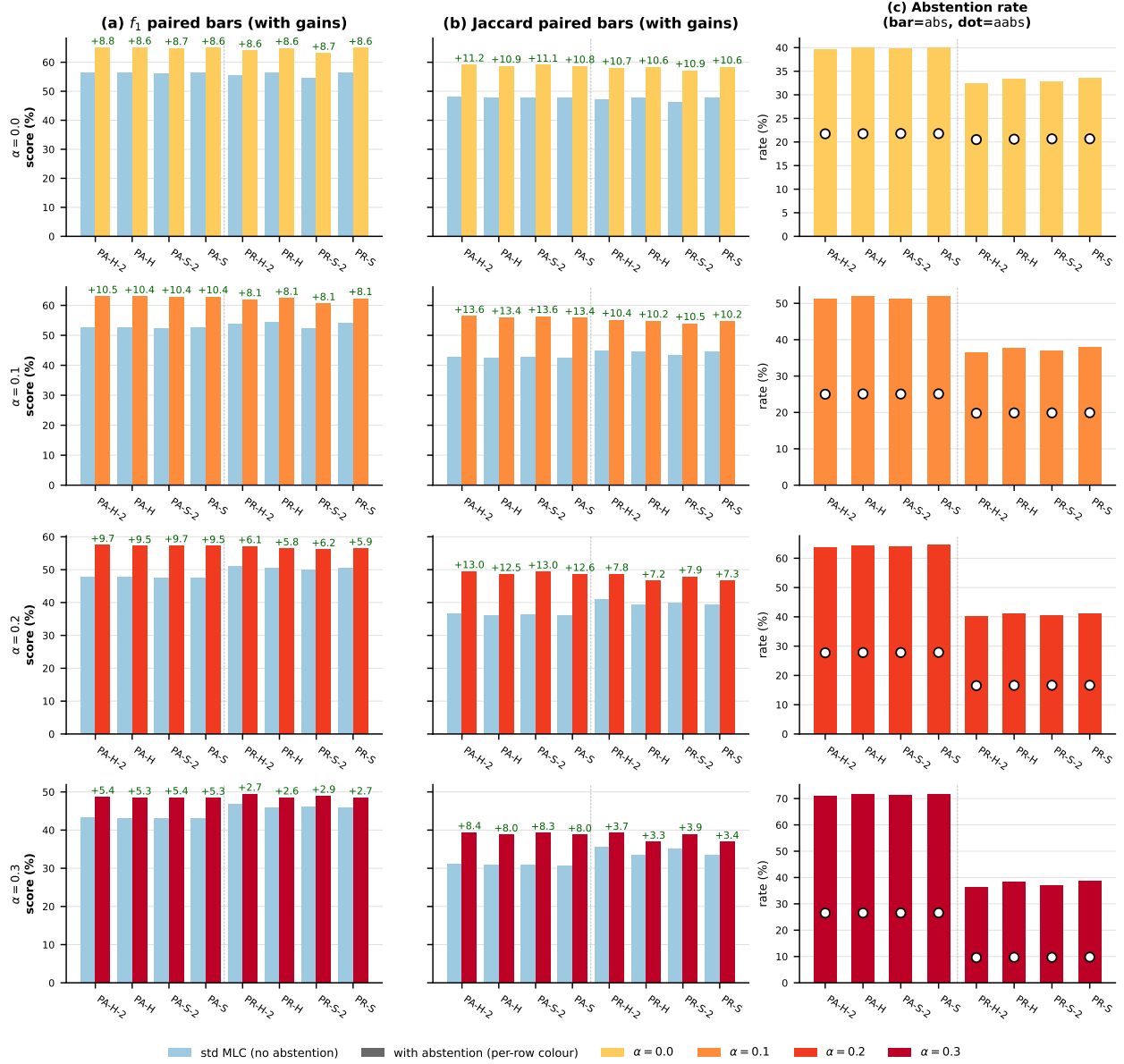


Figure 4: LGBM averaged across all datasets, 4×3 grid (rows = α , cols = f_1 / Jaccard / abstention rate) — (version 2) only the 8 PA/PR methods that can abstain; gain labels + $x.y$ printed above each with-abstention bar so the score-point benefit over the std-MLC bar at the same method is visible at a glance. Same colour / layout conventions as version 1. Color = $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$.

4 Main results: RF base learner, aggregated

RF • Aggregate • BinaryVector — Standard MLC predictions

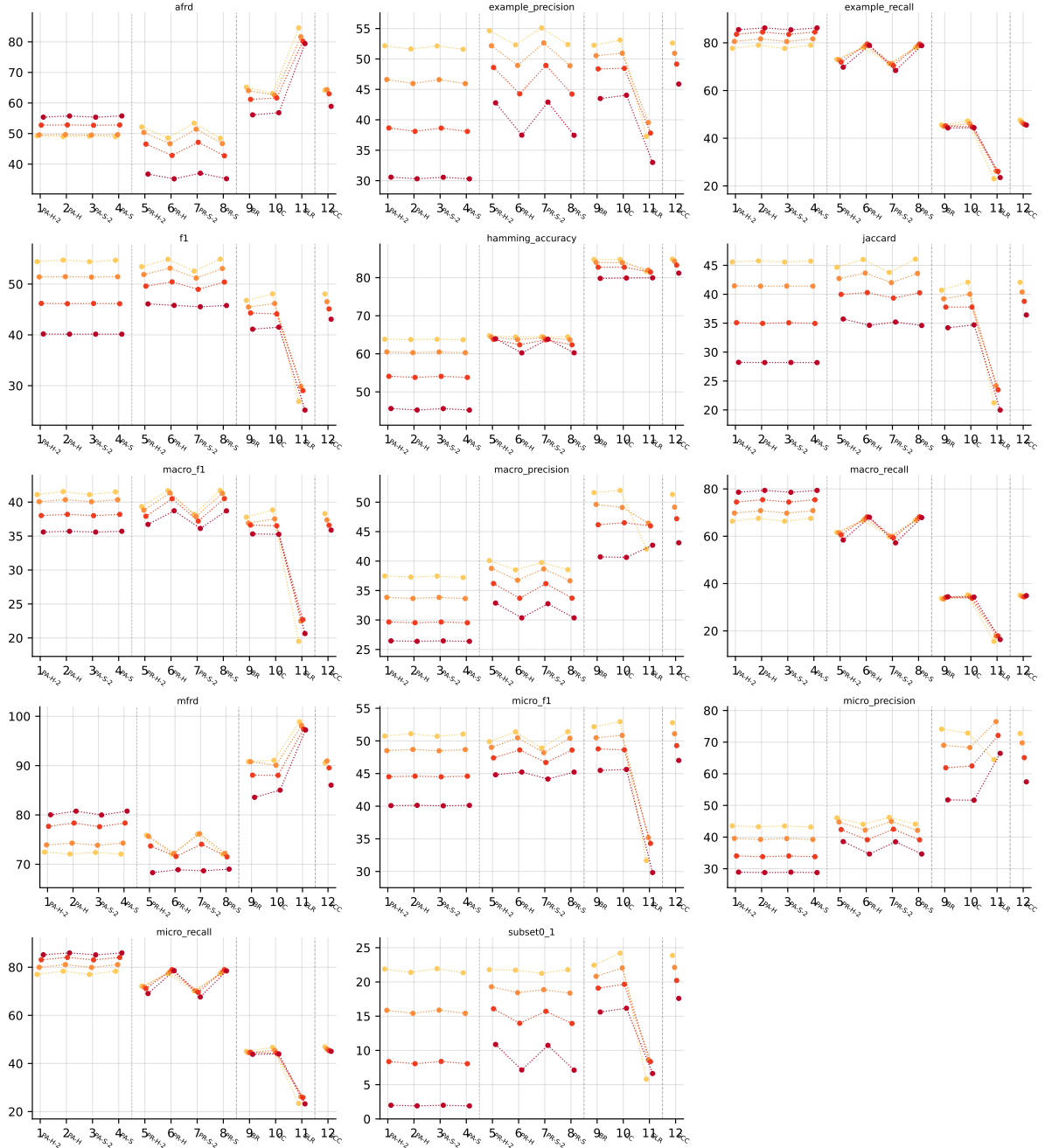


Table 11: Standard MLC predictions (BinaryVector), RF base learner, averaged across datasets. Method indices and line colors follow the encoding paragraph above.

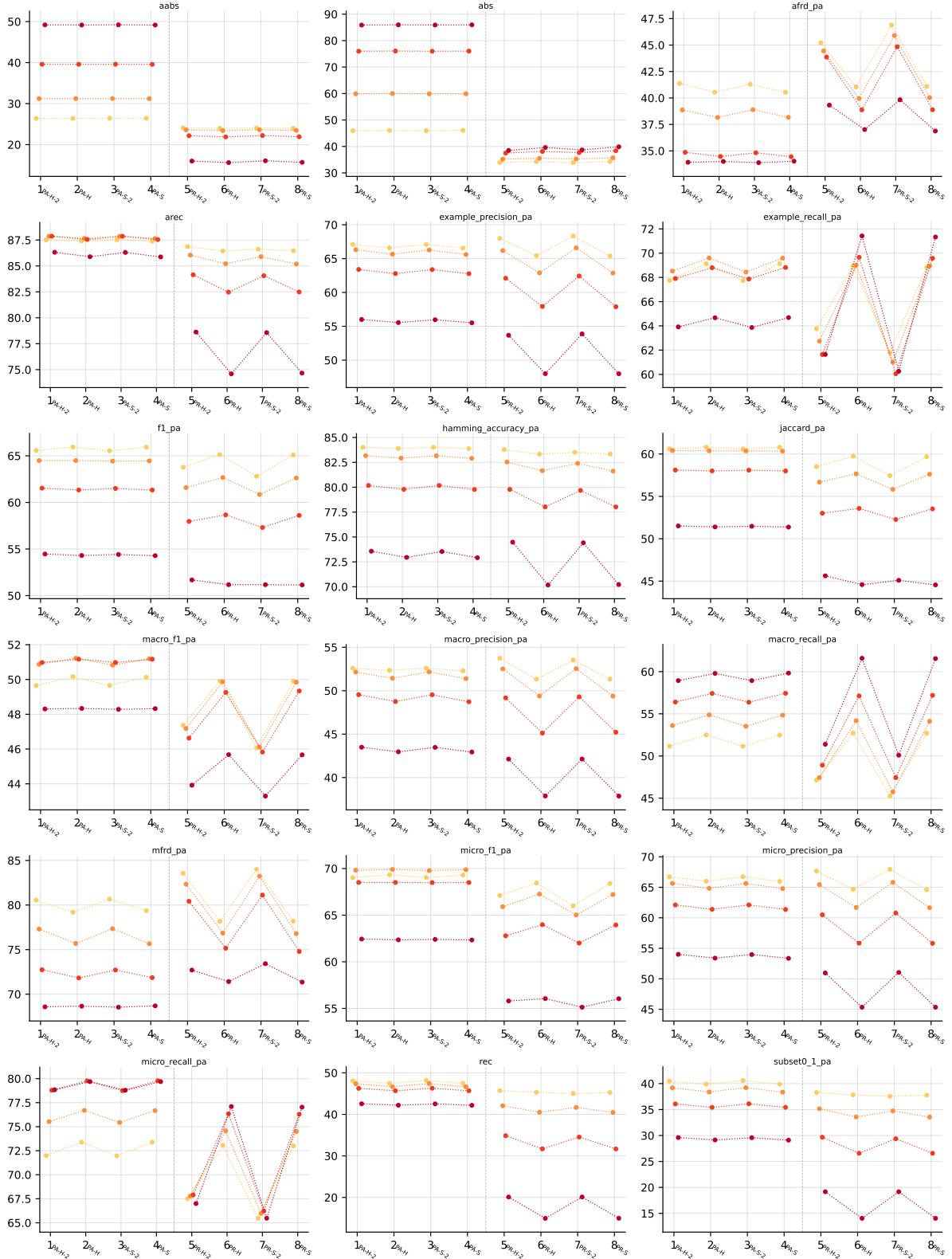


Table 12: Partial abstention (PartialAbstention), RF base learner, averaged across datasets. Method indices and line colors follow the encoding paragraph above.

RF • Aggregate • PartialAbstention — Partial abstention

RF • Aggregate • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

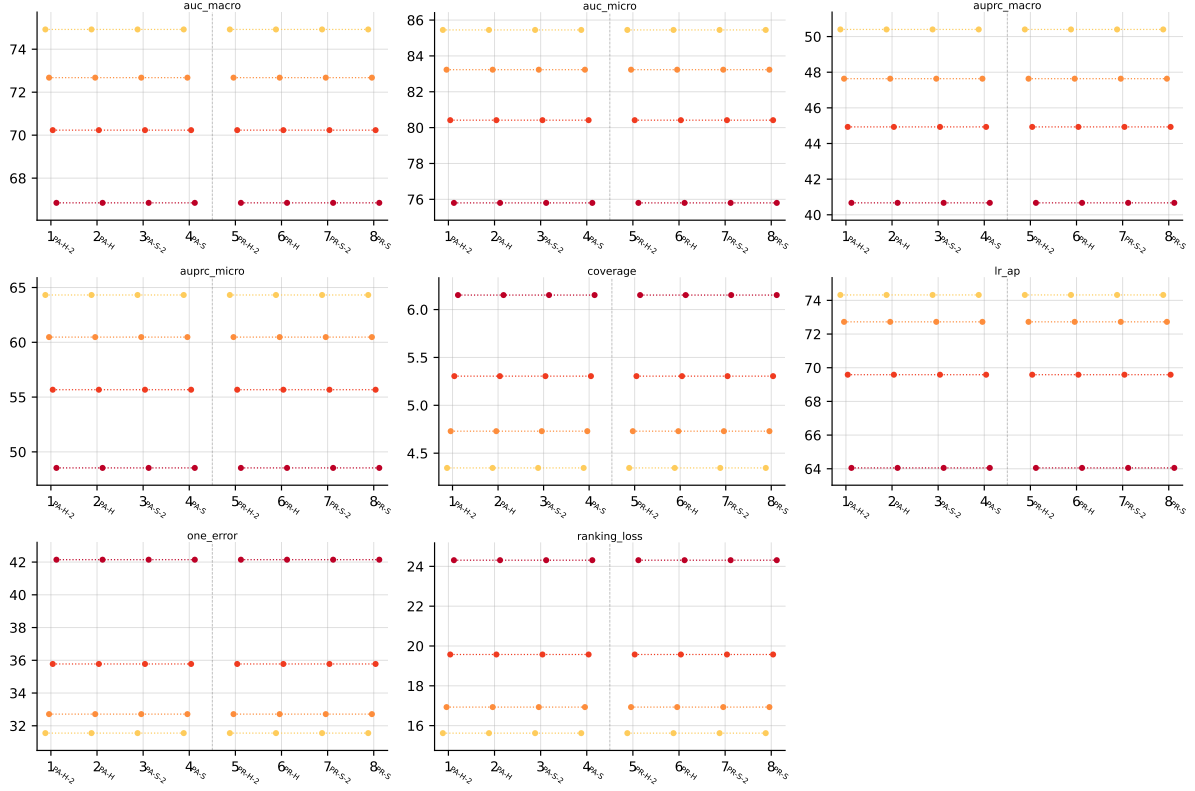


Table 13: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector), RF base learner, averaged across datasets. Method indices and line colors follow the encoding paragraph above.

5 Abstention vs standard MLC

Why this section exists. The paper’s central claim is that an order-based predictor can *abstain* on labels it is unsure about and, on the labels it does predict, achieve higher quality than a standard MLC method that is forced to predict every label. The two chart families below make that claim visible.

Terminology. ‘*Standard MLC*’ (a.k.a. **BinaryVector** / BV) means each method outputs a 0/1 prediction for every one of the K labels. ‘*With abstention*’ (a.k.a. **PartialAbstention** / PA) means an order-based predictor may return $\perp$ (abstain) on some labels; the PA quality metrics (**f1_pa**, **jaccard_pa**, ...) are computed only over the non-abstained labels. Only the 8 PA/PR methods (indices 1–8) can abstain; BR/CC/CLR/ECC cannot.

Reading the coverage-risk chart. Each point is one (PA/PR method, noise level α) pair. The x-axis is the abstention rate (fraction of labels the method skipped); the y-axis is the quality metric on the labels that *were* predicted (higher is better). Marker shape encodes method, colour encodes noise level. The dashed grey line is the best standard-MLC baseline (BR / CC / CLR / ECC) on the BV quality metric — it is the bar that “no abstention” has to clear. **Any point above the dashed line is direct evidence that abstaining improved quality beyond what any standard MLC method achieves.** Points further right traded coverage for that gain.

Reading the paired-bars chart. Same data, different view. The four sub-panels correspond to the four noise levels. Inside each sub-panel: 12 method positions, two bars per position — blue is the standard-MLC quality (**f1**, no abstention), warm-colour is the with-abstention quality on retained labels. Only the 8 PA/PR methods (1–8) have a warm bar; the standard MLC baselines (9–12) only have a blue bar by construction. The green “+ $x.y$ ” label above each warm bar is the gain in score-points over that method’s own BV bar.

5.1 RF, metric f1_pa — aggregate



Figure 5: RF (**f1_pa**) averaged across all datasets: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

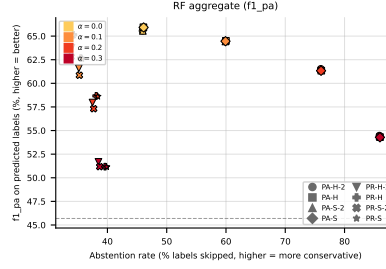


Figure 6: RF ($f1_{pa}$) averaged across all datasets: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



Figure 7: RF ($f1_{pa}$) averaged across all datasets: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).



Figure 8: RF ($f1_{pa}$) averaged across all datasets — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - abs$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - abs) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.2 RF, metric $f1_{pa}$ — per dataset



Figure 9: RF ($f1_{pa}$) on **GpositivePseAAC**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

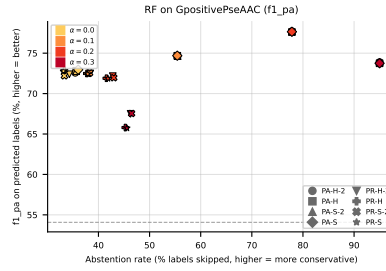


Figure 10: RF ($f1_{pa}$) on **GpositivePseAAC**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



Figure 11: RF ($f1_{pa}$) on **GpositivePseAAC**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).

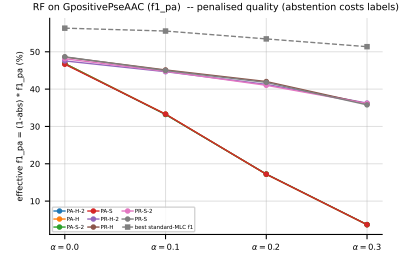
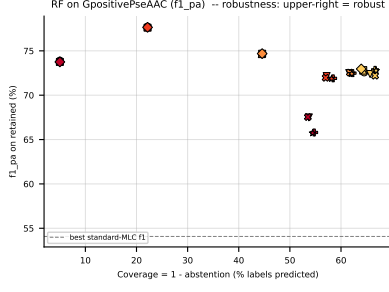


Figure 12: RF (f1_pa) on **GpositivePseAAC** — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

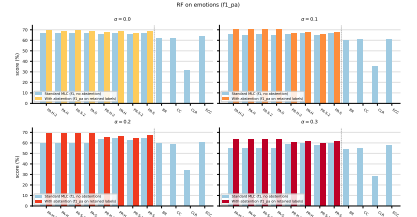
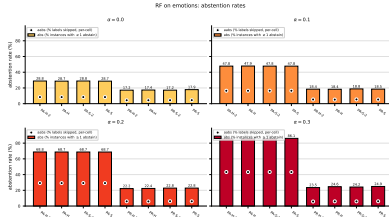


Figure 13: RF (f1_pa) on **emotions**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

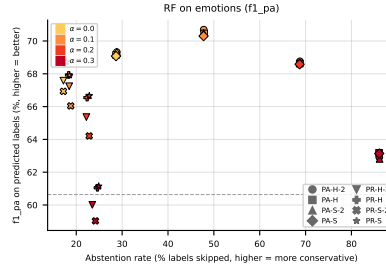


Figure 14: RF (f1_pa) on **emotions**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



Figure 15: RF (f1_pa) on **emotions**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).



Figure 16: RF (f1_pa) on **emotions** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

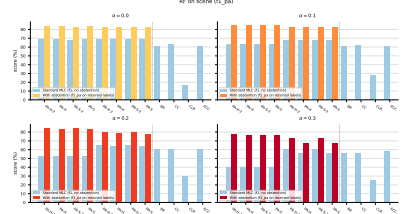
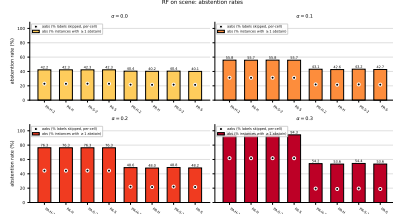


Figure 17: RF ($f1_{pa}$) on **scene**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

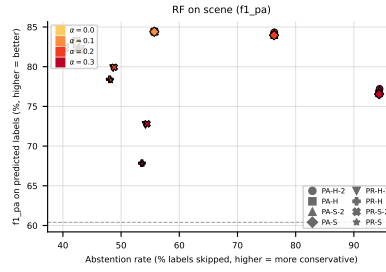


Figure 18: RF ($f1_{pa}$) on **scene**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

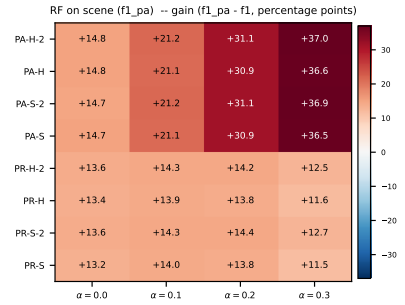
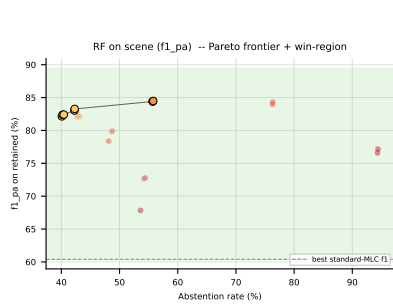


Figure 19: RF ($f1_{pa}$) on **scene**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).

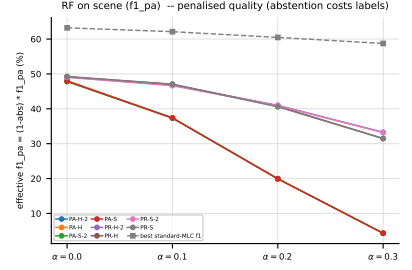
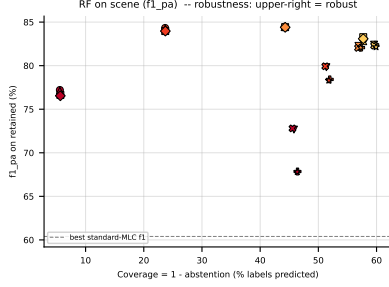


Figure 20: RF (f1_{pa}) on **scene** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_{pa} = $(1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

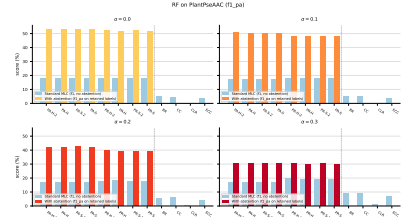
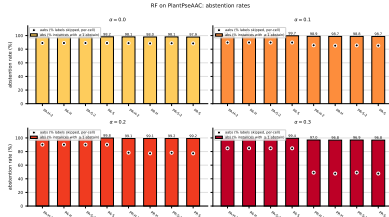


Figure 21: RF (f1_{pa}) on **PlantPseAAC**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

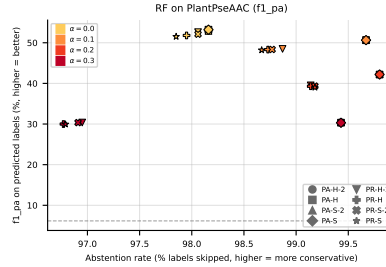


Figure 22: RF (f1_{pa}) on **PlantPseAAC**: coverage-risk scatter. Each point is one (method, noise) pair; $x = \text{abstention rate}$, $y = \text{quality on retained labels}$. Points above the dashed line beat the best standard-MLC baseline.

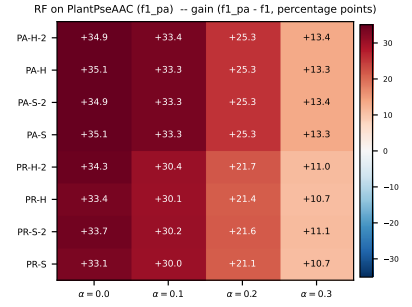
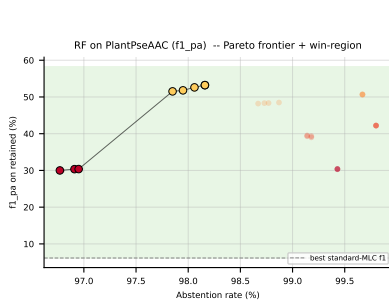


Figure 23: RF ($f1_{pa}$) on PlantPseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).

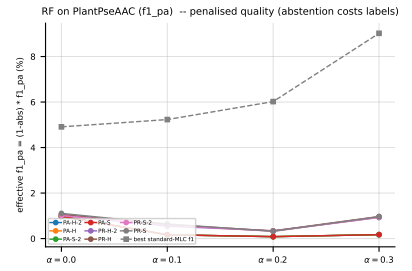
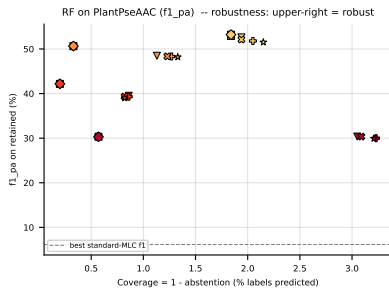


Figure 24: RF ($f1_{pa}$) on PlantPseAAC — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

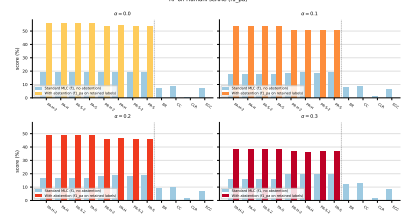
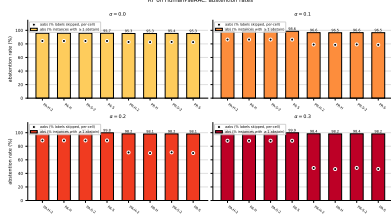


Figure 25: RF ($f1_{pa}$) on HumanPseAAC: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

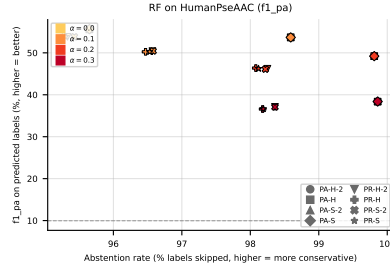


Figure 26: RF ($f1_{pa}$) on HumanPseAAC: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

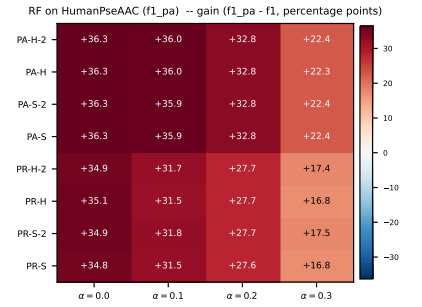
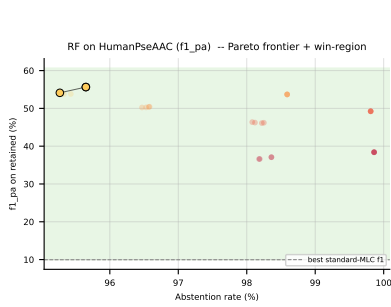


Figure 27: RF ($f1_{pa}$) on HumanPseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f_1$) heatmap (right).

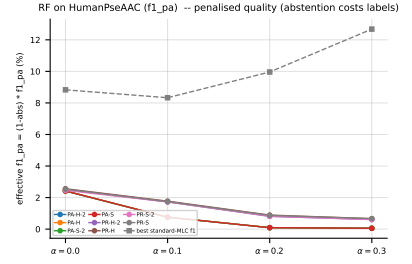
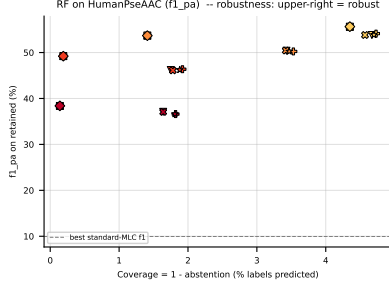


Figure 28: RF ($f1_{pa}$) on **HumanPseAAC** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

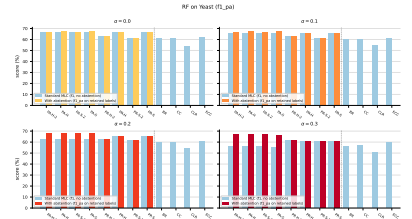
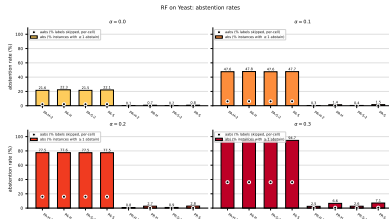


Figure 29: RF ($f1_{pa}$) on **Yeast**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

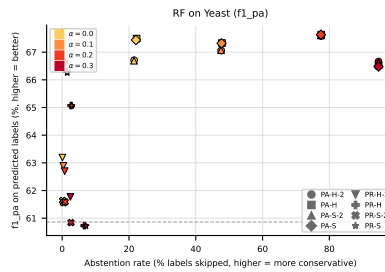


Figure 30: RF ($f1_{pa}$) on **Yeast**: coverage-risk scatter. Each point is one (method, noise) pair; $x = \text{abstention rate}$, $y = \text{quality on retained labels}$. Points above the dashed line beat the best standard-MLC baseline.



Figure 31: RF (f1_pa) on Yeast: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_pa - f1$) heatmap (right).



Figure 32: RF (f1_pa) on Yeast — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_pa = (1 - \text{abs}) \cdot f1_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.



Figure 33: RF (f1_pa) on CHD_49: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

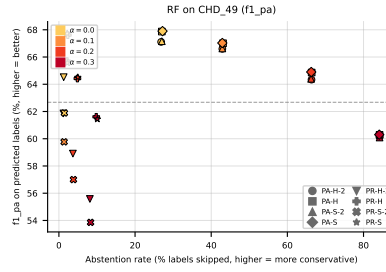


Figure 34: RF (f1_pa) on CHD_49: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



Figure 35: RF (f1_pa) on CHD_49: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).

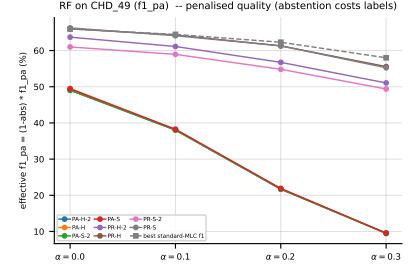
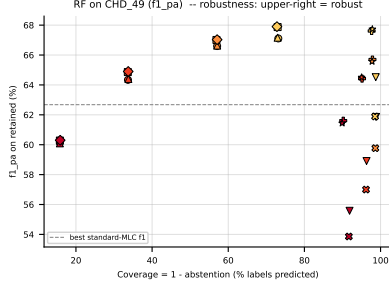


Figure 36: RF (f1_{pa}) on CHD₄₉ — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_{pa} = $(1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

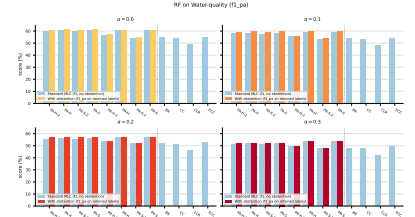
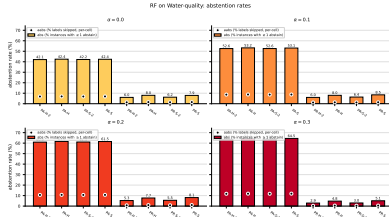


Figure 37: RF (f1_{pa}) on Water-quality: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

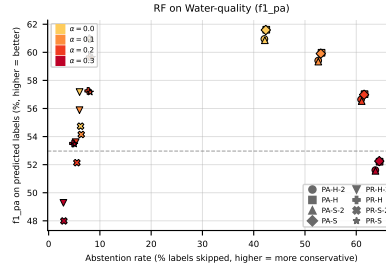


Figure 38: RF (f1_{pa}) on Water-quality: coverage-risk scatter. Each point is one (method, noise) pair; $x = \text{abstention rate}$, $y = \text{quality on retained labels}$. Points above the dashed line beat the best standard-MLC baseline.



Figure 39: RF (f1_pa) on Water-quality: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).



Figure 40: RF (f1_pa) on Water-quality — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

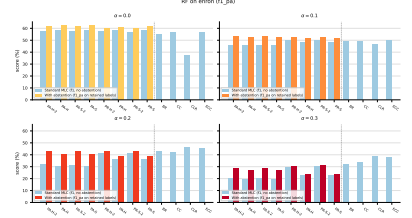
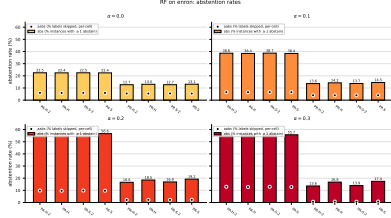


Figure 41: RF ($f1_{pa}$) on **enron**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

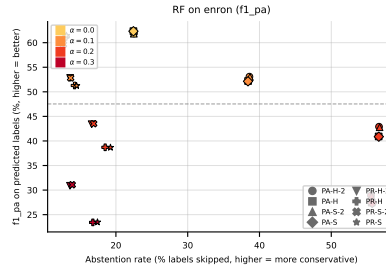


Figure 42: RF ($f1_{pa}$) on **enron**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

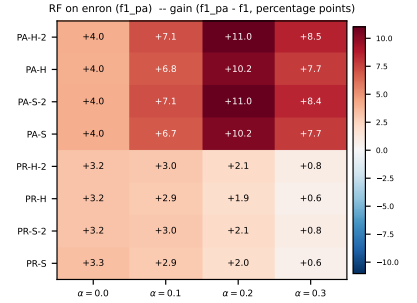
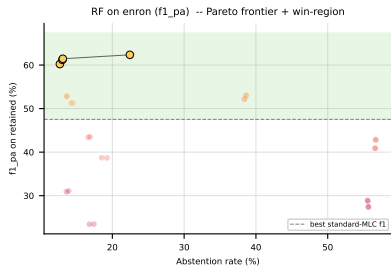


Figure 43: RF ($f1_{pa}$) on **enron**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).

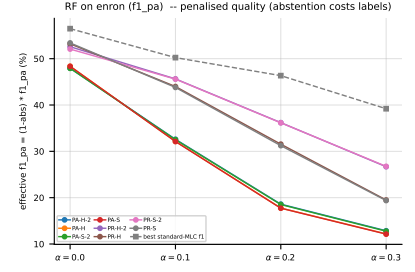
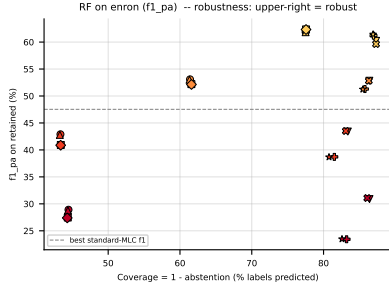


Figure 44: RF ($f1_{pa}$) on **enron** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.3 RF, metric jaccard_pa — aggregate



Figure 45: RF (jaccard_pa) averaged across all datasets: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

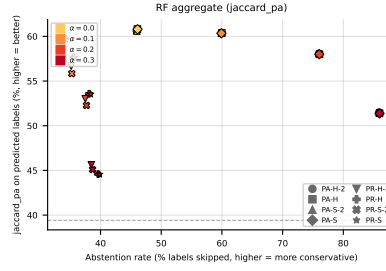


Figure 46: RF (jaccard_pa) averaged across all datasets: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



Figure 47: RF (jaccard_pa) averaged across all datasets: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).

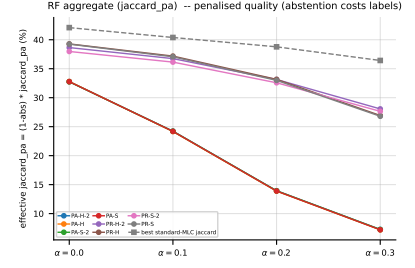
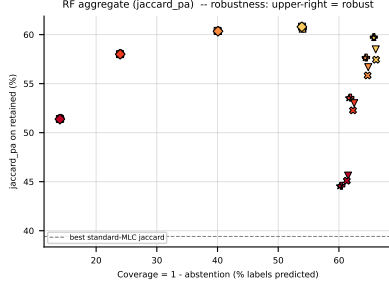


Figure 48: RF (jaccard_{pa}) averaged across all datasets — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_{pa} = $(1 - \text{abs}) \cdot \text{jaccard}_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.4 RF, metric jaccard_{pa} — per dataset

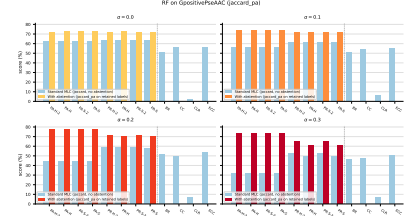
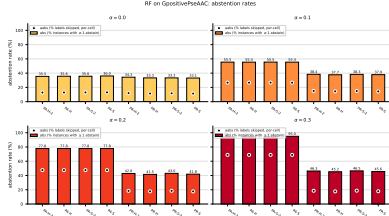


Figure 49: RF (jaccard_{pa}) on GpositivePseAAC: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

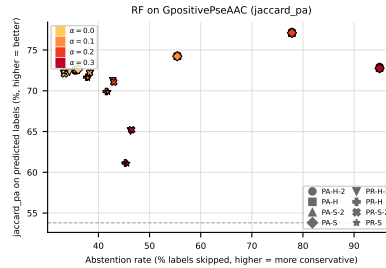


Figure 50: RF (jaccard_{pa}) on GpositivePseAAC: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

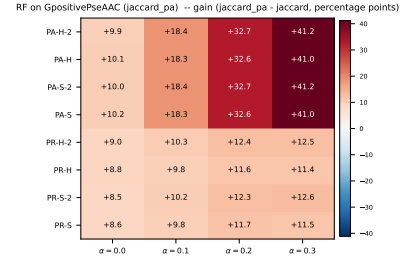
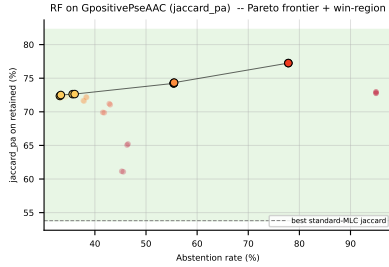


Figure 51: RF (jaccard_{pa}) on GpositivePseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_{pa} − f_1) heatmap (right).

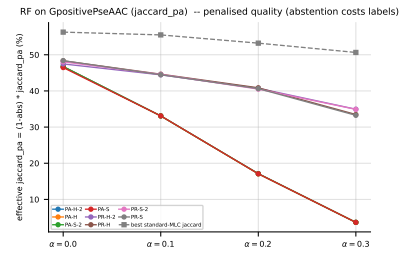
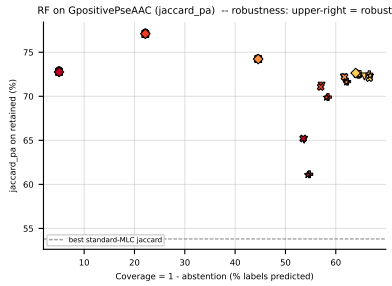


Figure 52: RF (jaccard_{pa}) on GpositivePseAAC — robustness view. Left: efficiency-quality scatter (x = coverage = 1 − abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_{pa} = (1 − abs) · jaccard_{pa}, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

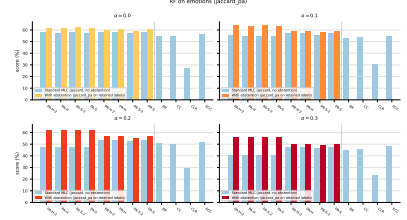
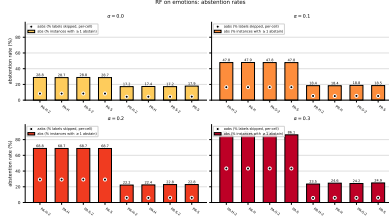


Figure 53: RF (jaccard_pa) on **emotions**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

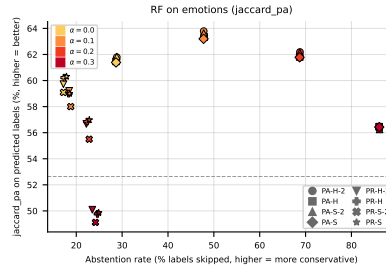


Figure 54: RF (jaccard_pa) on **emotions**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

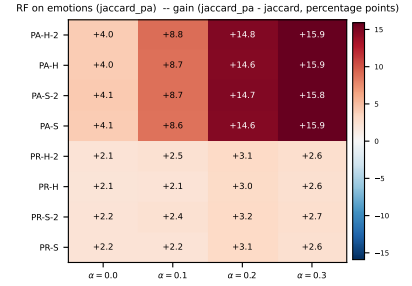
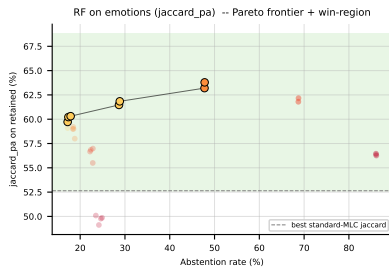


Figure 55: RF (jaccard_pa) on **emotions**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).

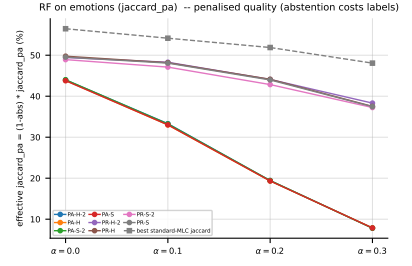
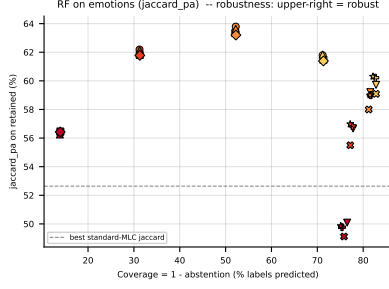


Figure 56: RF (jaccard_pa) on **emotions** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

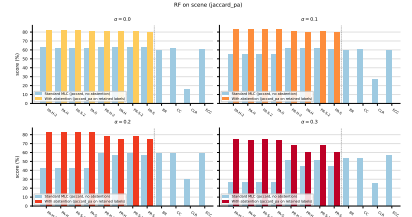
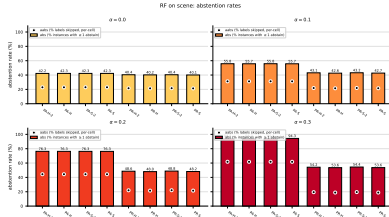


Figure 57: RF (jaccard_pa) on **scene**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

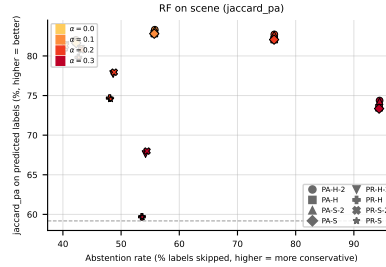


Figure 58: RF (jaccard_pa) on **scene**: coverage-risk scatter. Each point is one (method, noise) pair; $x = \text{abstention rate}$, $y = \text{quality on retained labels}$. Points above the dashed line beat the best standard-MLC baseline.

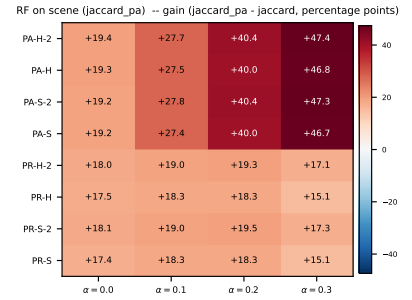
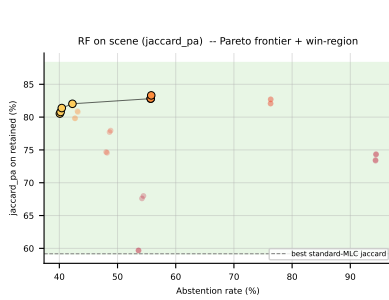


Figure 59: RF (jaccard_{pa}) on **scene**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_{pa} − f_1) heatmap (right).

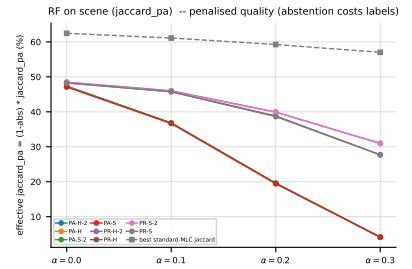
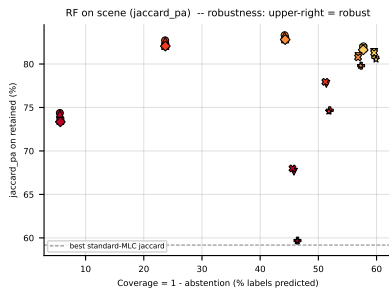


Figure 60: RF (jaccard_{pa}) on **scene** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_{pa} = $(1 - \text{abs}) \cdot \text{jaccard}_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

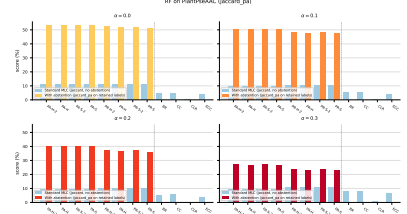
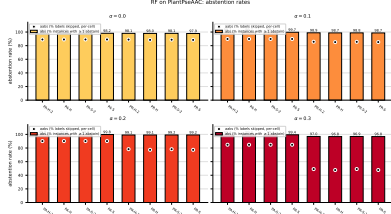


Figure 61: RF (jaccard_pa) on PlantPseAAC: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

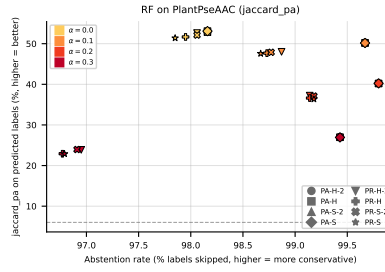


Figure 62: RF (jaccard_pa) on PlantPseAAC: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

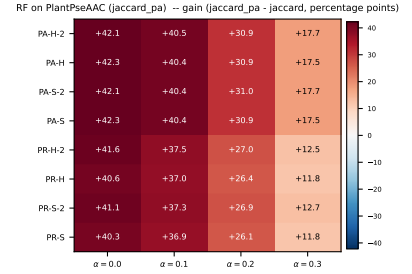
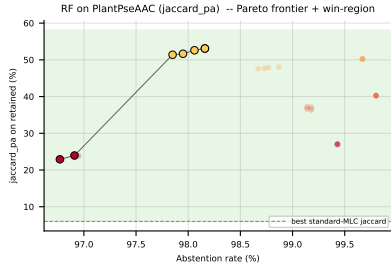


Figure 63: RF (jaccard_pa) on PlantPseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($jaccard_pa - f_1$) heatmap (right).

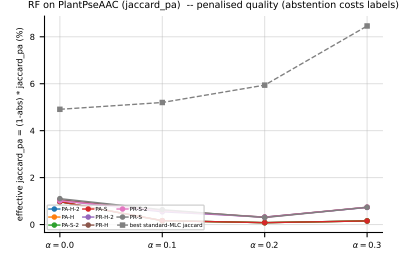
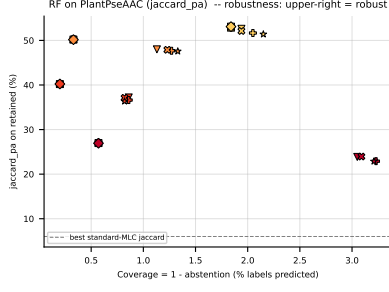


Figure 64: RF (jaccard_{pa}) on **PlantPseAAC** — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_{pa} = $(1 - \text{abs}) \cdot \text{jaccard}_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

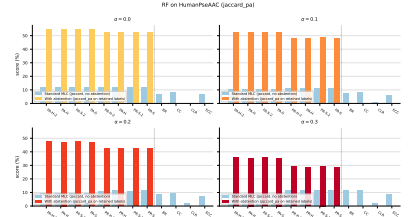
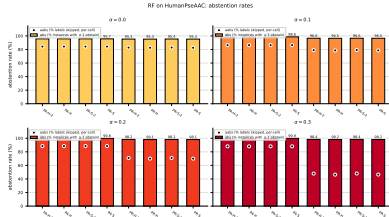


Figure 65: RF (jaccard_{pa}) on **HumanPseAAC**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

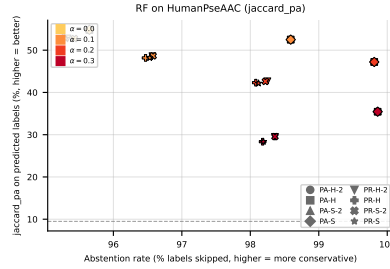


Figure 66: RF (jaccard_{pa}) on **HumanPseAAC**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

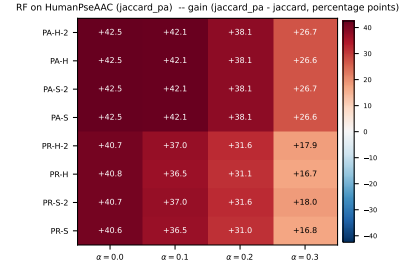
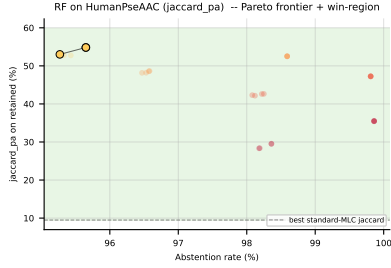


Figure 67: RF (jaccard_pa) on HumanPseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($jaccard_pa - f_1$) heatmap (right).

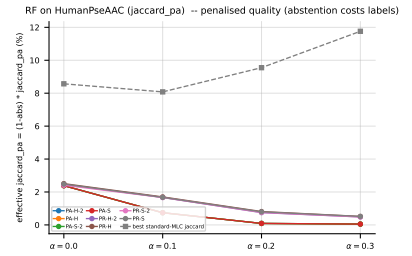
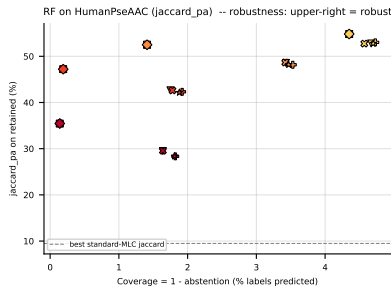


Figure 68: RF (jaccard_pa) on HumanPseAAC — robustness view. Left: efficiency-quality scatter ($x = coverage = 1 - abs$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - abs) \cdot jaccard_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

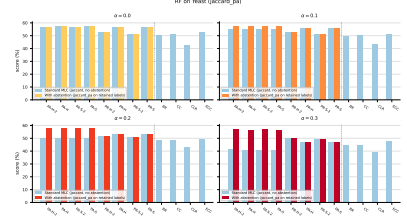
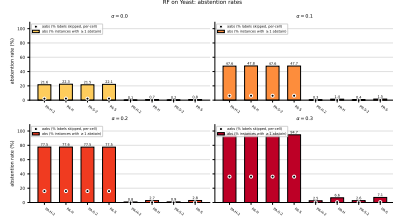


Figure 69: RF (jaccard_pa) on **Yeast**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

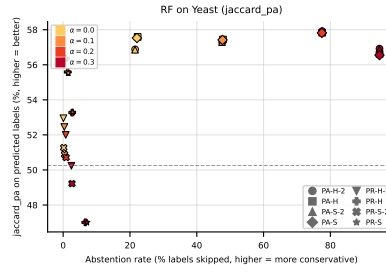


Figure 70: RF (jaccard_pa) on **Yeast**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

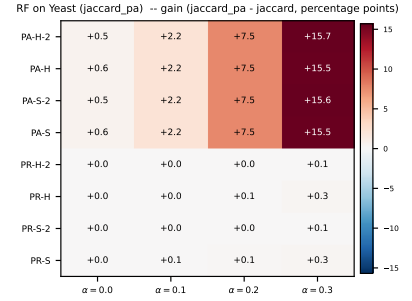
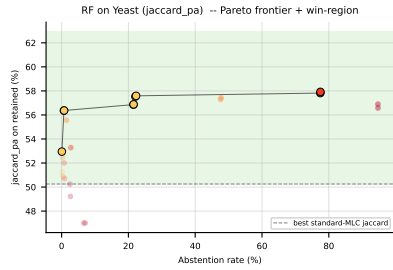


Figure 71: RF (jaccard_pa) on **Yeast**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).

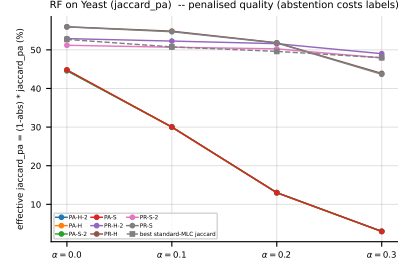
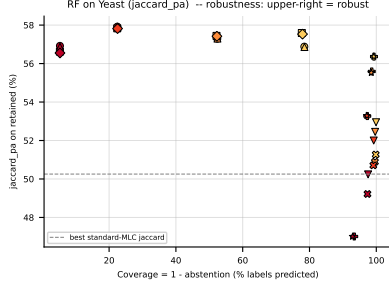


Figure 72: RF (jaccard_pa) on **Yeast** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

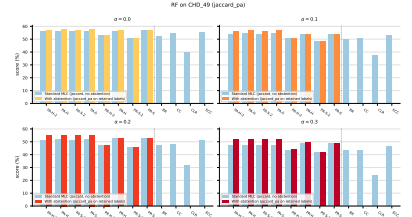
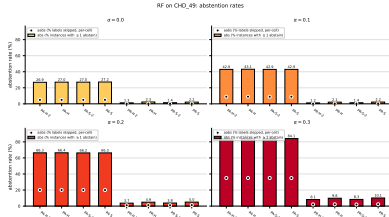


Figure 73: RF (jaccard_pa) on **CHD_49**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

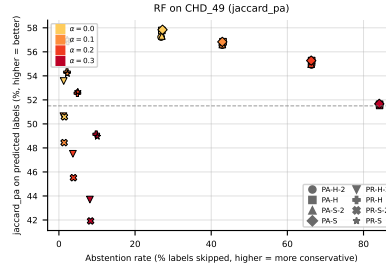


Figure 74: RF (jaccard_pa) on **CHD_49**: coverage-risk scatter. Each point is one (method, noise) pair; $x = \text{abstention rate}$, $y = \text{quality on retained labels}$. Points above the dashed line beat the best standard-MLC baseline.

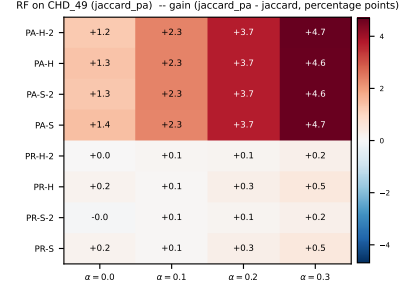
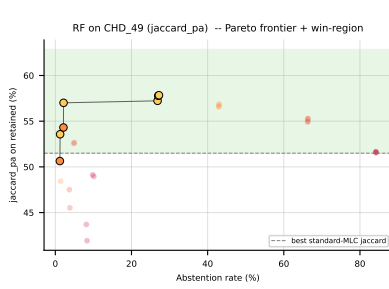


Figure 75: RF (jaccard.pa) on CHD_49: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($jaccard.pa - f_1$) heatmap (right).

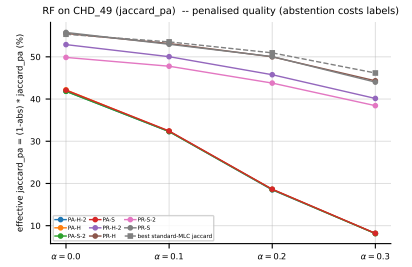
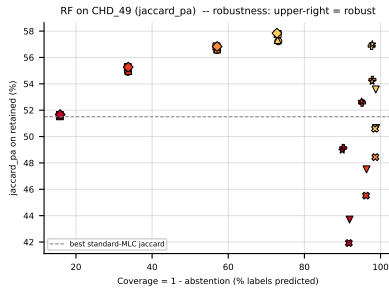


Figure 76: RF (jaccard.pa) on CHD_49 — robustness view. Left: efficiency-quality scatter ($x = coverage = 1 - abs$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard.pa = $(1 - abs) \cdot jaccard.pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

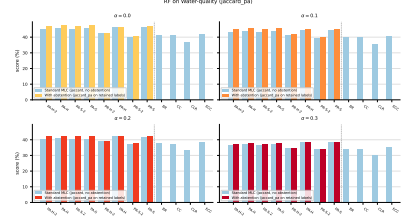
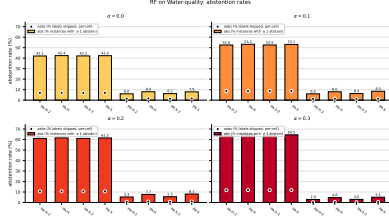


Figure 77: RF (jaccard_pa) on **Water-quality**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

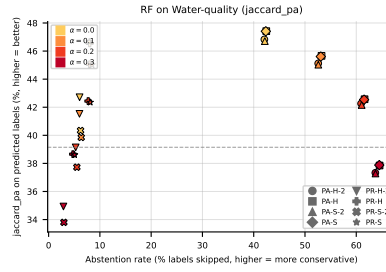


Figure 78: RF (jaccard_pa) on **Water-quality**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

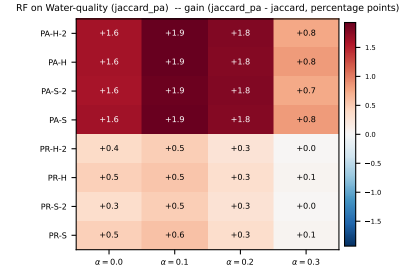
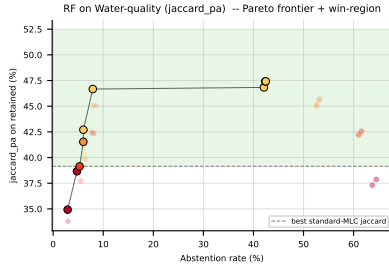


Figure 79: RF (jaccard_pa) on **Water-quality**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).

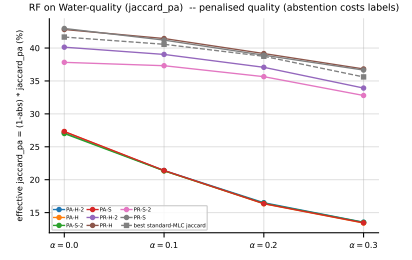
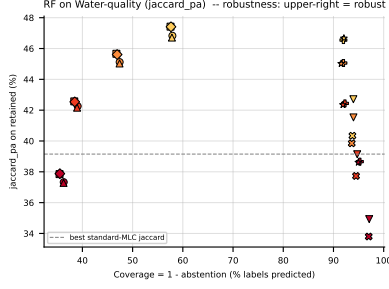


Figure 80: RF (jaccard_pa) on **Water-quality** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

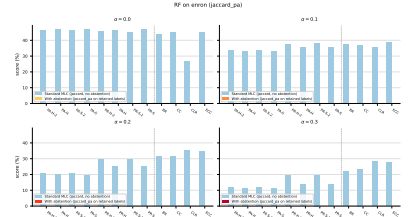
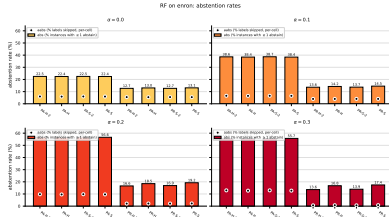


Figure 81: RF (jaccard_pa) on **enron**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

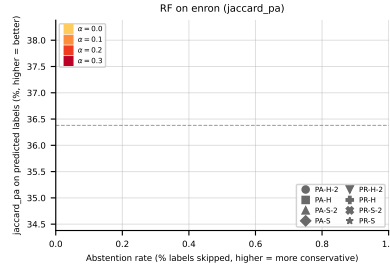


Figure 82: RF (jaccard_pa) on **enron**: coverage-risk scatter. Each point is one (method, noise) pair; $x = \text{abstention rate}$, $y = \text{quality on retained labels}$. Points above the dashed line beat the best standard-MLC baseline.



Figure 83: RF (jaccard_pa) on **enron**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $- f_1$) heatmap (right).



Figure 84: RF (jaccard_pa) on **enron** — robustness view. Left: efficiency-quality scatter (x = coverage = 1 - abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = (1 - abs) $\cdot$ jaccard_pa, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.5 LGBM, metric f1_pa — aggregate



Figure 85: LGBM (f1_pa) averaged across all datasets: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

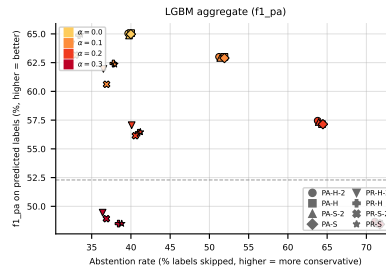


Figure 86: LGBM (f1_pa) averaged across all datasets: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



Figure 87: LGBM (f1_pa) averaged across all datasets: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).

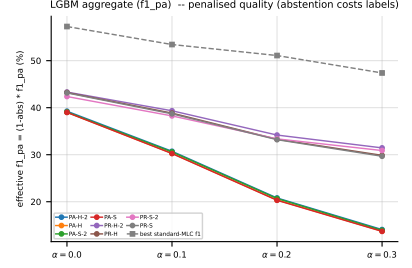
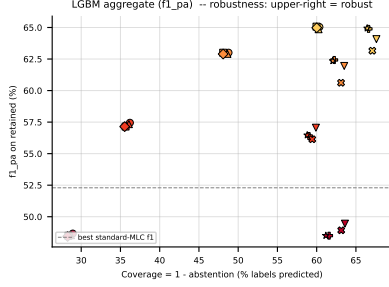


Figure 88: LGBM ($f1_{pa}$) averaged across all datasets — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.6 LGBM, metric $f1_{pa}$ — per dataset

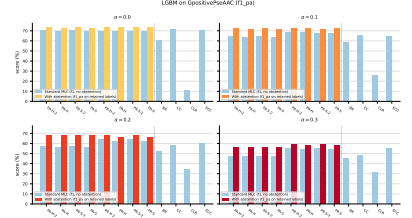
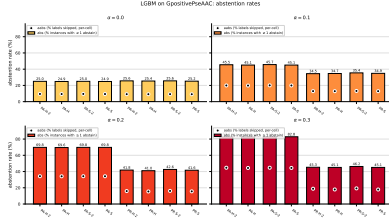


Figure 89: LGBM ($f1_{pa}$) on **GpositivePseAAC**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

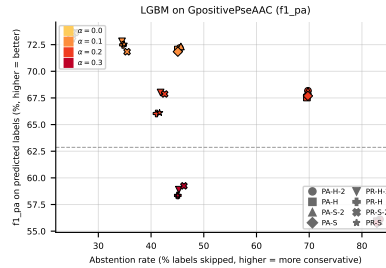


Figure 90: LGBM ($f1_{pa}$) on **GpositivePseAAC**: coverage-risk scatter. Each point is one (method, noise) pair; $x = \text{abstention rate}$, $y = \text{quality on retained labels}$. Points above the dashed line beat the best standard-MLC baseline.



Figure 91: LGBM ($f1_{pa}$) on **GpositivePseAAC**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).



Figure 92: LGBM ($f1_{pa}$) on **GpositivePseAAC** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

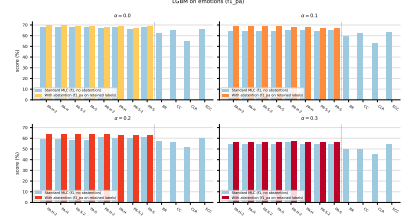
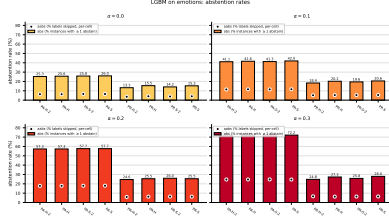


Figure 93: LGBM ($f1_{pa}$) on **emotions**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

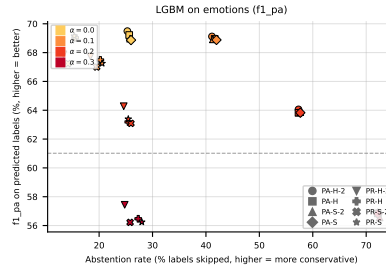


Figure 94: LGBM ($f1_{pa}$) on **emotions**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

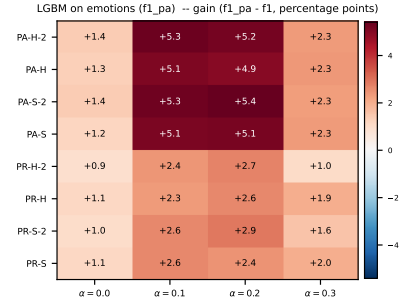
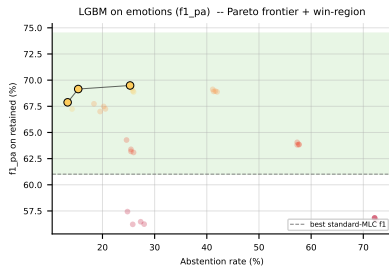


Figure 95: LGBM ($f1_{pa}$) on **emotions**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f_1$) heatmap (right).

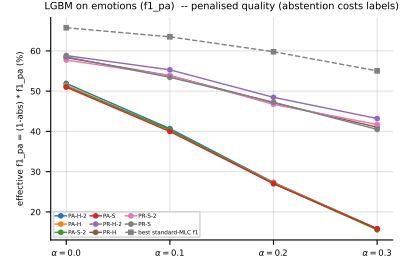
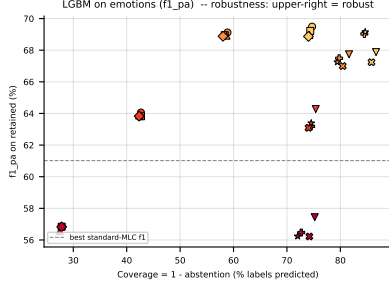


Figure 96: LGBM ($f1_{pa}$) on **emotions** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

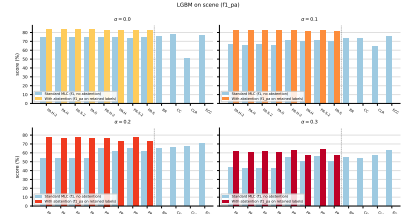
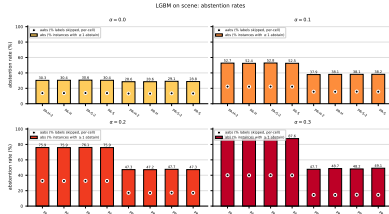


Figure 97: LGBM ($f1_{pa}$) on **scene**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

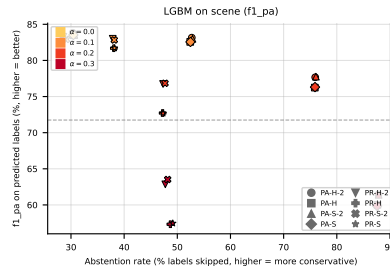


Figure 98: LGBM ($f1_{pa}$) on **scene**: coverage-risk scatter. Each point is one (method, noise) pair; $x = \text{abstention rate}$, $y = \text{quality on retained labels}$. Points above the dashed line beat the best standard-MLC baseline.

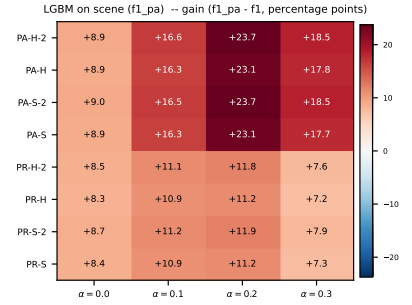
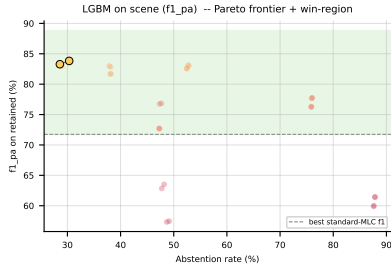


Figure 99: LGBM ($f1_{pa}$) on **scene**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).

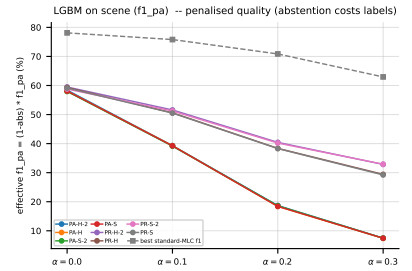
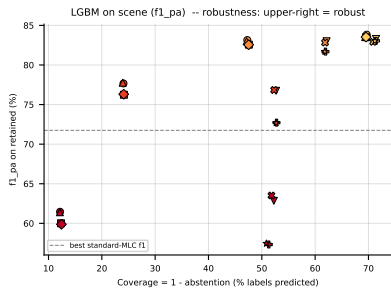


Figure 100: LGBM ($f1_{pa}$) on **scene** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

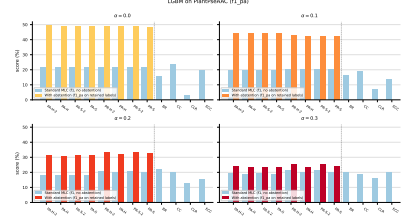
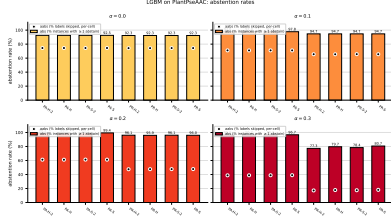


Figure 101: LGBM ($f1_pa$) on PlantPseAAC: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

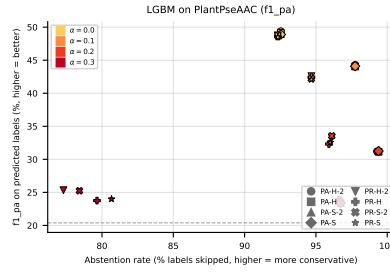


Figure 102: LGBM ($f1_pa$) on PlantPseAAC: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

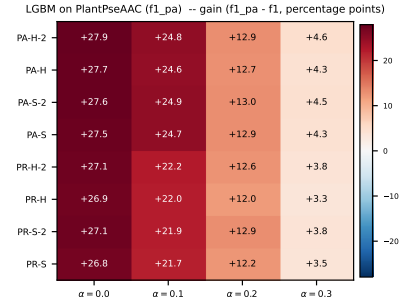
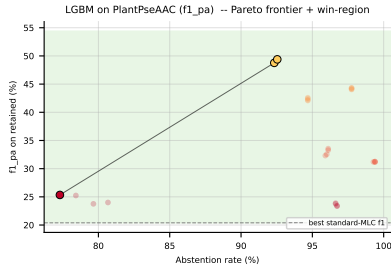


Figure 103: LGBM ($f1_pa$) on PlantPseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_pa - f1$) heatmap (right).

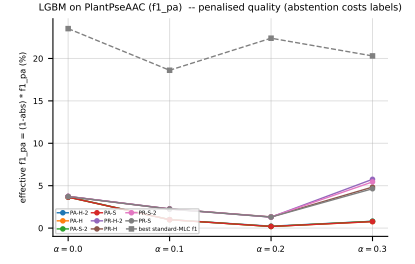
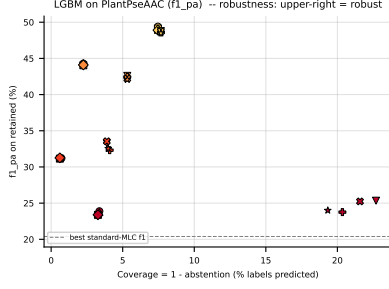


Figure 104: LGBM ($f1_{pa}$) on PlantPseAAC — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

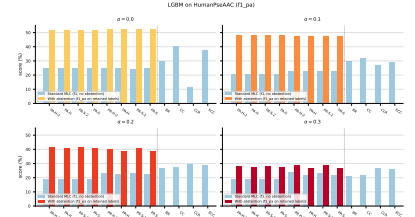
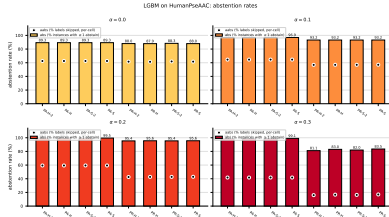


Figure 105: LGBM ($f1_{pa}$) on HumanPseAAC: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

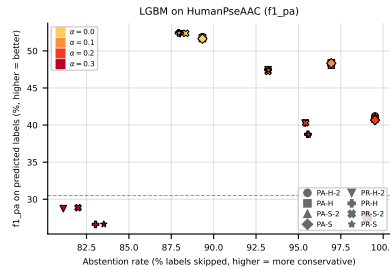


Figure 106: LGBM ($f1_{pa}$) on HumanPseAAC: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



Figure 107: LGBM ($f1_{pa}$) on HumanPseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).



Figure 108: LGBM ($f1_{pa}$) on HumanPseAAC — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

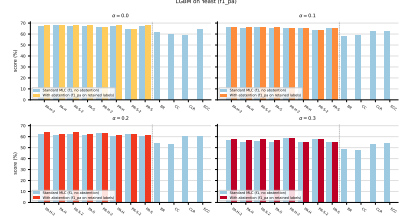
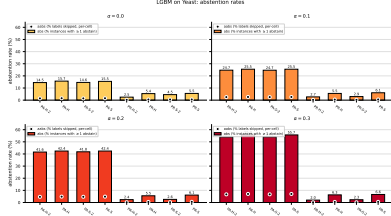


Figure 109: LGBM ($f1_{pa}$) on **Yeast**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

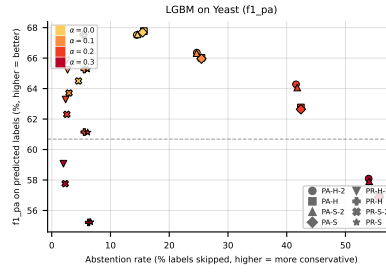


Figure 110: LGBM ($f1_{pa}$) on **Yeast**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

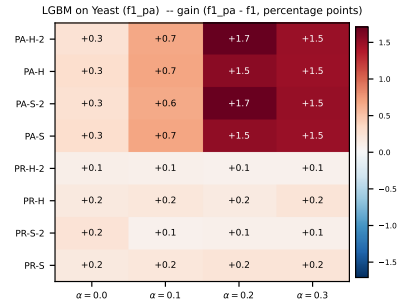
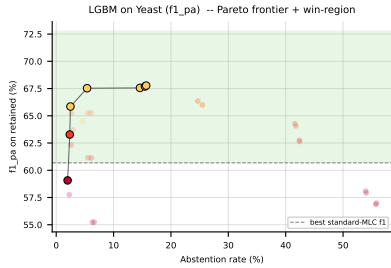


Figure 111: LGBM ($f1_{pa}$) on **Yeast**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f_1$) heatmap (right).

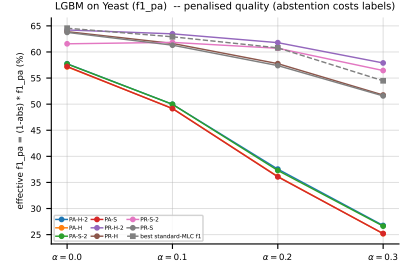
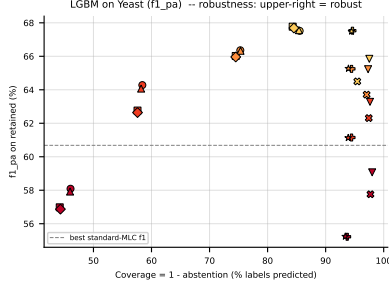


Figure 112: LGBM ($f1_{pa}$) on Yeast — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

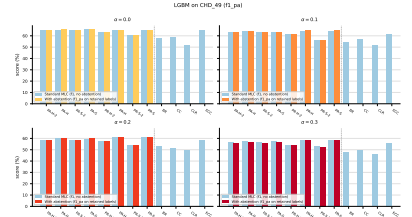
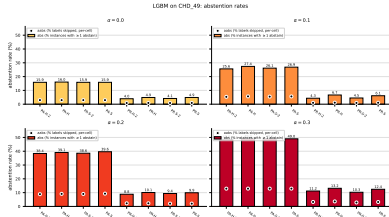


Figure 113: LGBM ($f1_{pa}$) on CHD_49: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

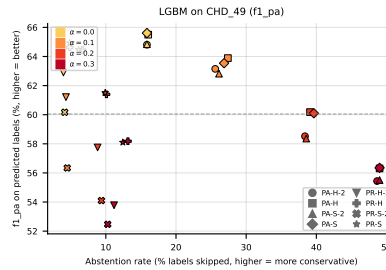


Figure 114: LGBM ($f1_{pa}$) on CHD_49: coverage-risk scatter. Each point is one (method, noise) pair; $x = \text{abstention rate}$, $y = \text{quality on retained labels}$. Points above the dashed line beat the best standard-MLC baseline.

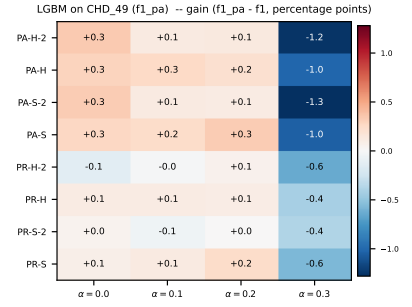
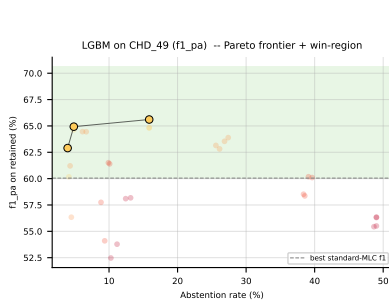


Figure 115: LGBM ($f1_{pa}$) on CHD_49: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).

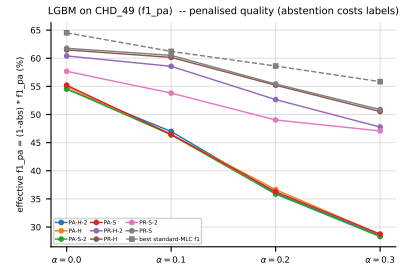
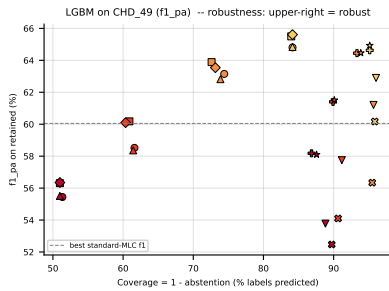


Figure 116: LGBM ($f1_{pa}$) on CHD_49 — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

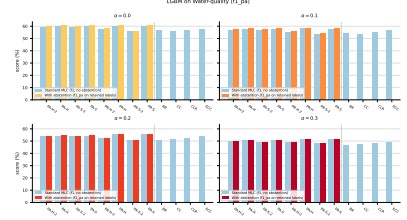
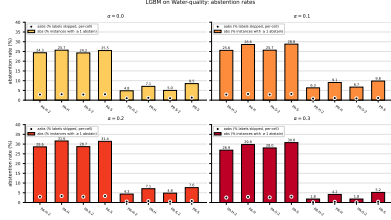


Figure 117: LGBM (f1_pa) on Water-quality: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

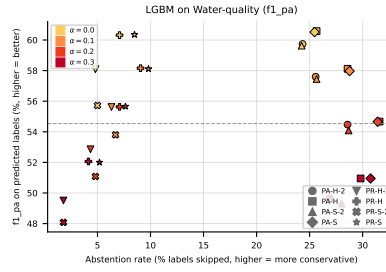


Figure 118: LGBM (f1_pa) on Water-quality: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

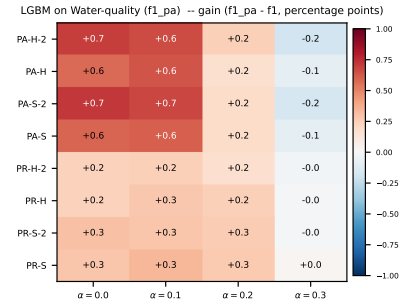
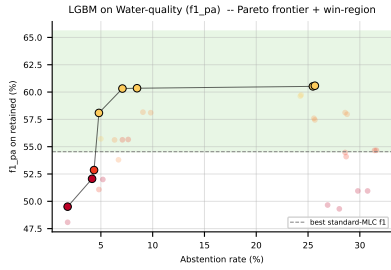


Figure 119: LGBM (f1_pa) on Water-quality: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).

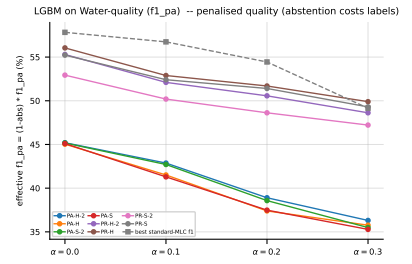
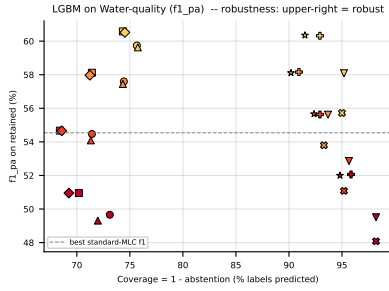


Figure 120: LGBM ($f1_{pa}$) on **Water-quality** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.7 LGBM, metric jaccard_pa — aggregate



Figure 121: LGBM (jaccard_pa) averaged across all datasets: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

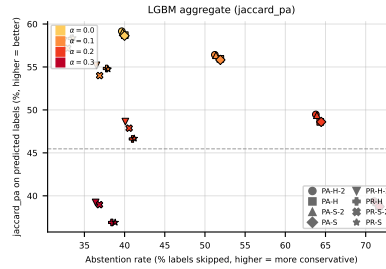


Figure 122: LGBM (jaccard_pa) averaged across all datasets: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



Figure 123: LGBM (jaccard_pa) averaged across all datasets: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).

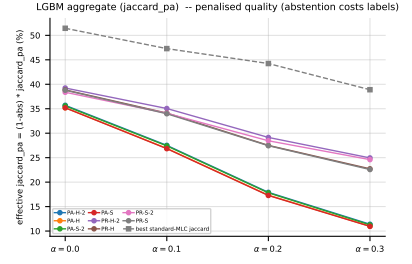
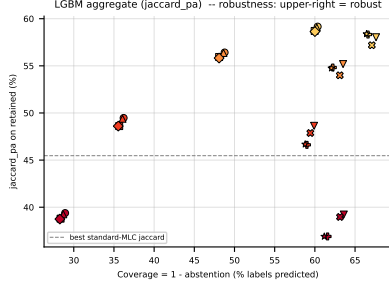


Figure 124: LGBM (`jaccard_pa`) averaged across all datasets — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective `jaccard_pa` = $(1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.8 LGBM, metric `jaccard_pa` — per dataset

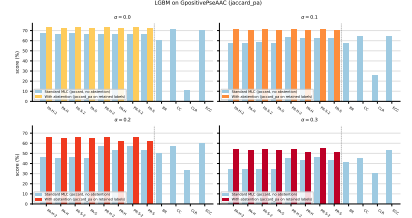
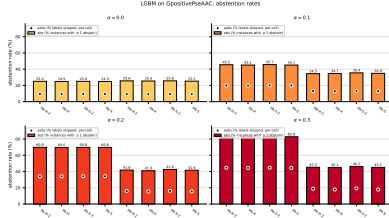


Figure 125: LGBM (`jaccard_pa`) on `GpositivePseAAC`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

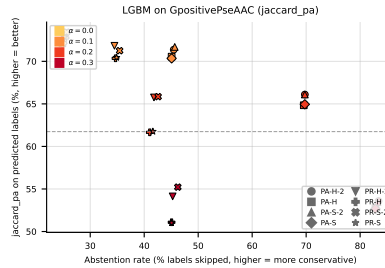


Figure 126: LGBM (`jaccard_pa`) on `GpositivePseAAC`: coverage-risk scatter. Each point is one (method, noise) pair; $x = \text{abstention rate}$, $y = \text{quality on retained labels}$. Points above the dashed line beat the best standard-MLC baseline.

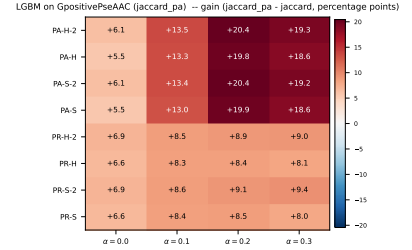
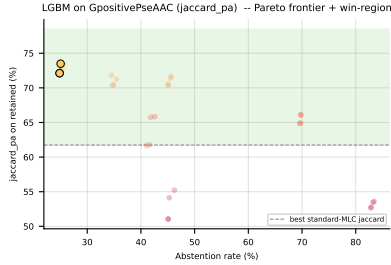


Figure 127: LGBM (jaccard_pa) on GpositivePseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa - f_1) heatmap (right).

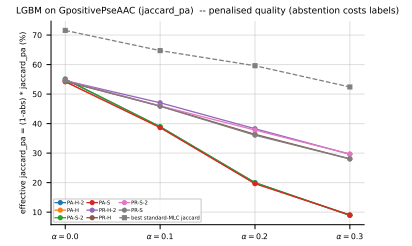
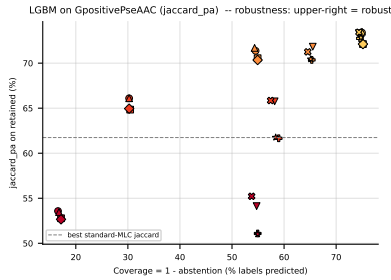


Figure 128: LGBM (jaccard_pa) on GpositivePseAAC — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

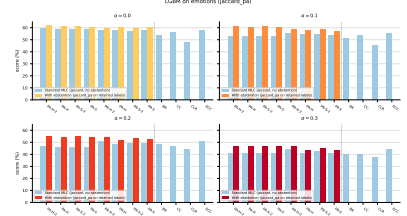
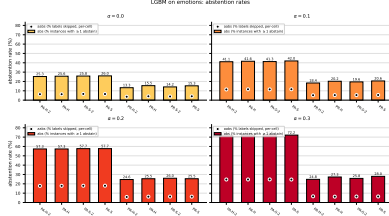


Figure 129: LGBM (jaccard_pa) on **emotions**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

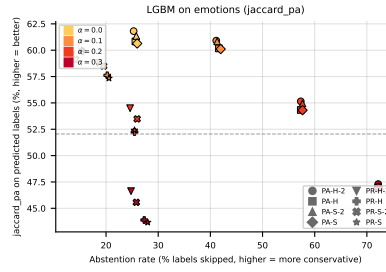


Figure 130: LGBM (jaccard_pa) on **emotions**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

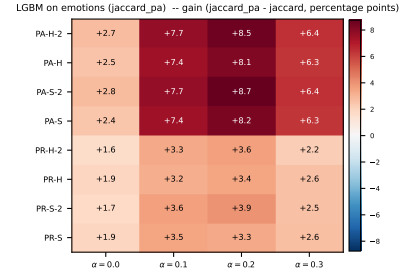
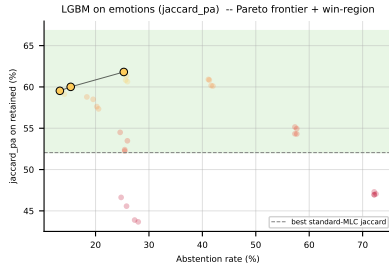


Figure 131: LGBM (jaccard_pa) on **emotions**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).

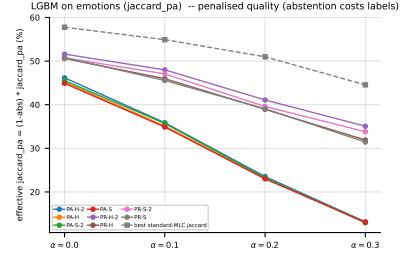
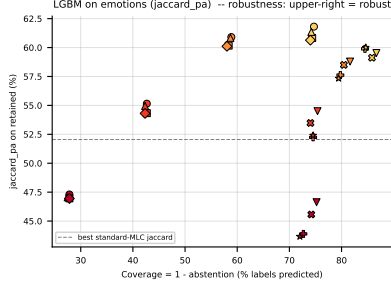


Figure 132: LGBM ($jaccard_pa$) on **emotions** — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - abs$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $jaccard_pa = (1 - abs) \cdot jaccard_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

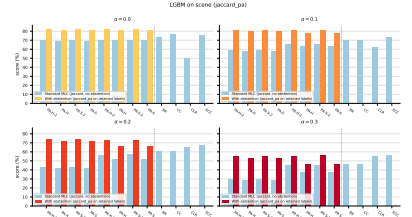
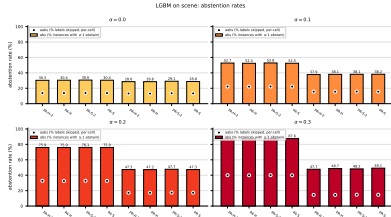


Figure 133: LGBM ($jaccard_pa$) on **scene**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

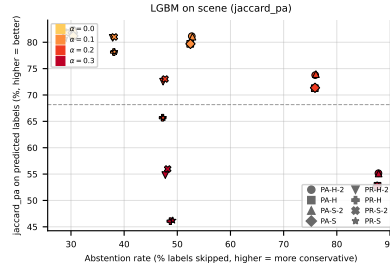


Figure 134: LGBM ($jaccard_pa$) on **scene**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

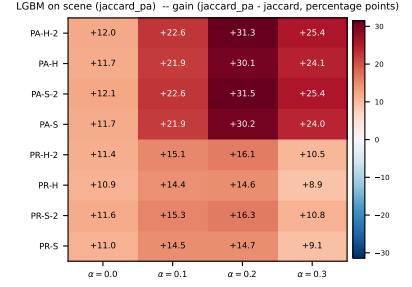
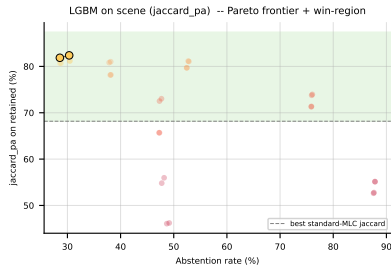


Figure 135: LGBM (jaccard_pa) on **scene**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).

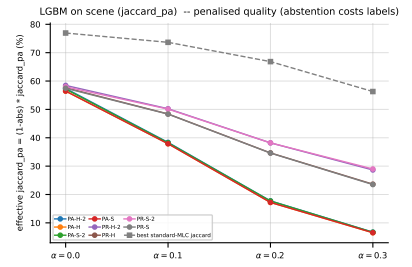
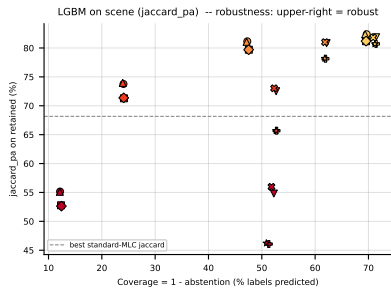


Figure 136: LGBM (jaccard_pa) on **scene** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $\text{jaccard_pa} = (1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

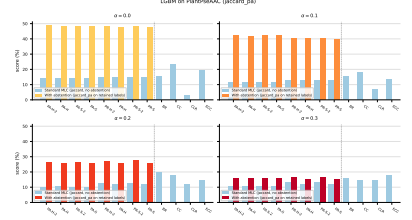
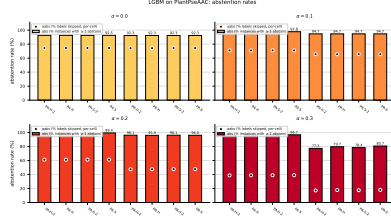


Figure 137: LGBM (jaccard_pa) on PlantPseAAC: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

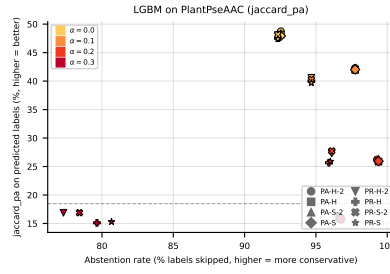


Figure 138: LGBM (jaccard_pa) on PlantPseAAC: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

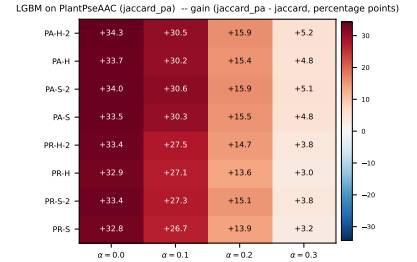
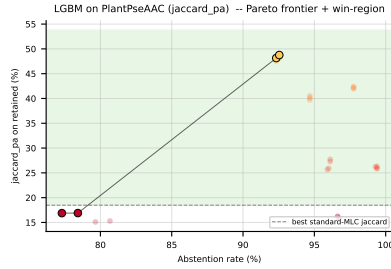


Figure 139: LGBM (jaccard_pa) on PlantPseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).

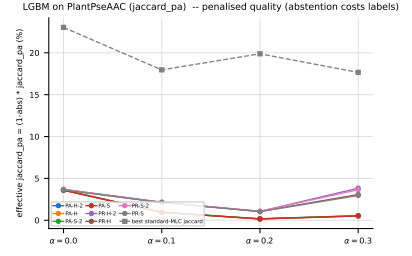
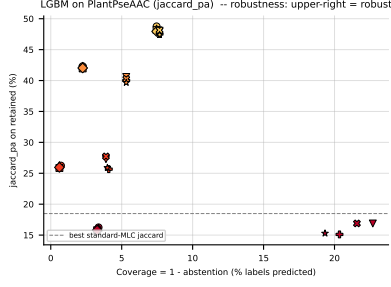


Figure 140: LGBM (jaccard_pa) on PlantPseAAC — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

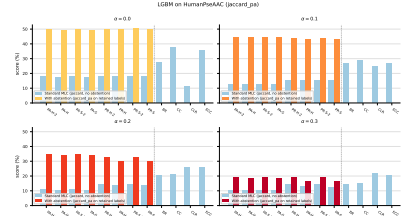
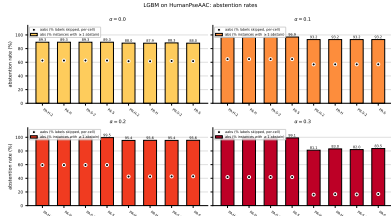


Figure 141: LGBM (jaccard_pa) on HumanPseAAC: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

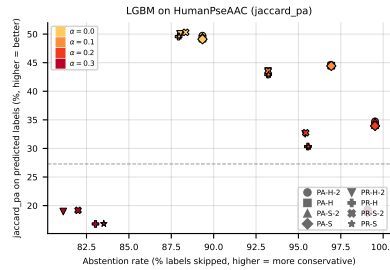


Figure 142: LGBM (jaccard_pa) on HumanPseAAC: coverage-risk scatter. Each point is one (method, noise) pair; $x = \text{abstention rate}$, $y = \text{quality on retained labels}$. Points above the dashed line beat the best standard-MLC baseline.

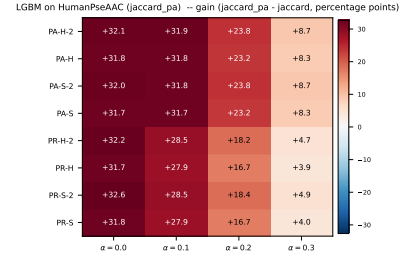
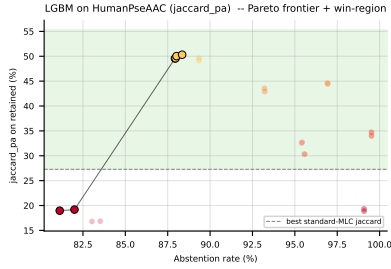


Figure 143: LGBM (jaccard_pa) on HumanPseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).

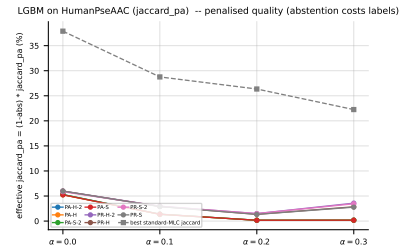
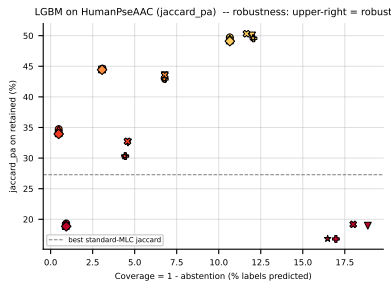


Figure 144: LGBM (jaccard_pa) on HumanPseAAC — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $\text{jaccard_pa} = (1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

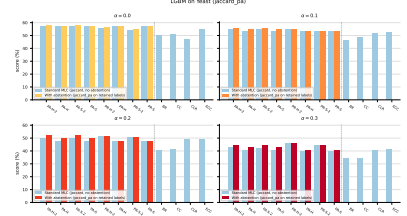
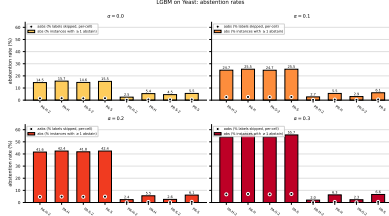


Figure 145: LGBM (jaccard_pa) on **Yeast**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

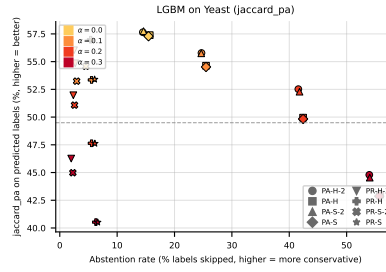


Figure 146: LGBM (jaccard_pa) on **Yeast**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

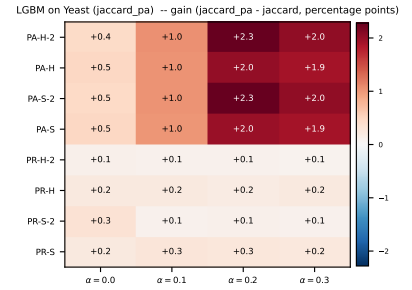
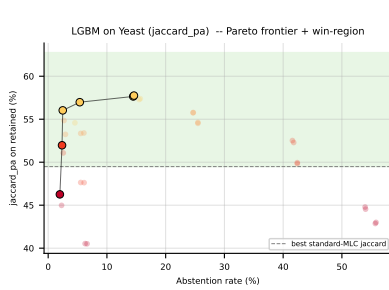


Figure 147: LGBM (jaccard_pa) on **Yeast**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).

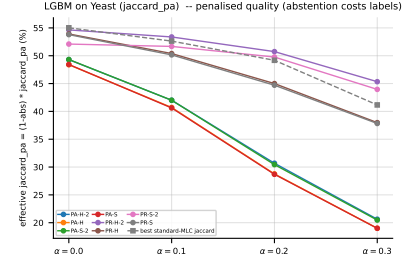
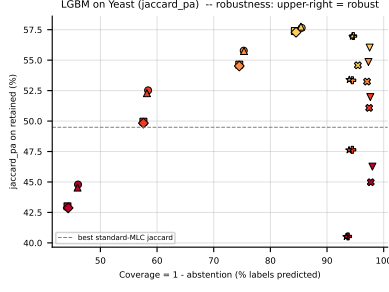


Figure 148: LGBM (jaccard_pa) on **Yeast** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

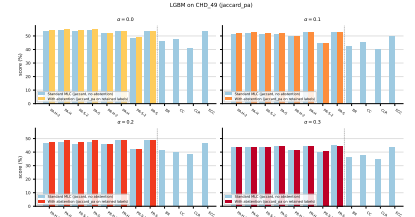
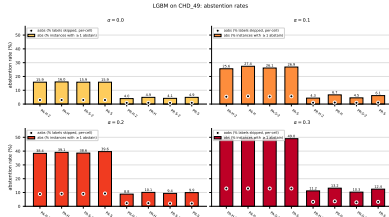


Figure 149: LGBM (jaccard_pa) on **CHD_49**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

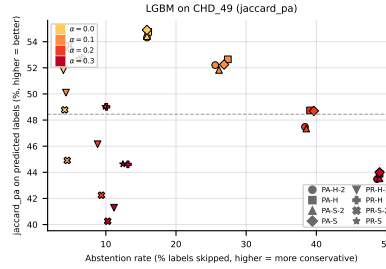


Figure 150: LGBM (jaccard_pa) on **CHD_49**: coverage-risk scatter. Each point is one (method, noise) pair; $x = \text{abstention rate}$, $y = \text{quality on retained labels}$. Points above the dashed line beat the best standard-MLC baseline.

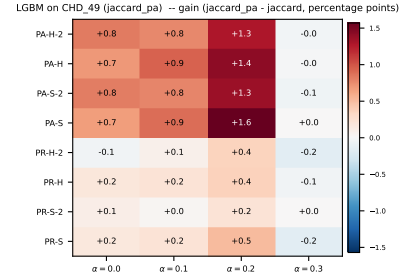
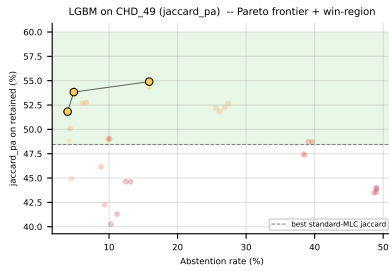


Figure 151: LGBM (jaccard_pa) on CHD_49: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($jaccard_pa - f_1$) heatmap (right).

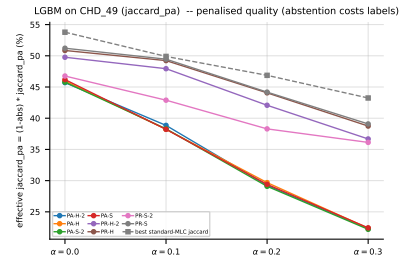
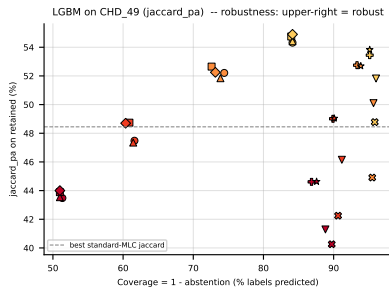


Figure 152: LGBM (jaccard_pa) on CHD_49 — robustness view. Left: efficiency-quality scatter ($x = coverage = 1 - abs$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $jaccard_pa = (1 - abs) \cdot jaccard_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

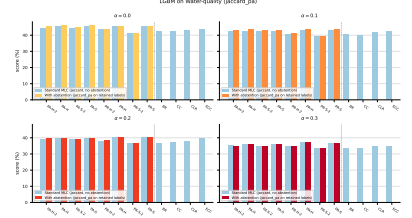
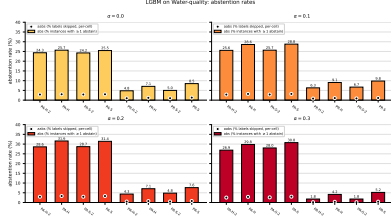


Figure 153: LGBM (jaccard_pa) on **Water-quality**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

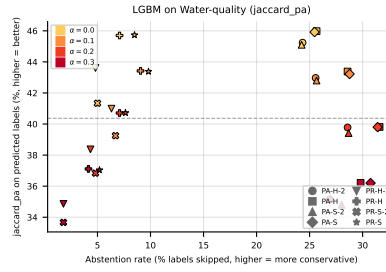


Figure 154: LGBM (jaccard_pa) on **Water-quality**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

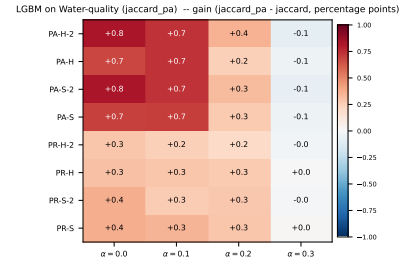
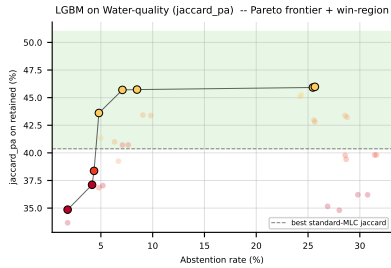


Figure 155: LGBM (jaccard_pa) on **Water-quality**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).

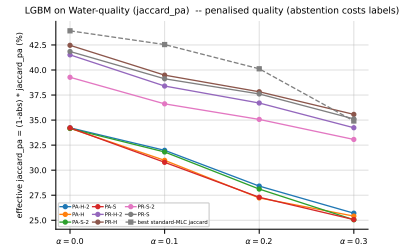
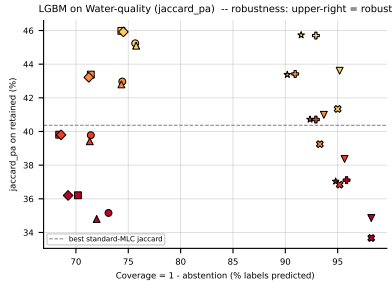
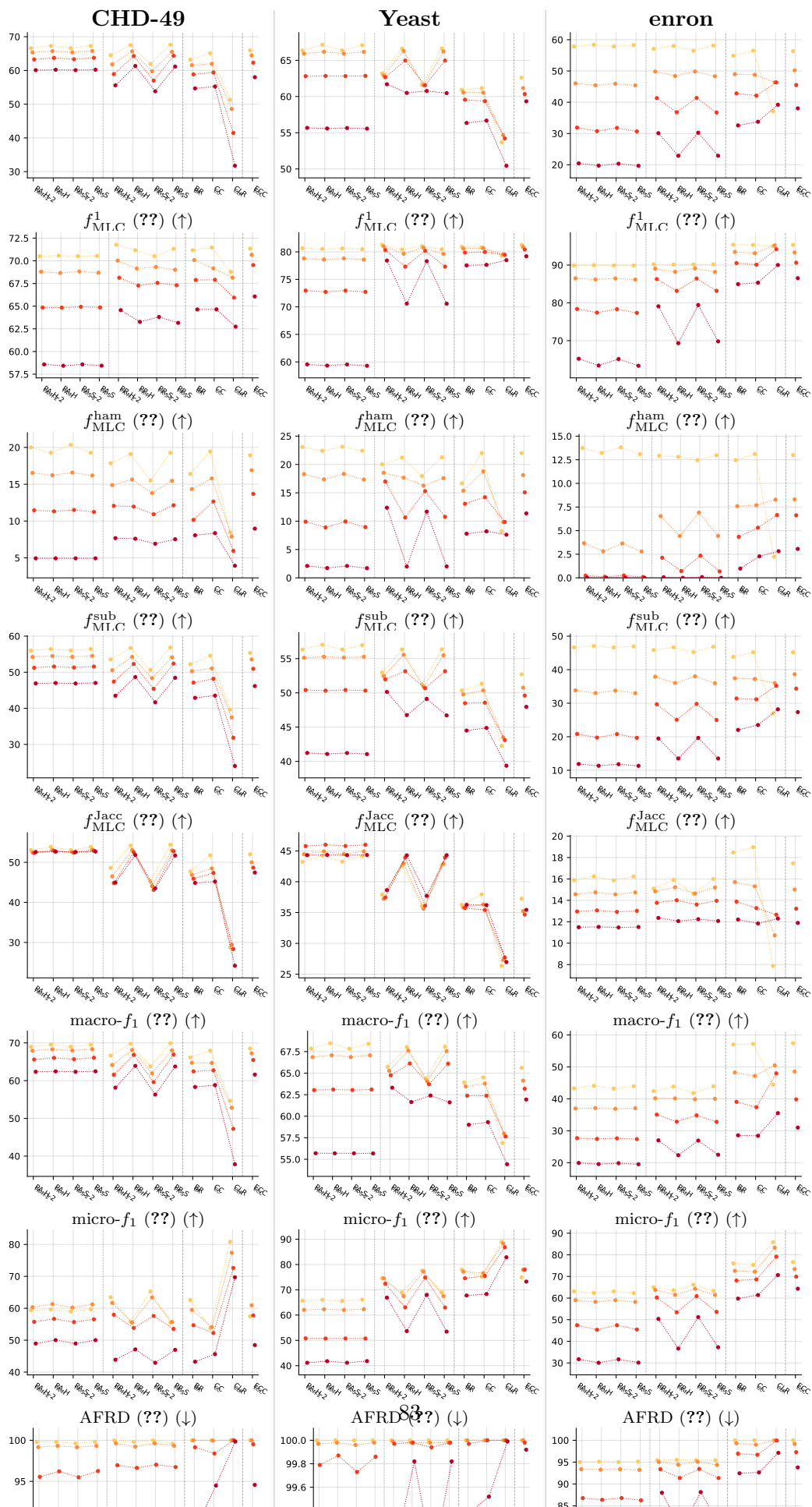


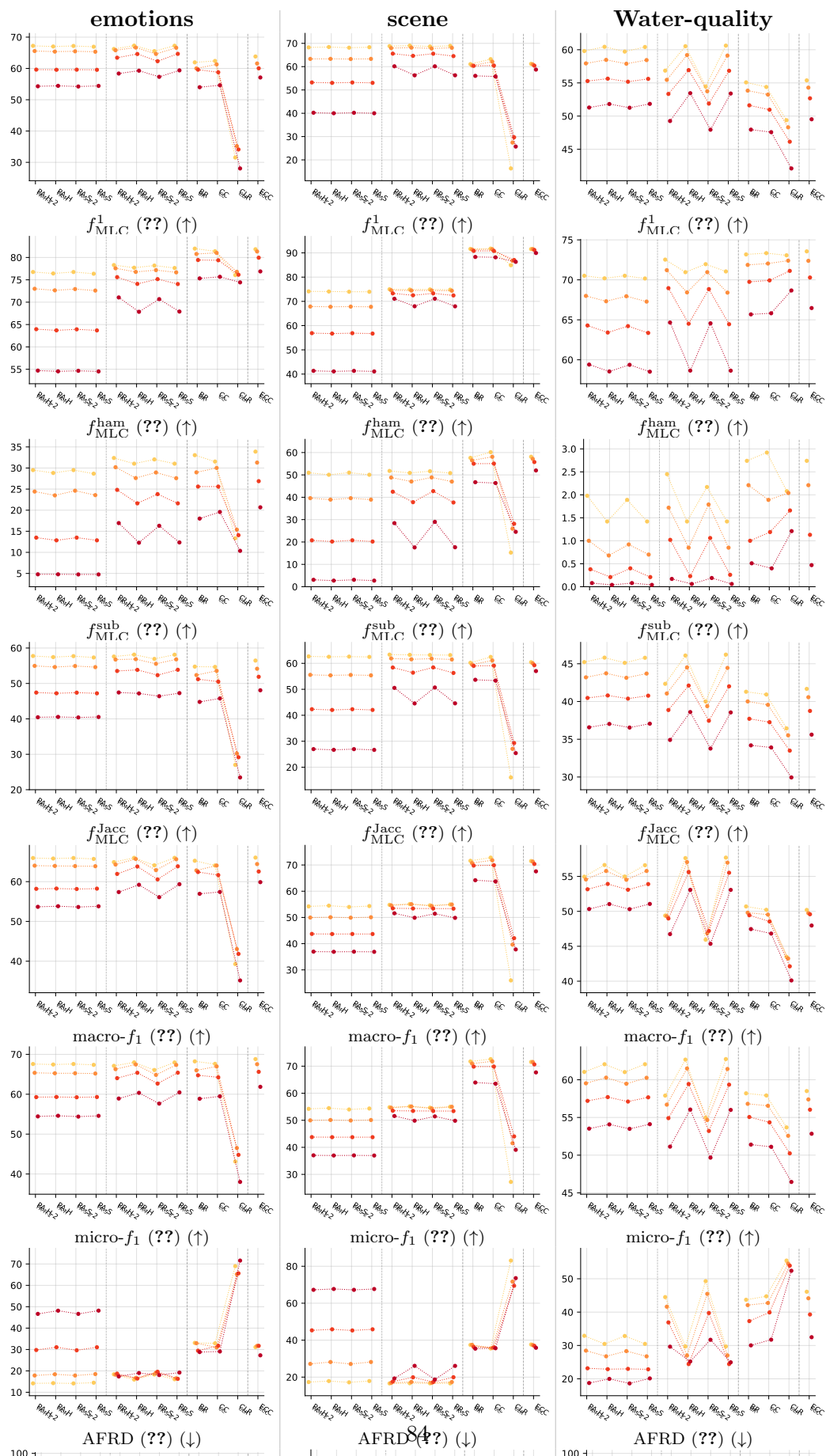
Figure 156: LGBM (jaccard_pa) on **Water-quality** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $\text{jaccard_pa} = (1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

A Experimental results — appendix tables (RF base learner)

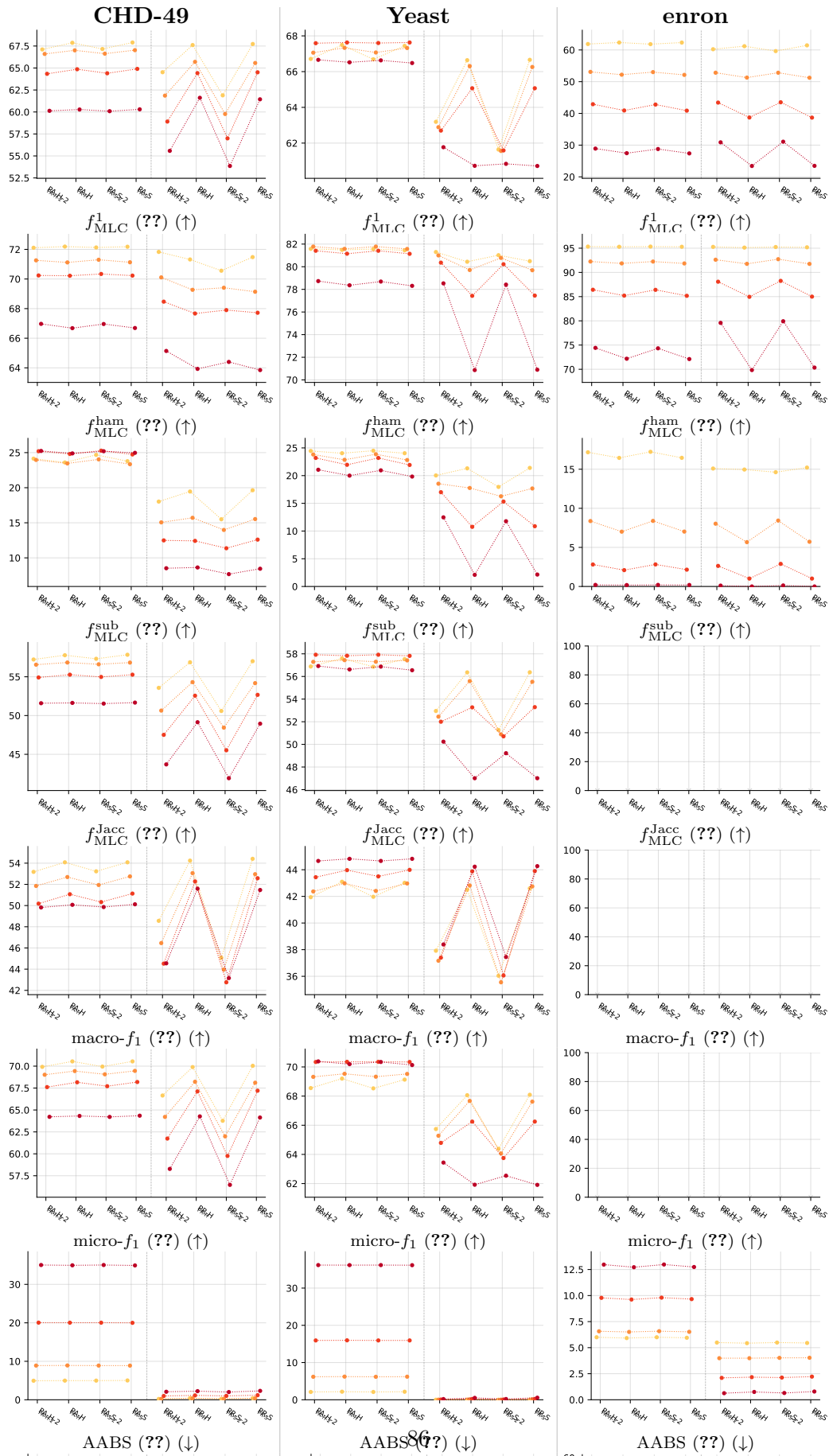
Appendix counterpart of the main Paper results section. Tables All tables share the same 8-metric set (original 5 + 3 revision additions: $f_{\text{MLC}}^{\text{Jacc}}$, macro- f_1 , micro- f_1). G2/G4 = imbalanced datasets, G3/G5 = balanced datasets (split follows the original paper’s appendix). G6/G7 = main-paper datasets (GpositivePse, PlantPse, HumanPse) shown with the full metric set; the main body only carries the revised compact Tables 4/5.

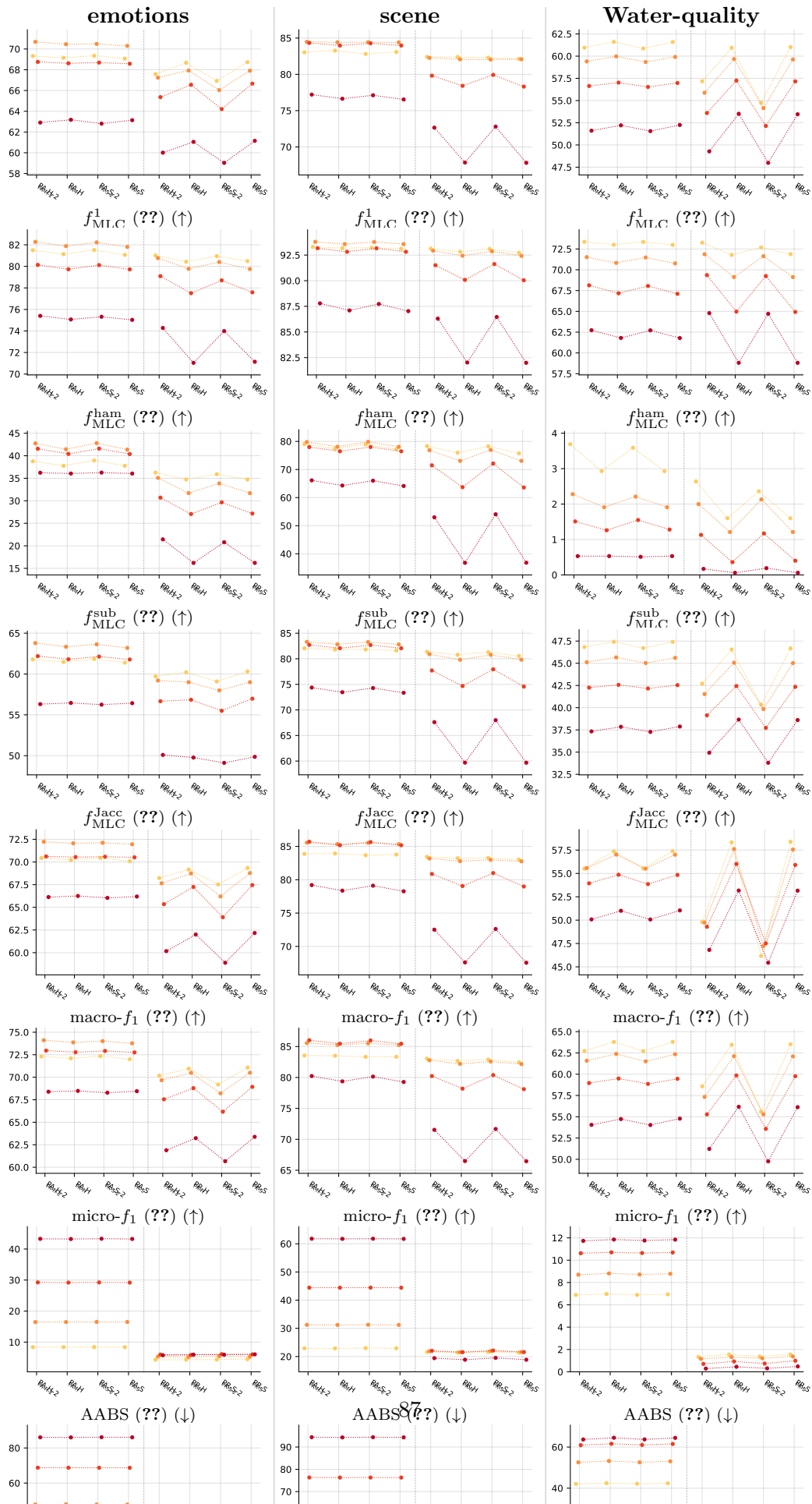
A.1 Standard MLC predictions (BinaryVector)





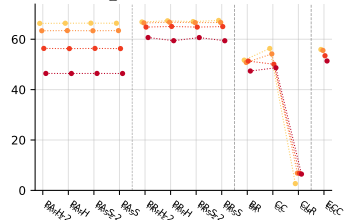
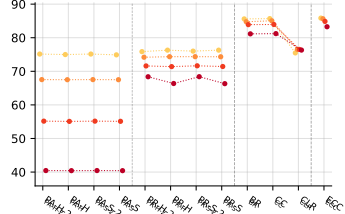
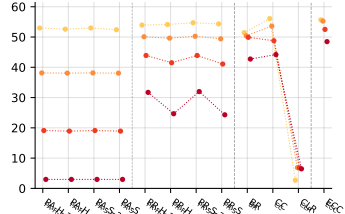
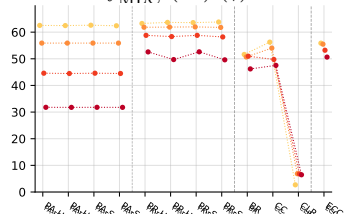
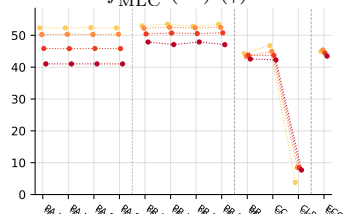
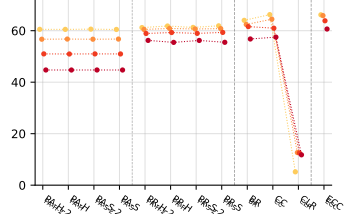
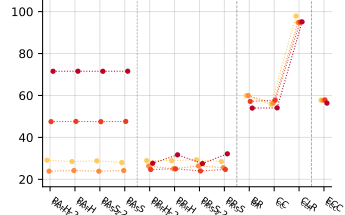
A.2 Partial abstention (PartialAbstention)





A.3 Standard MLC predictions (BinaryVector)

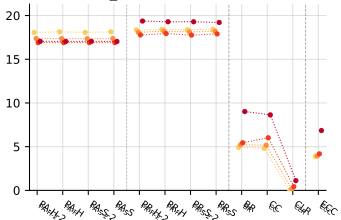
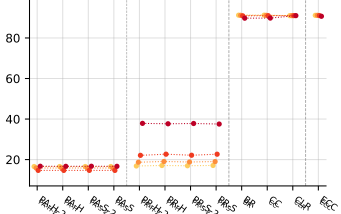
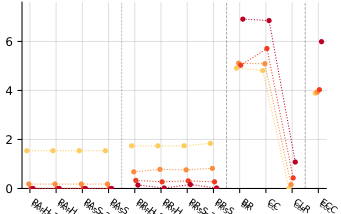
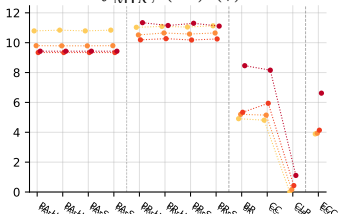
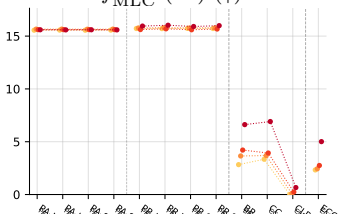
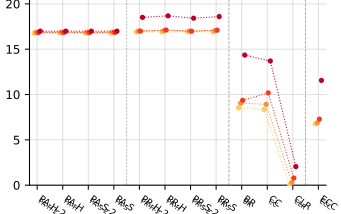
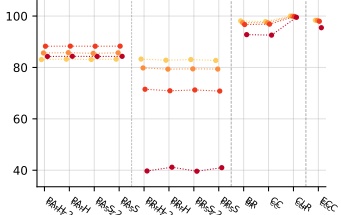
GpositivePse

 f_{MLC}^1 (??) (↑) f_{MLC}^{ham} (??) (↑) f_{MLC}^{sub} (??) (↑) f_{MLC}^{Jacc} (??) (↑)macro- f_1 (??) (↑)micro- f_1 (??) (↑)

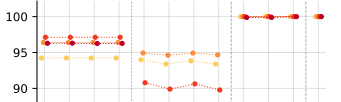
AFRD (??) (↓)



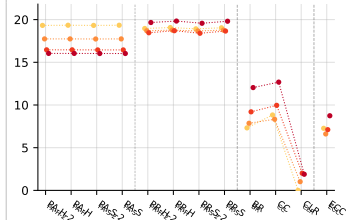
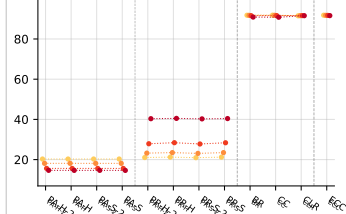
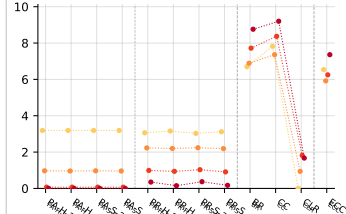
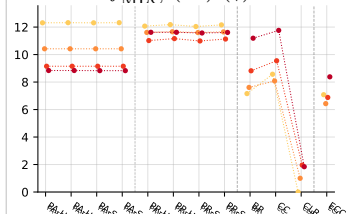
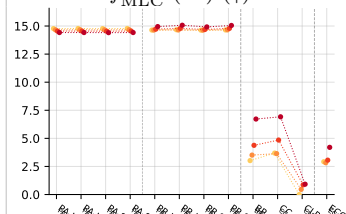
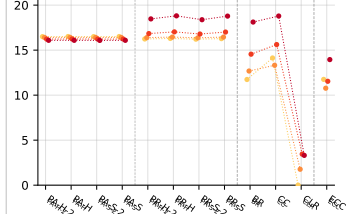
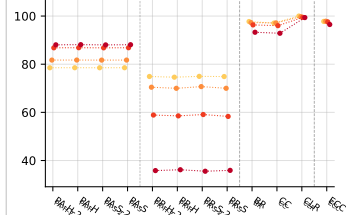
plantPse

 f_{MLC}^1 (??) (↑) f_{MLC}^{ham} (??) (↑) f_{MLC}^{sub} (??) (↑) f_{MLC}^{Jacc} (??) (↑)macro- f_1 (??) (↑)micro- f_1 (??) (↑)

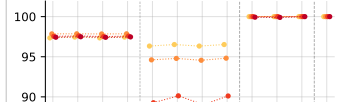
AFRD (??) (↓)



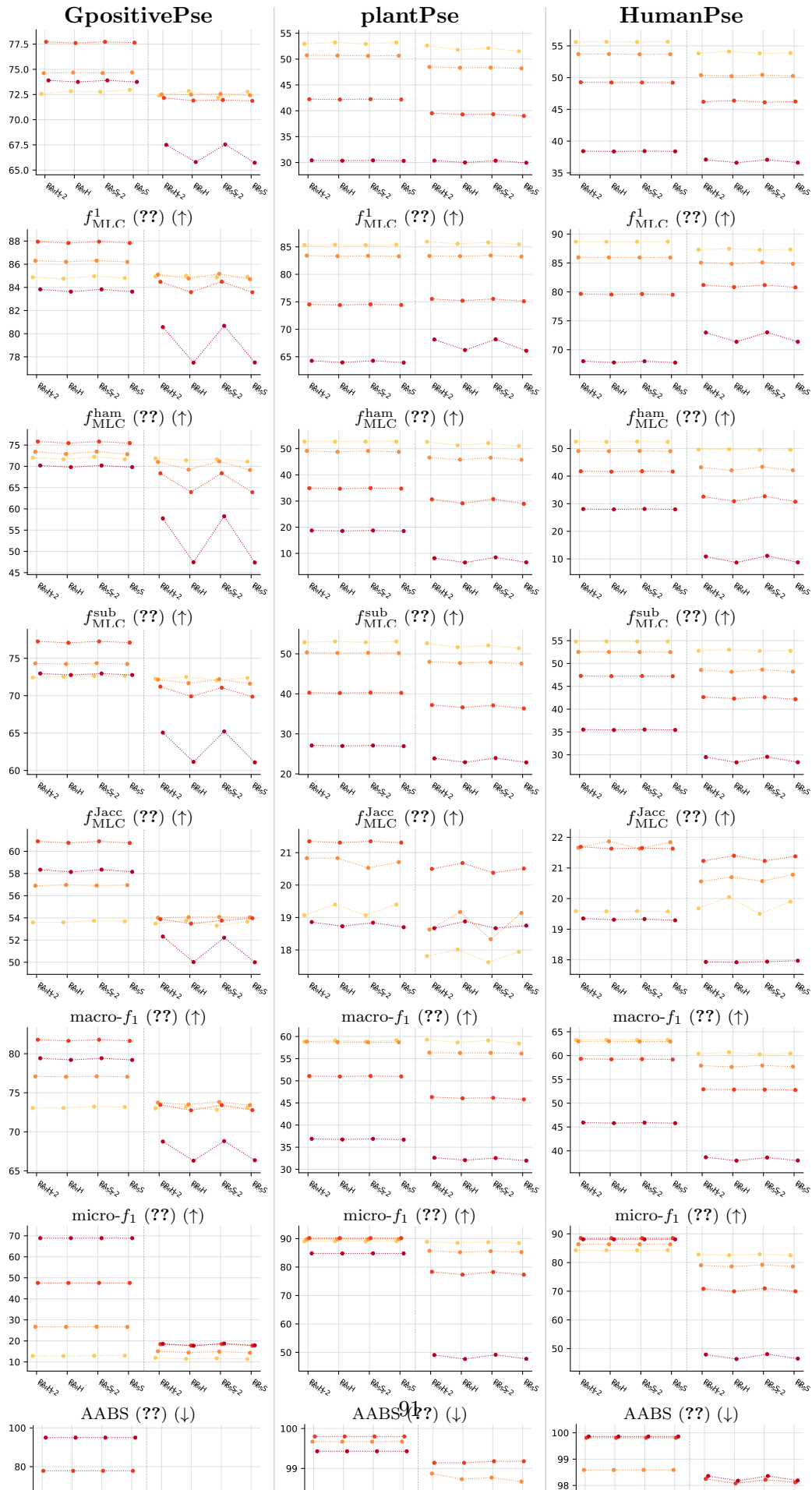
HumanPse

 f_{MLC}^1 (??) (↑) f_{MLC}^{ham} (??) (↑) f_{MLC}^{sub} (??) (↑) f_{MLC}^{Jacc} (??) (↑)macro- f_1 (??) (↑)micro- f_1 (??) (↑)

AFRD (??) (↓)



A.4 Partial abstention (PartialAbstention)



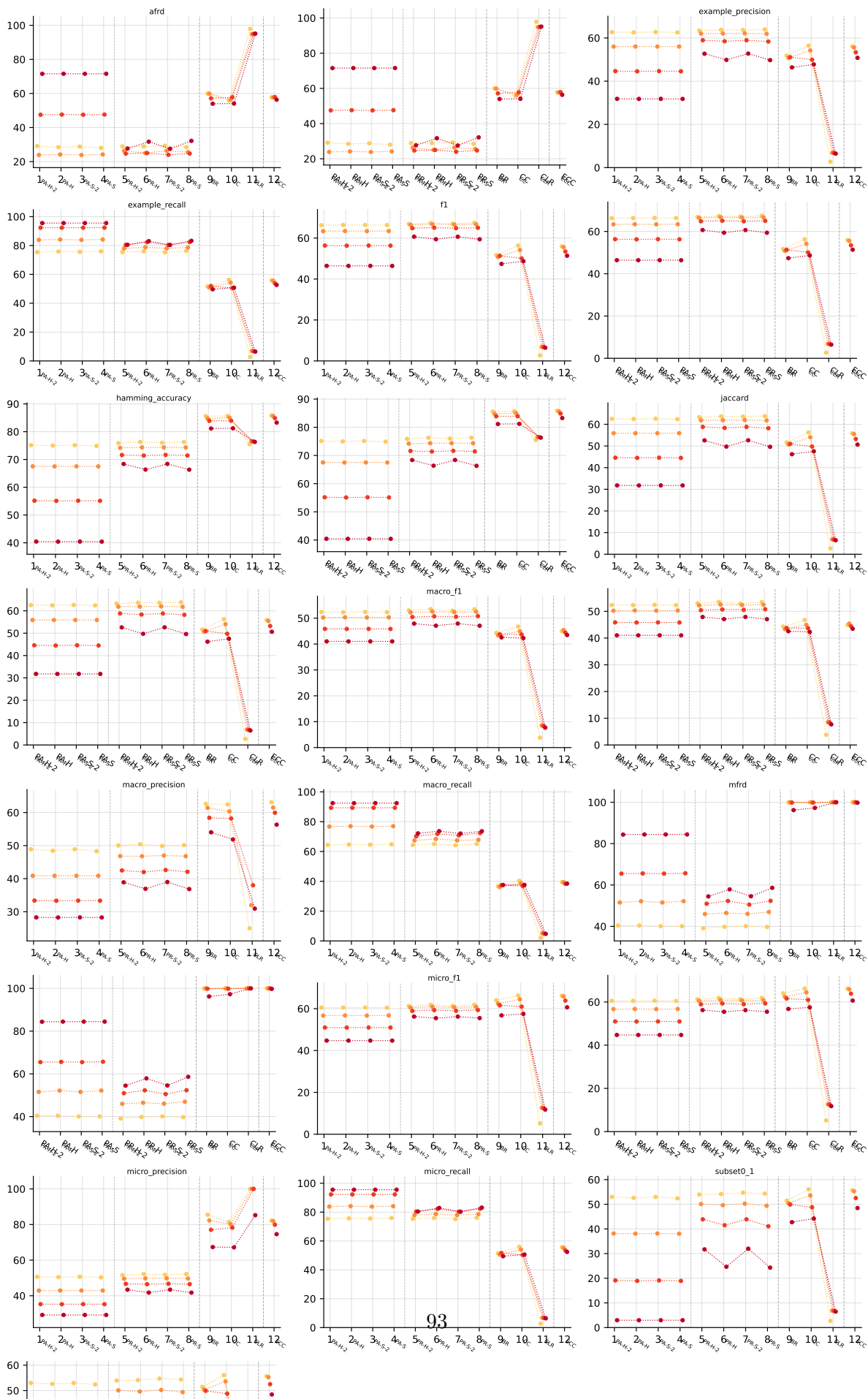
B Per-dataset results (RF base learner)

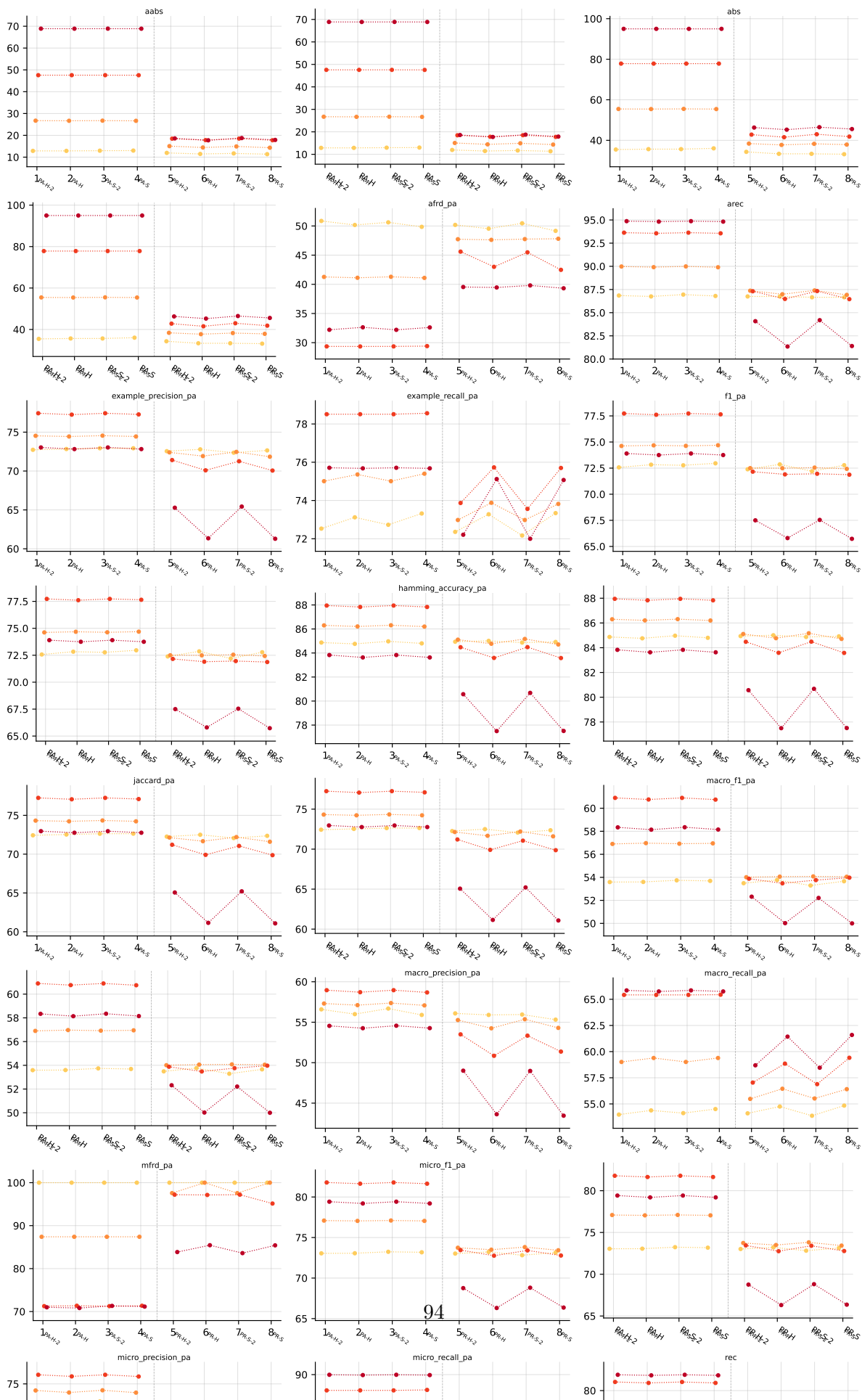
B.1 GpositivePseAAC ($K = 4$)

RF • GpositivePseAAC • BinaryVector — Standard MLC predictions

RF • GpositivePseAAC • PartialAbstention — Partial abstention

RF • GpositivePseAAC • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)





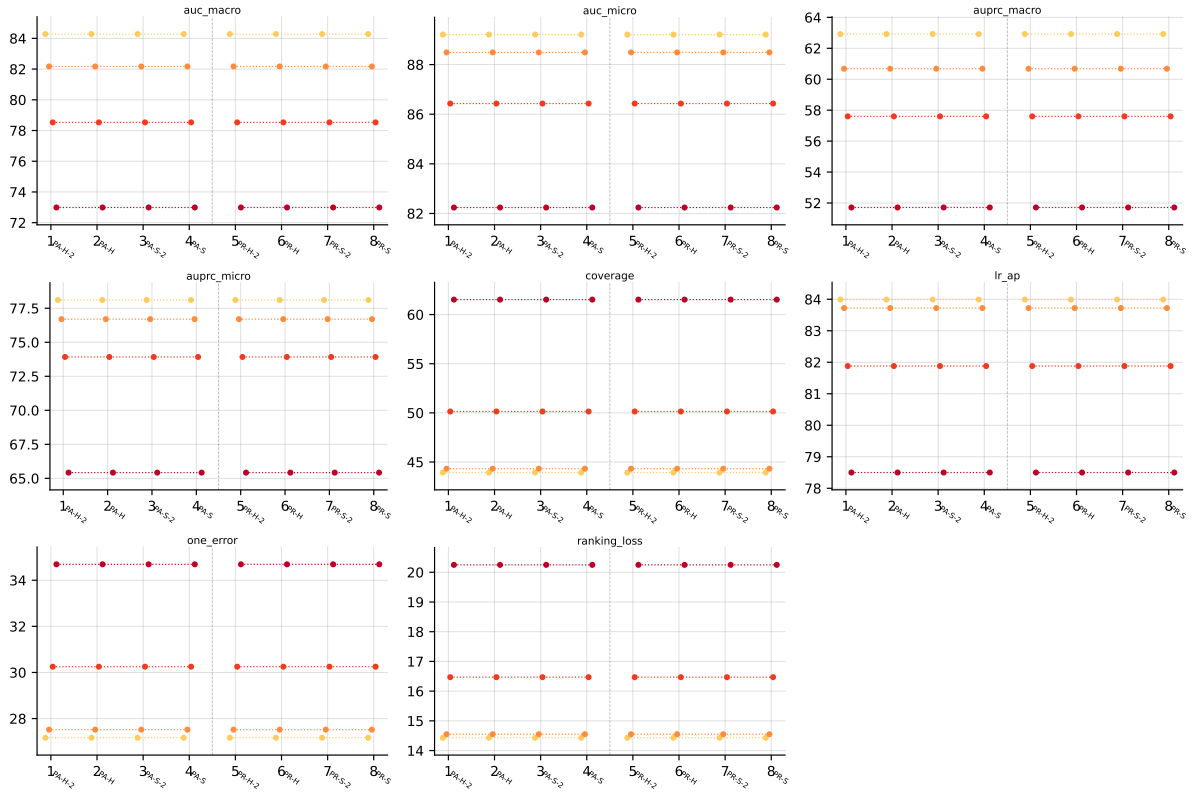


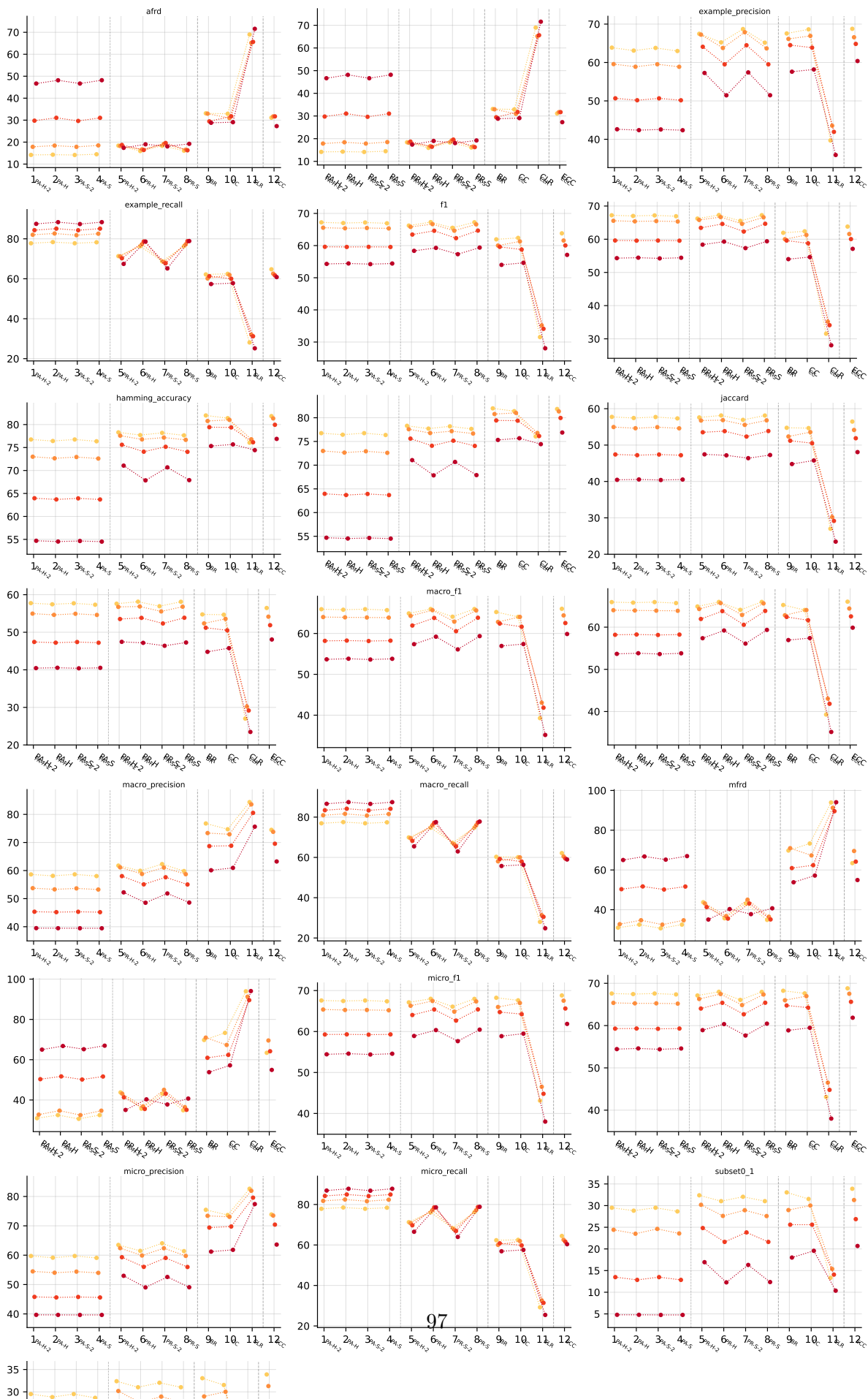
Table 22: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on GpositivePseAAC (RF base learner).

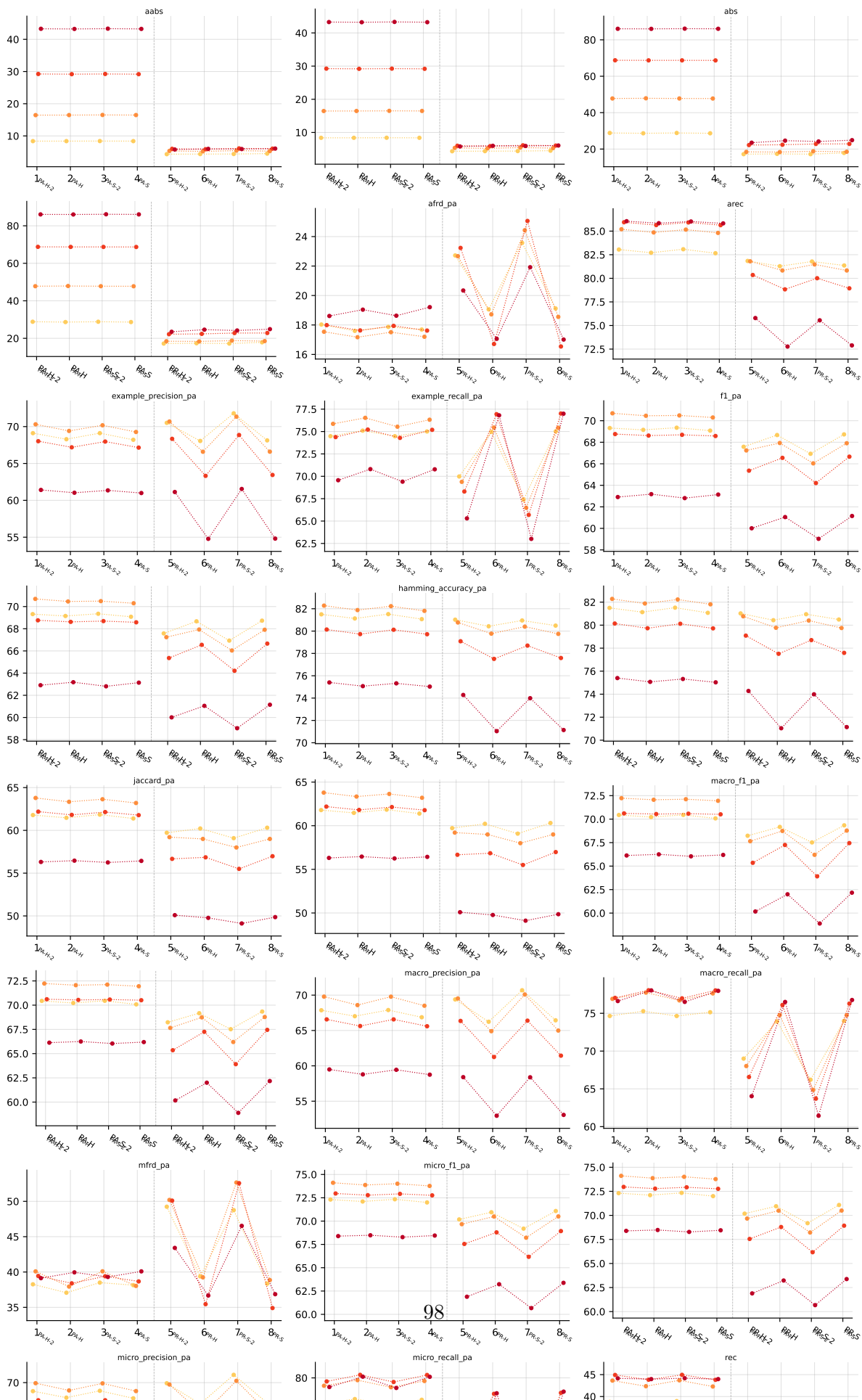
B.2 Emotions ($K = 6$)

RF • emotions • BinaryVector — Standard MLC predictions

RF • emotions • PartialAbstention — Partial abstention

RF • emotions • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)





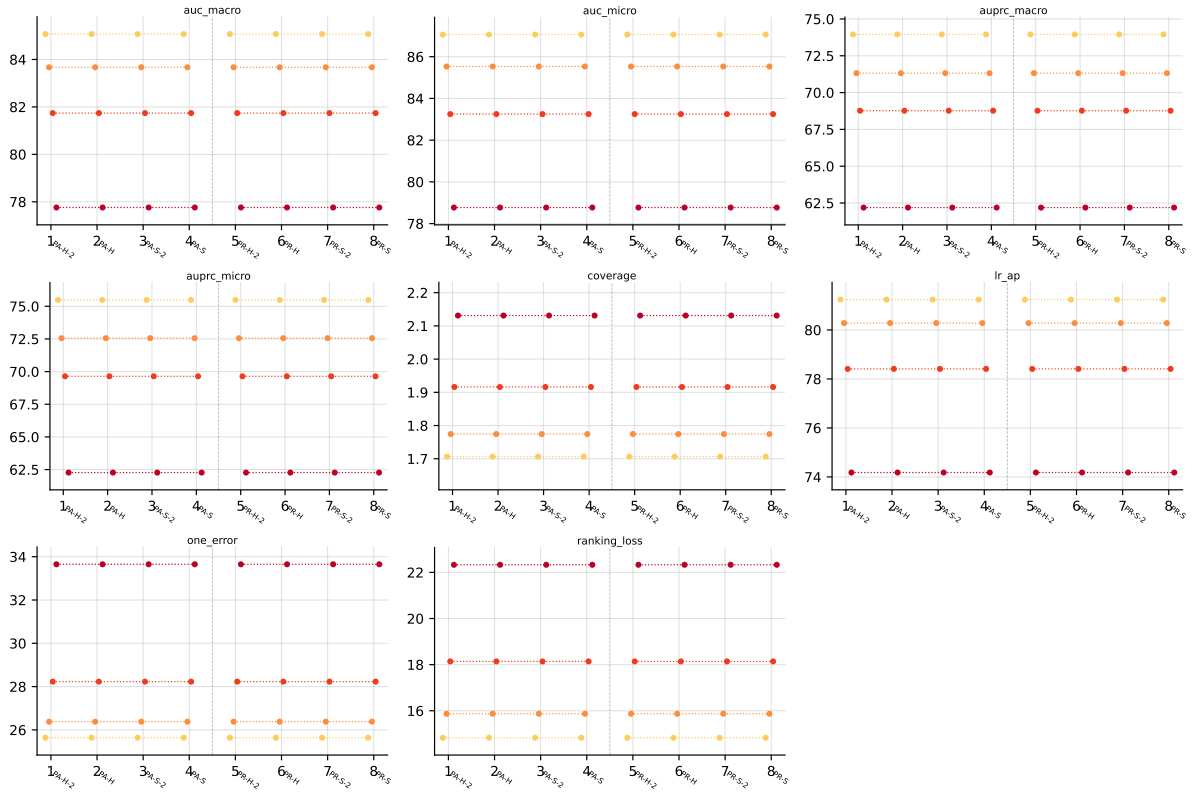


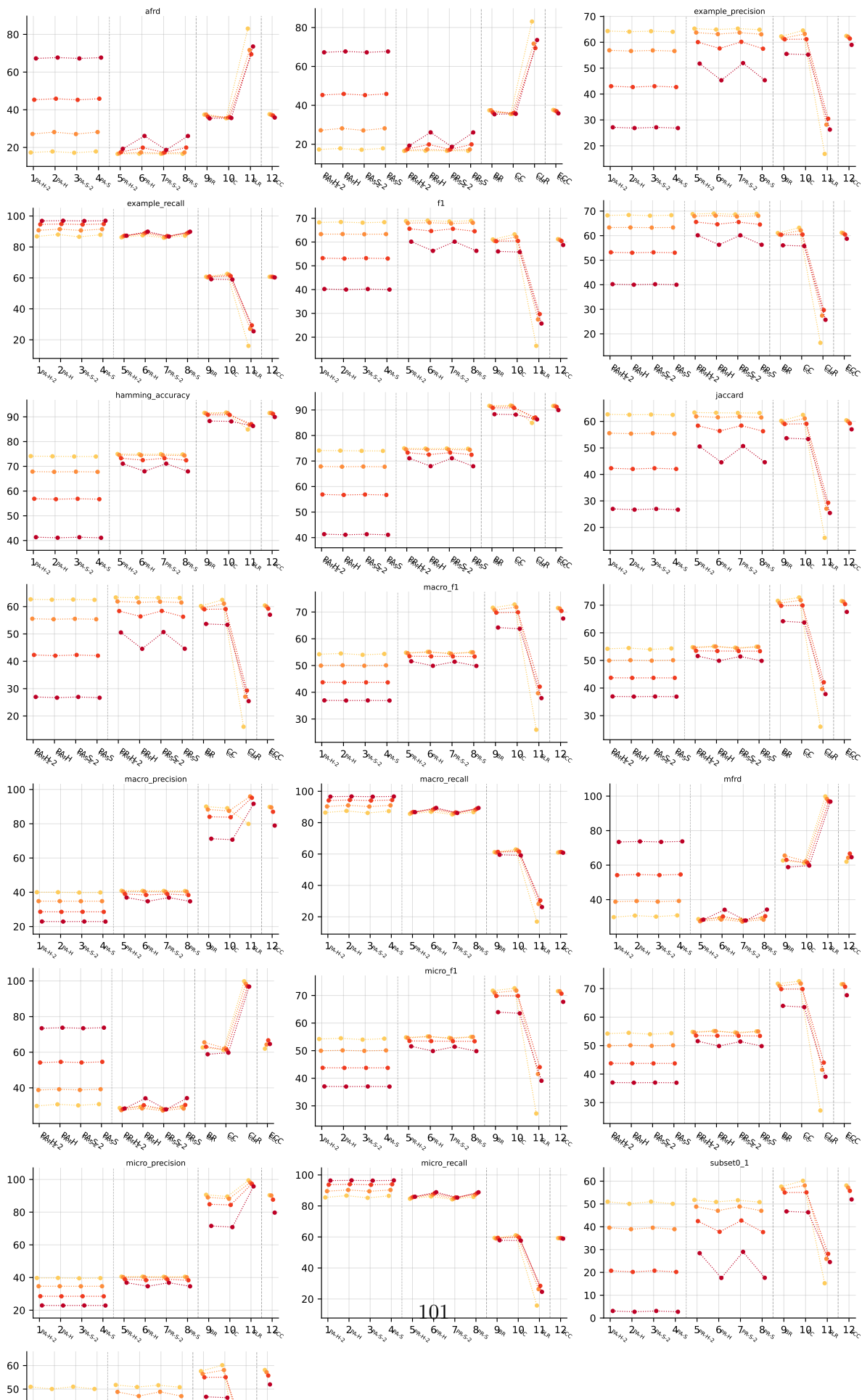
Table 25: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on **emotions** (RF base learner).

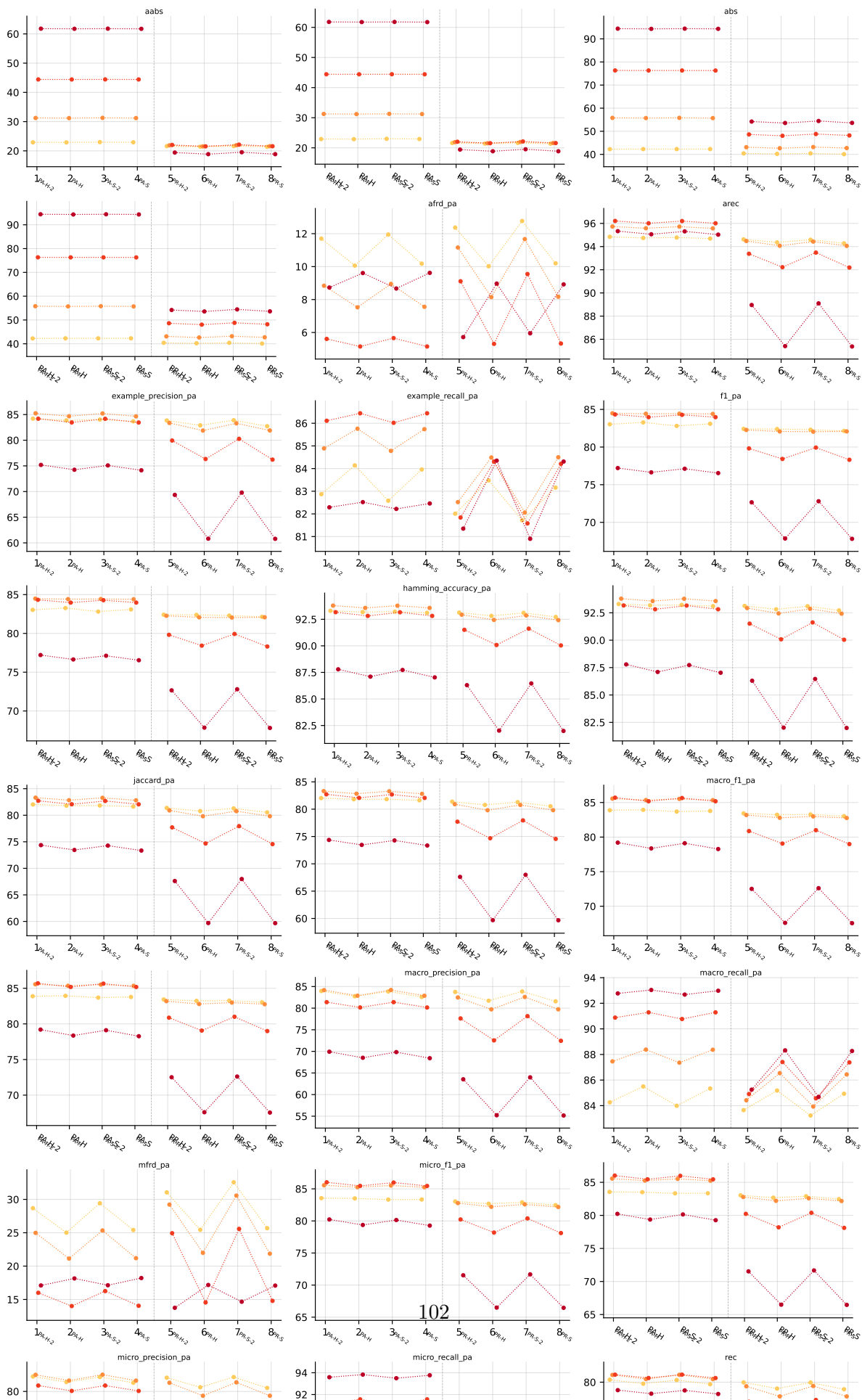
B.3 Scene ($K = 6$)

RF • scene • BinaryVector — Standard MLC predictions

RF • scene • PartialAbstention — Partial abstention

RF • scene • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)





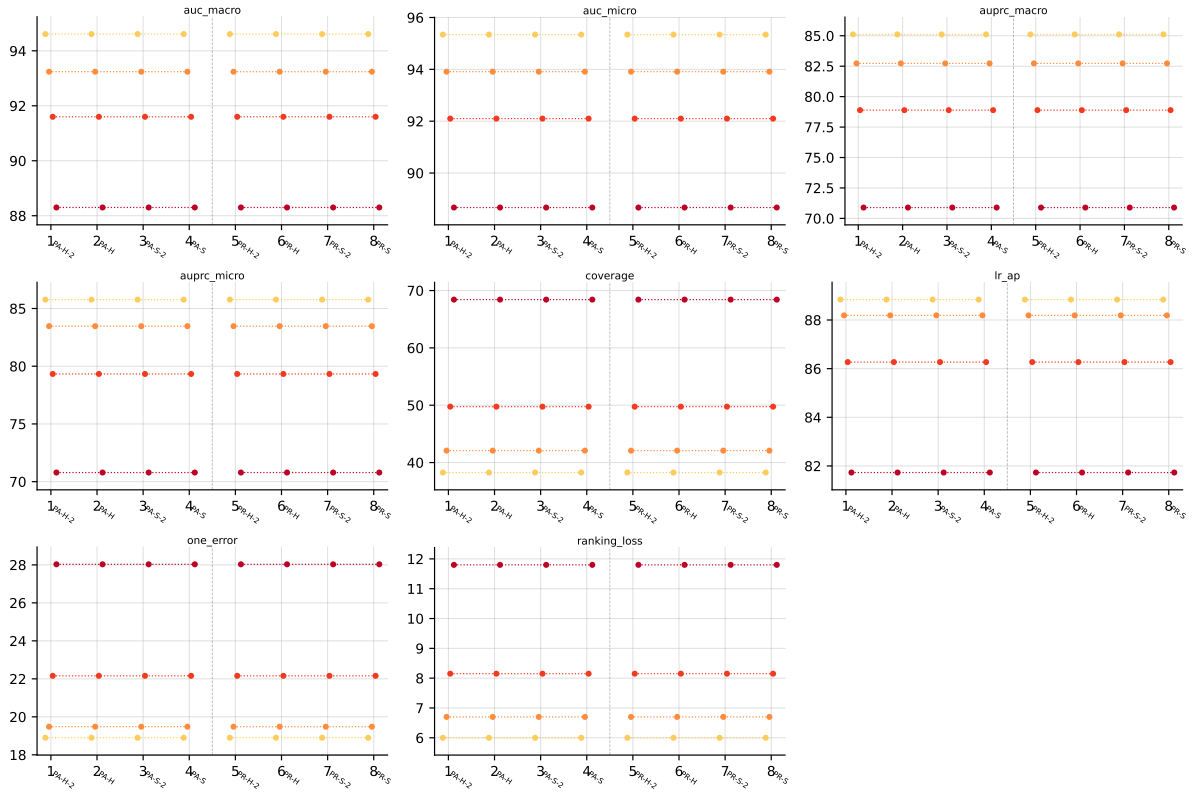


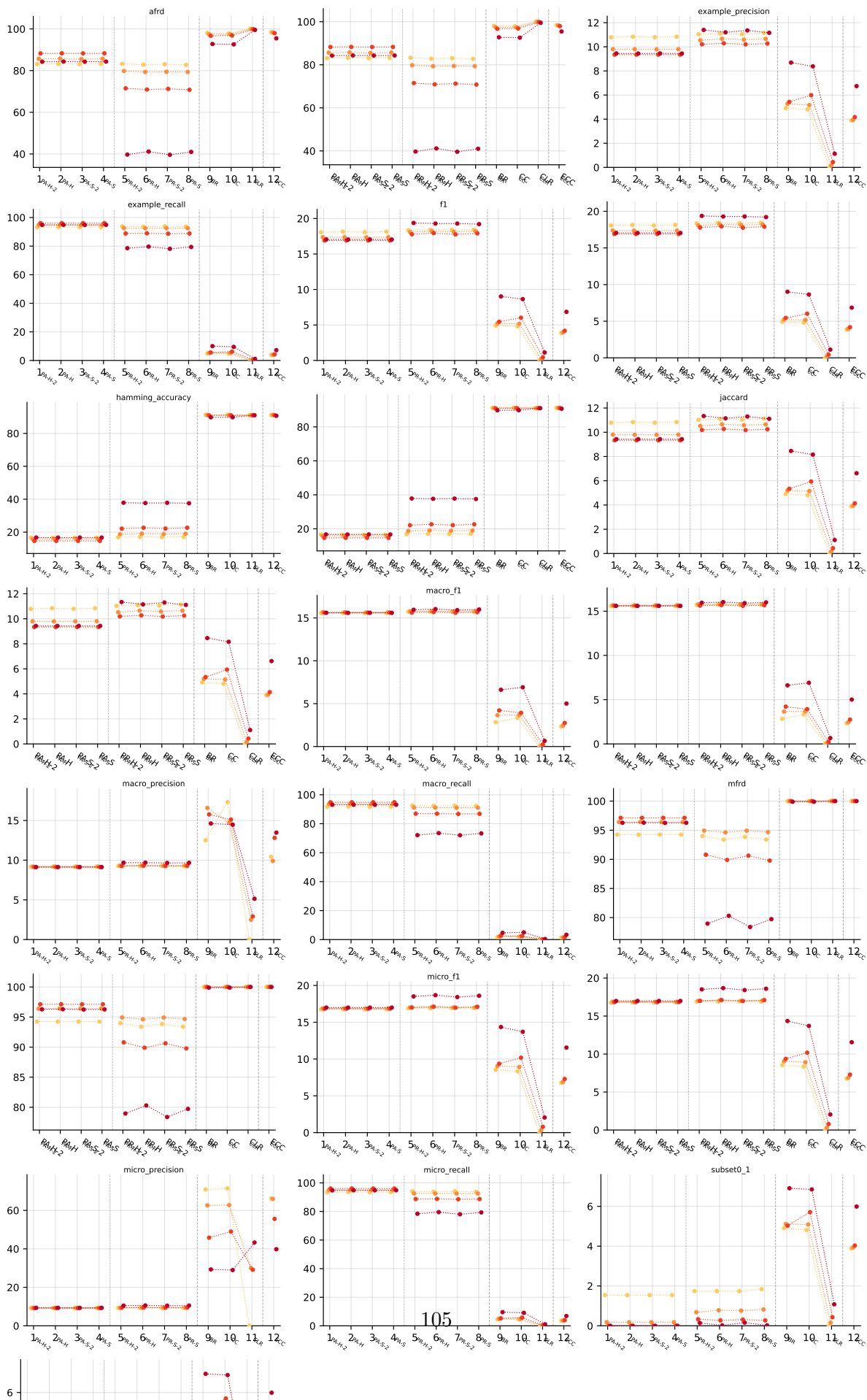
Table 28: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `scene` (RF base learner).

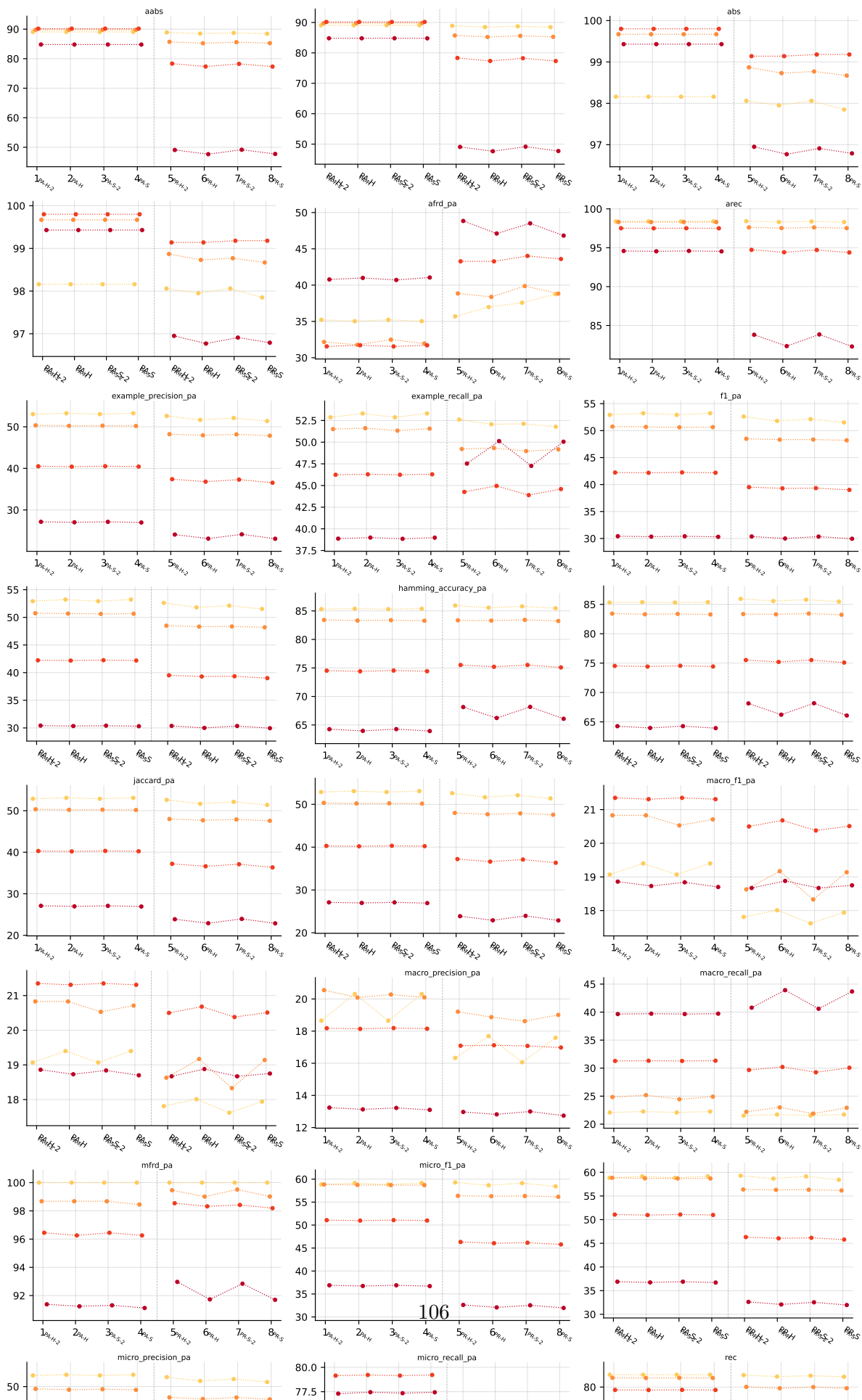
B.4 PlantPseAAC ($K = 12$)

RF • PlantPseAAC • BinaryVector — Standard MLC predictions

RF • PlantPseAAC • PartialAbstention — Partial abstention

RF • PlantPseAAC • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)





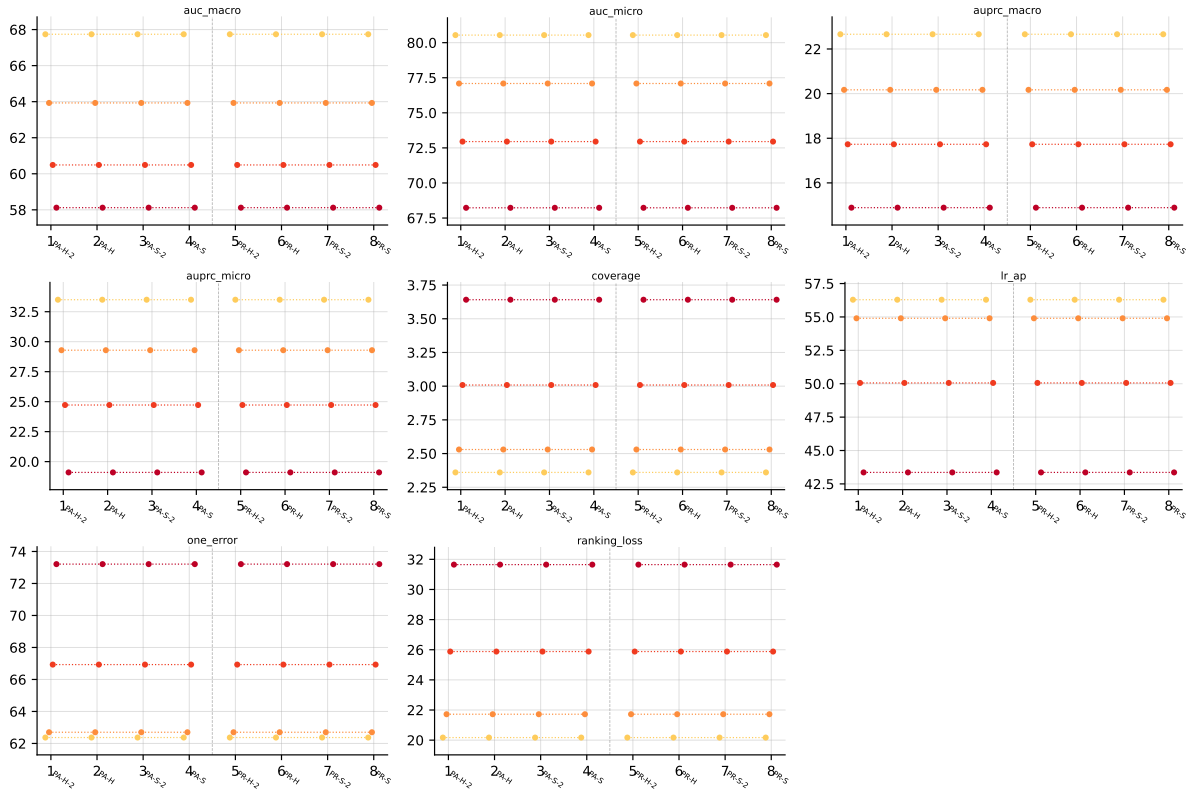


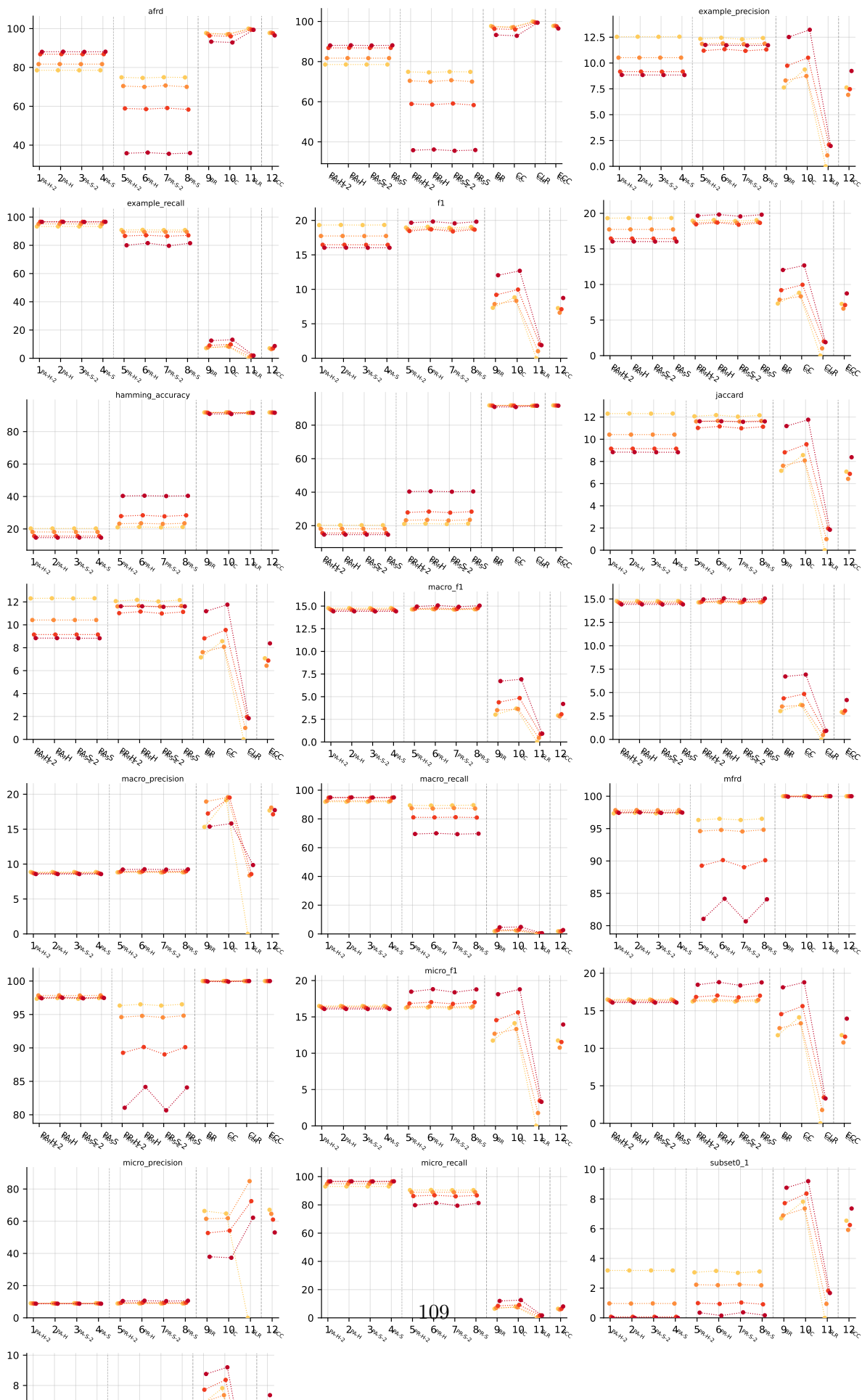
Table 31: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on PlantPseAAC (RF base learner).

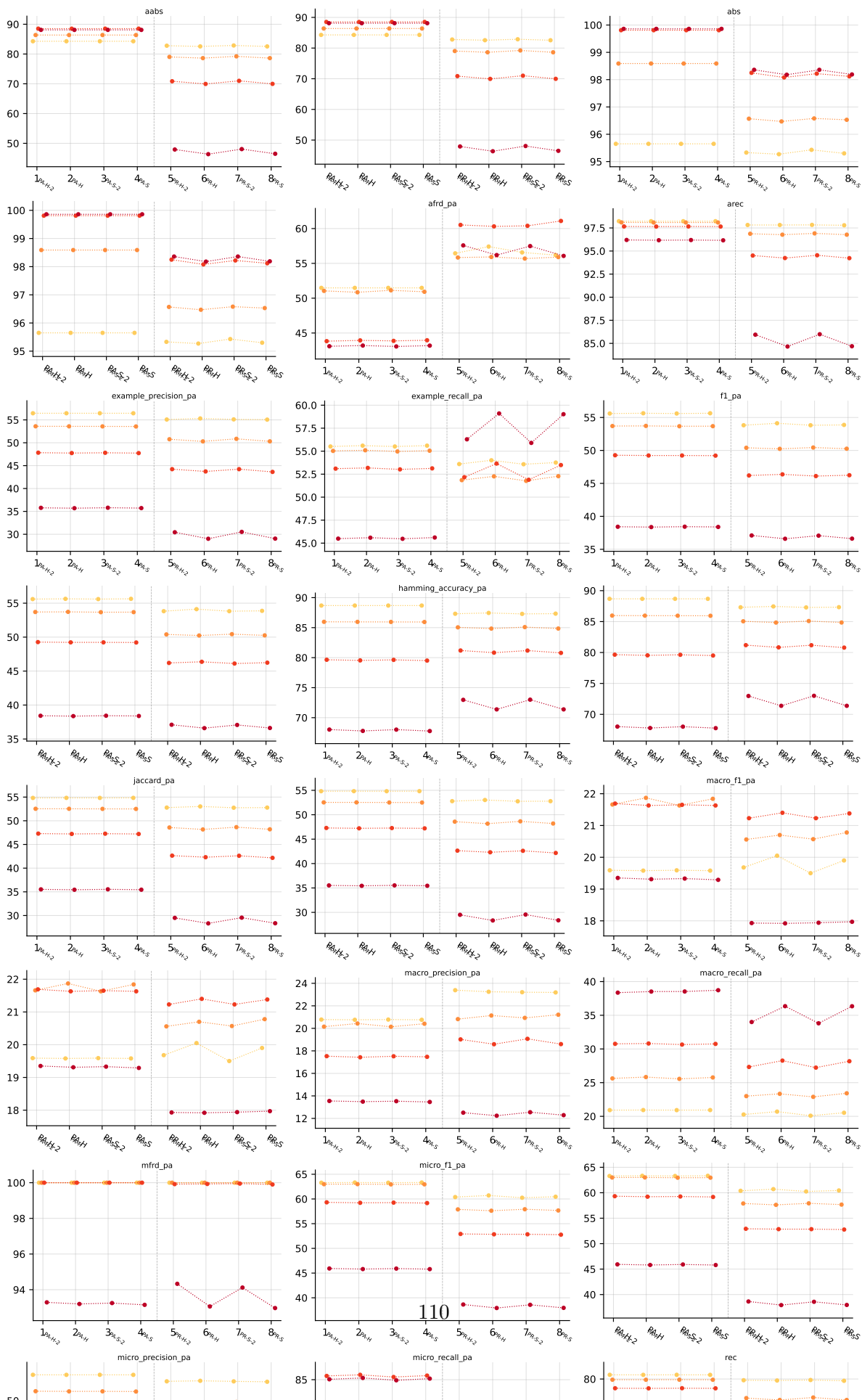
B.5 HumanPseAAC ($K = 14$)

RF • HumanPseAAC • BinaryVector — Standard MLC predictions

RF • HumanPseAAC • PartialAbstention — Partial abstention

RF • HumanPseAAC • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)





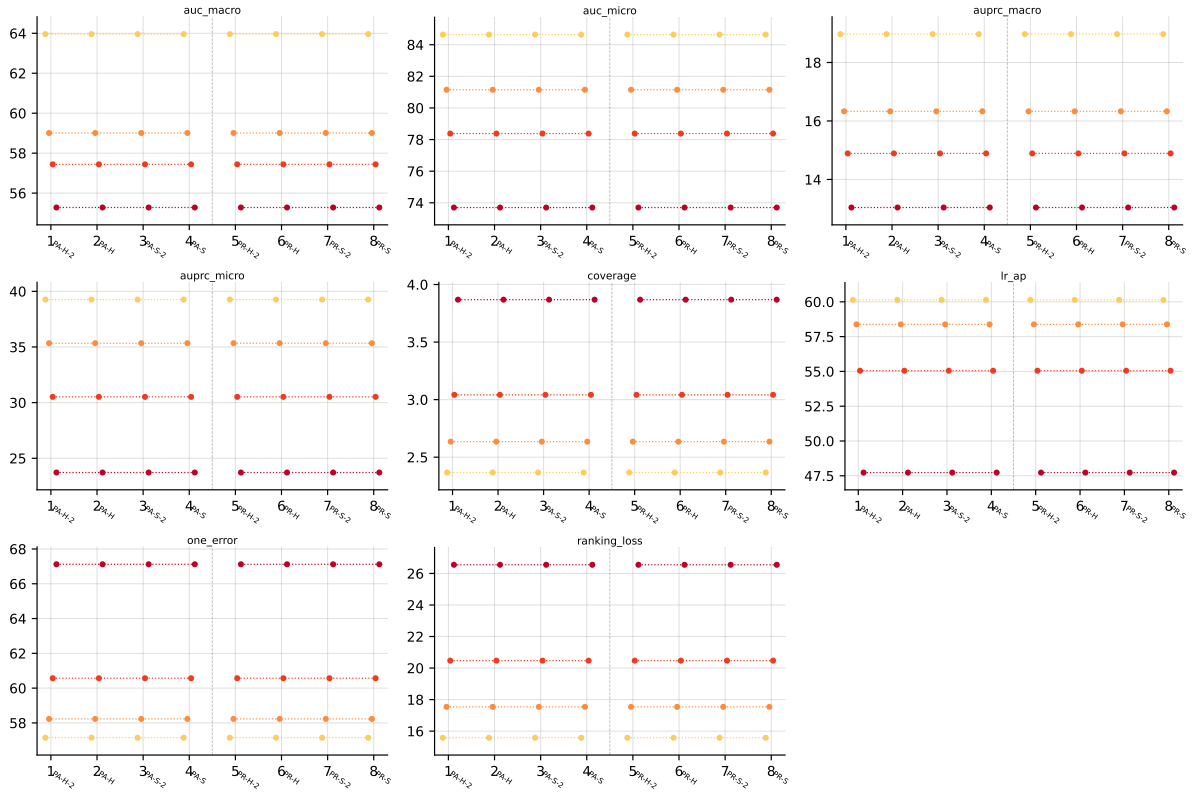


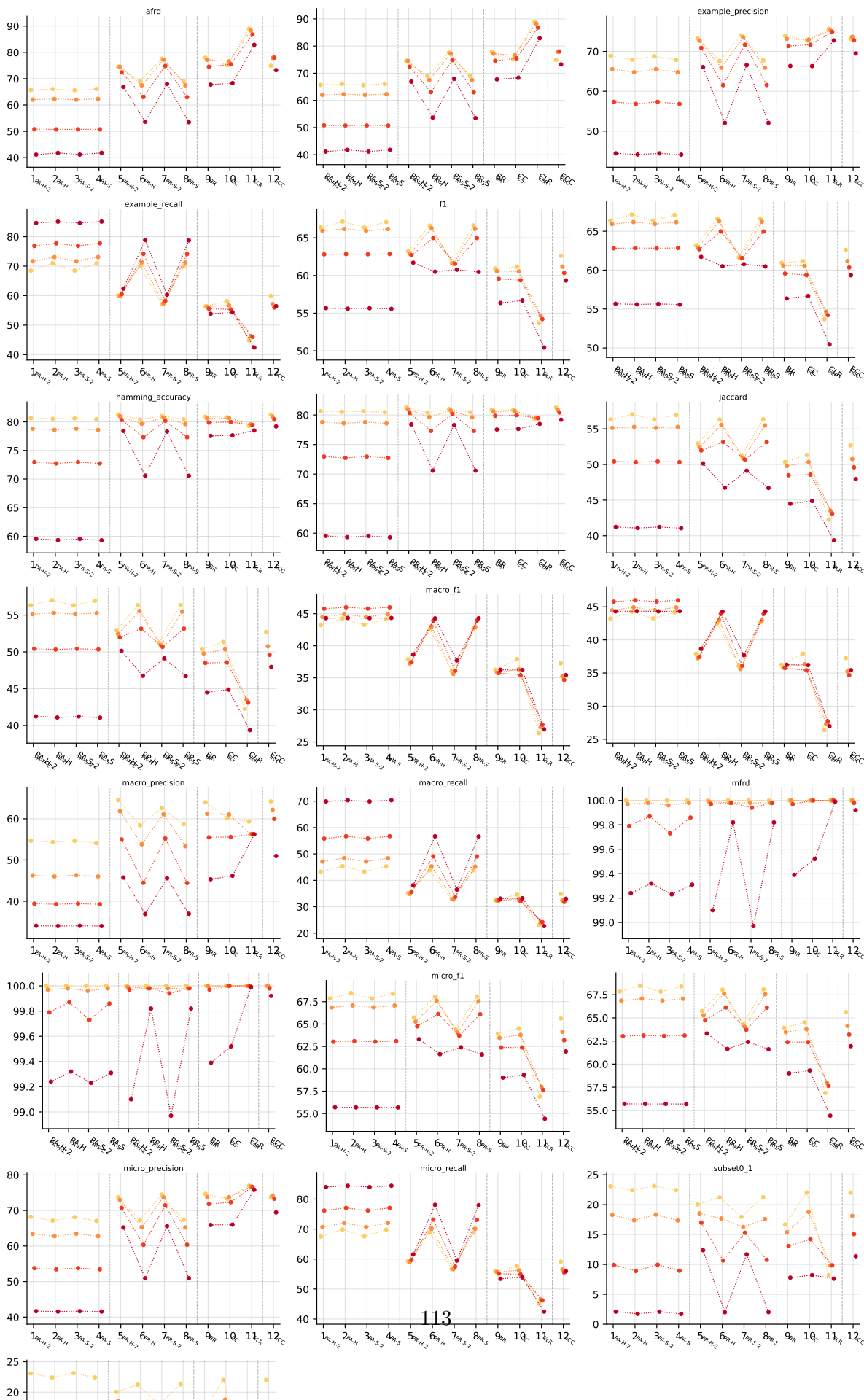
Table 34: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on HumanPseAAC (RF base learner).

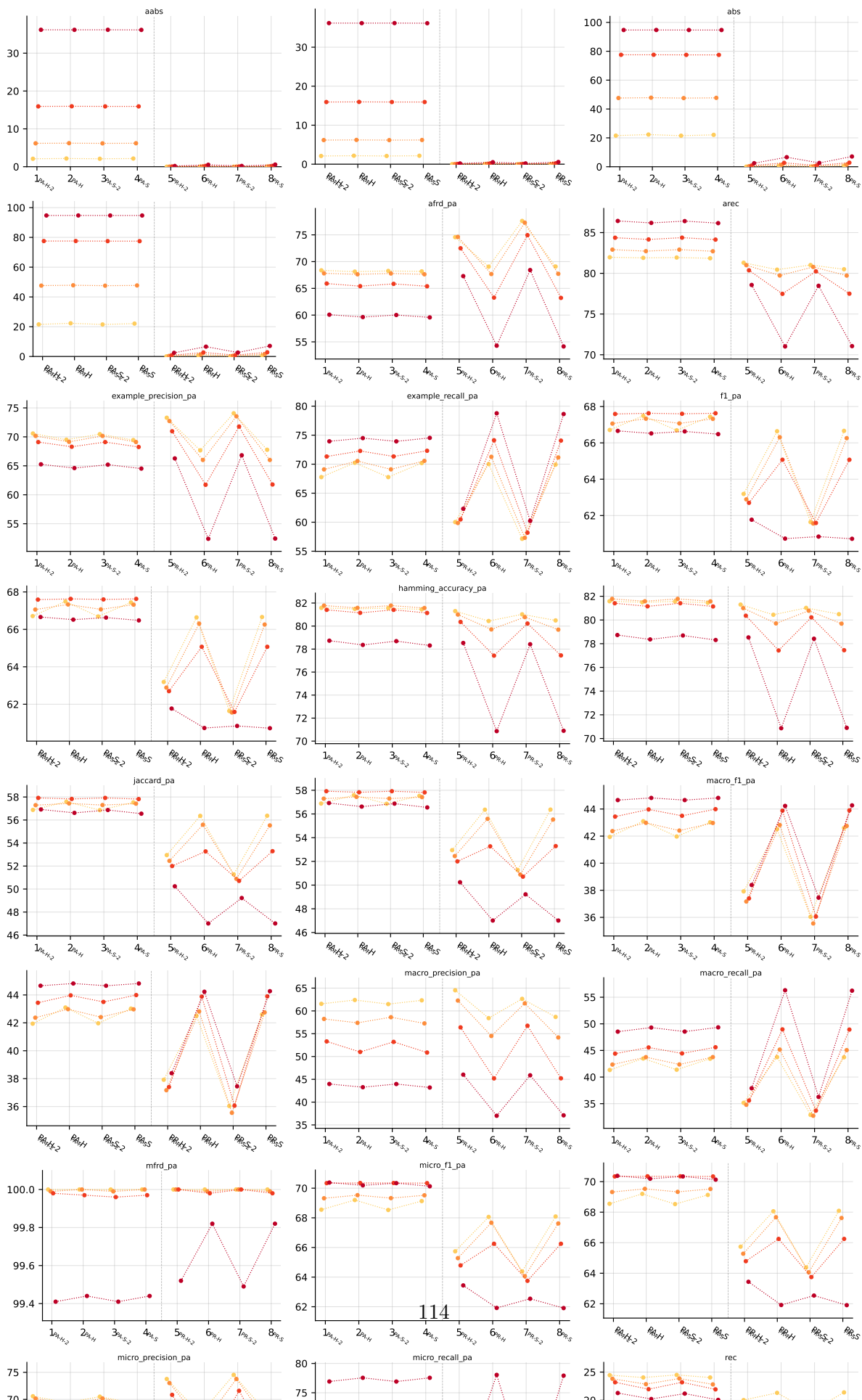
B.6 Yeast ($K = 14$)

RF • Yeast • BinaryVector — Standard MLC predictions

RF • Yeast • PartialAbstention — Partial abstention

RF • Yeast • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)





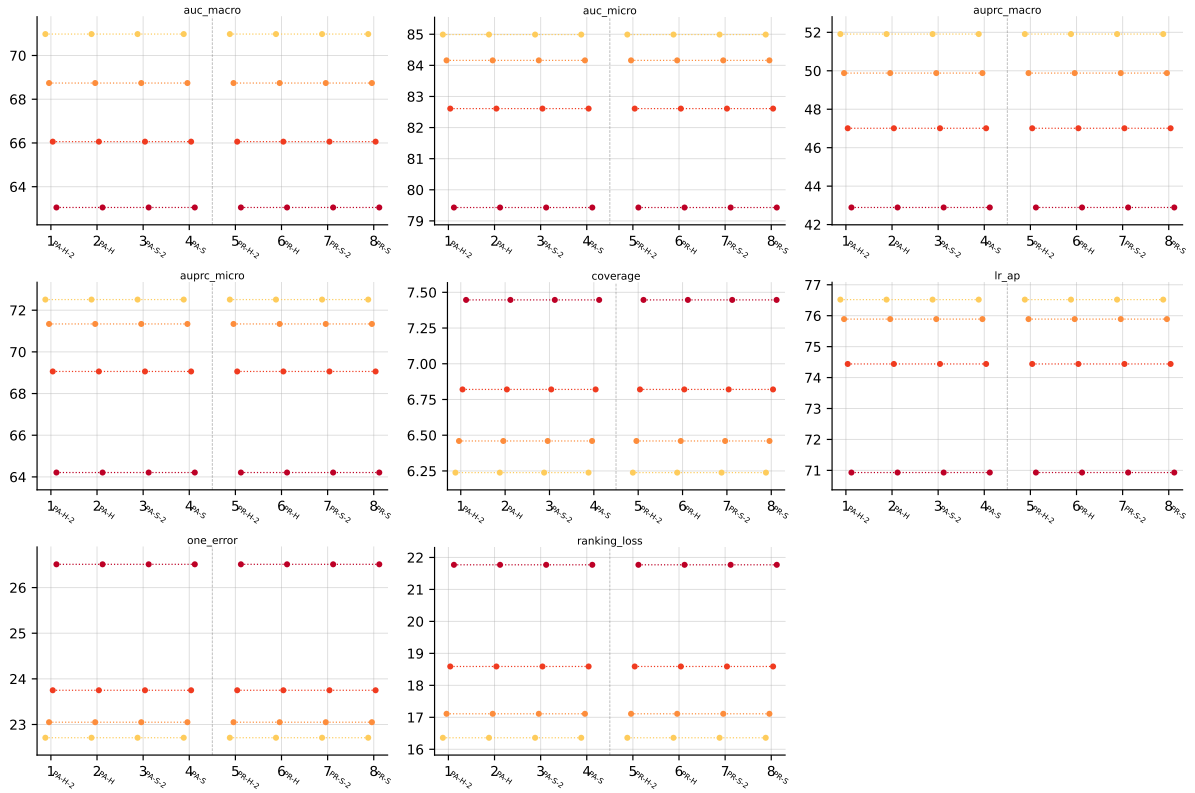


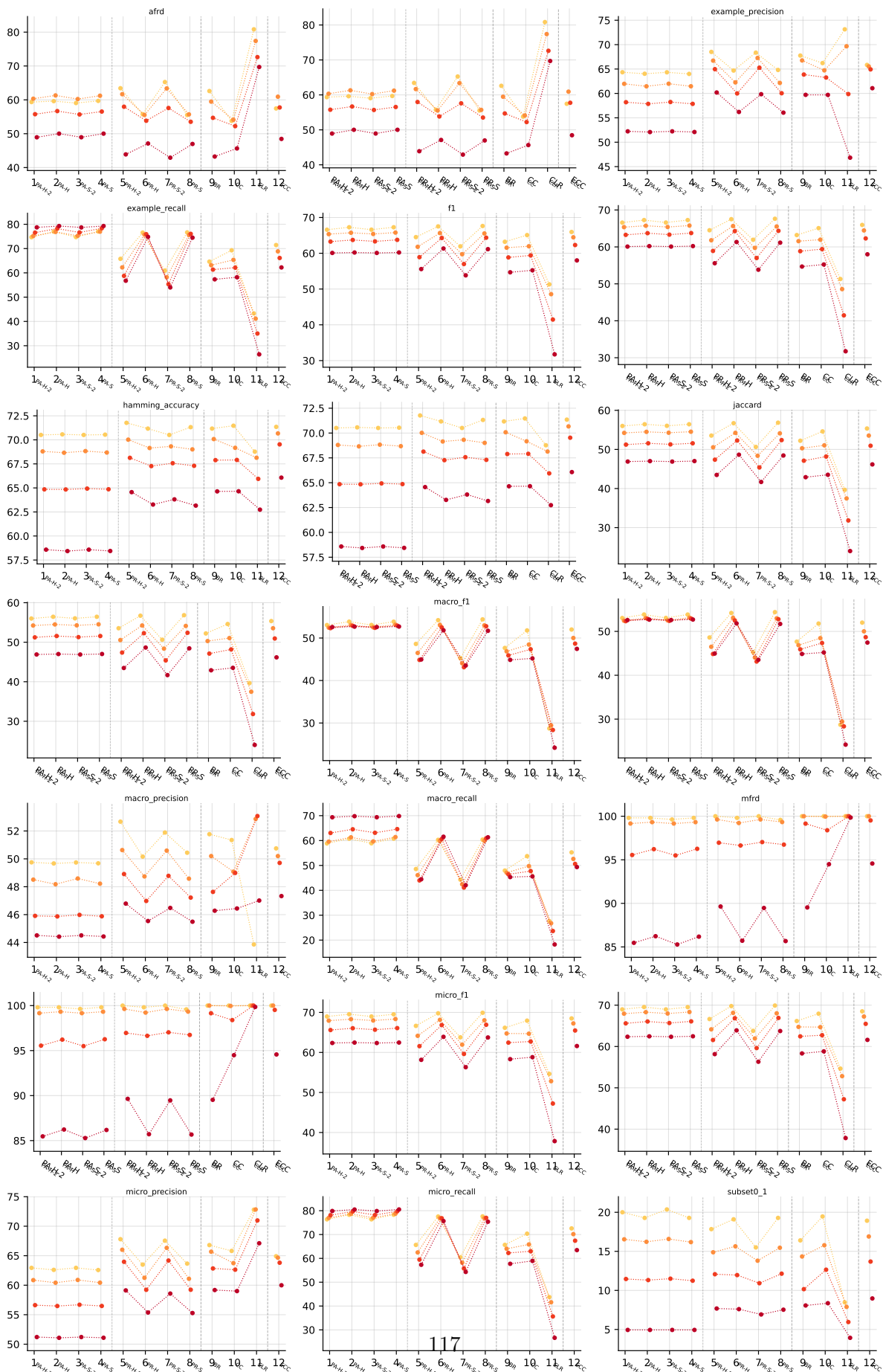
Table 37: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on Yeast (RF base learner).

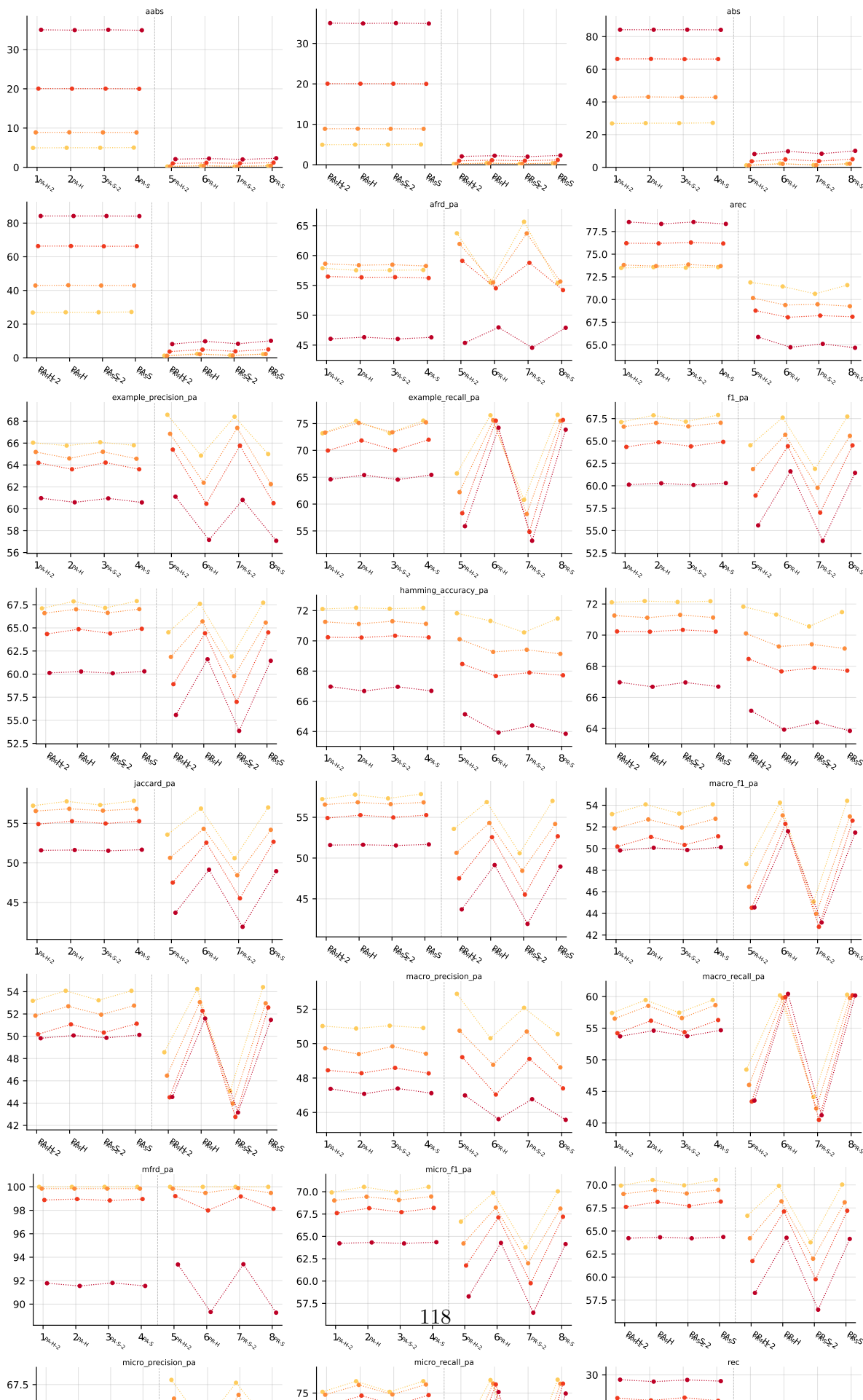
B.7 CHD_49 ($K = 6$)

RF • CHD_49 • BinaryVector — Standard MLC predictions

RF • CHD_49 • PartialAbstention — Partial abstention

RF • CHD_49 • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)





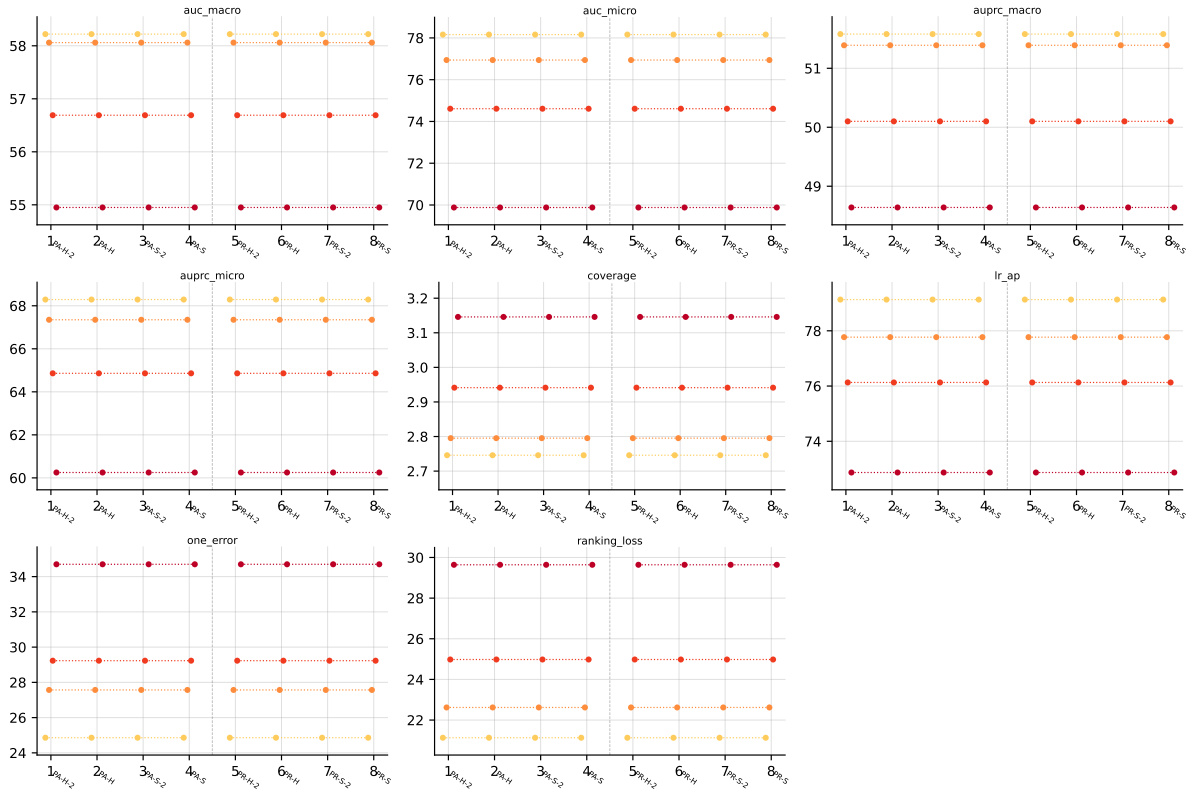


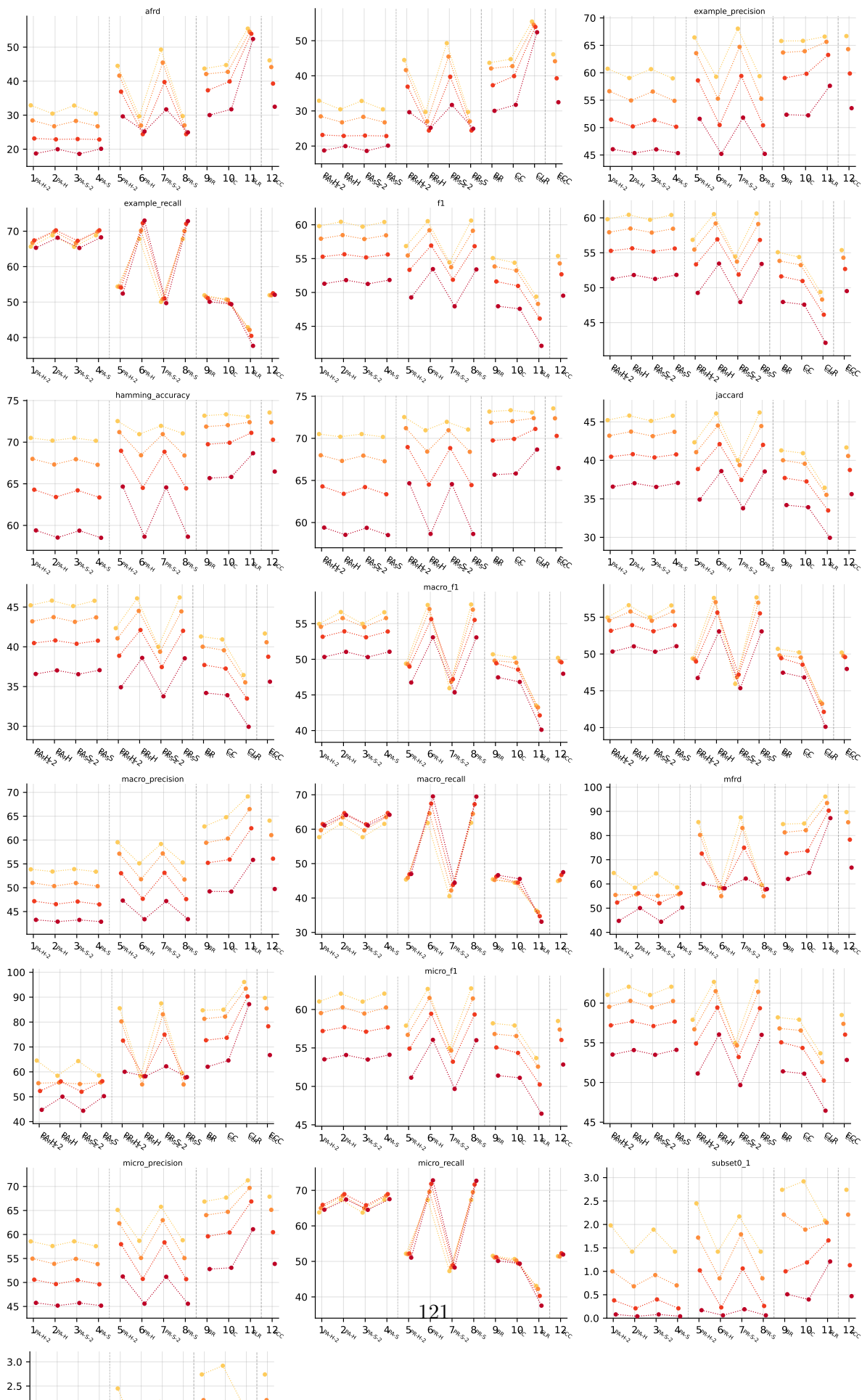
Table 40: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on CHD_49 (RF base learner).

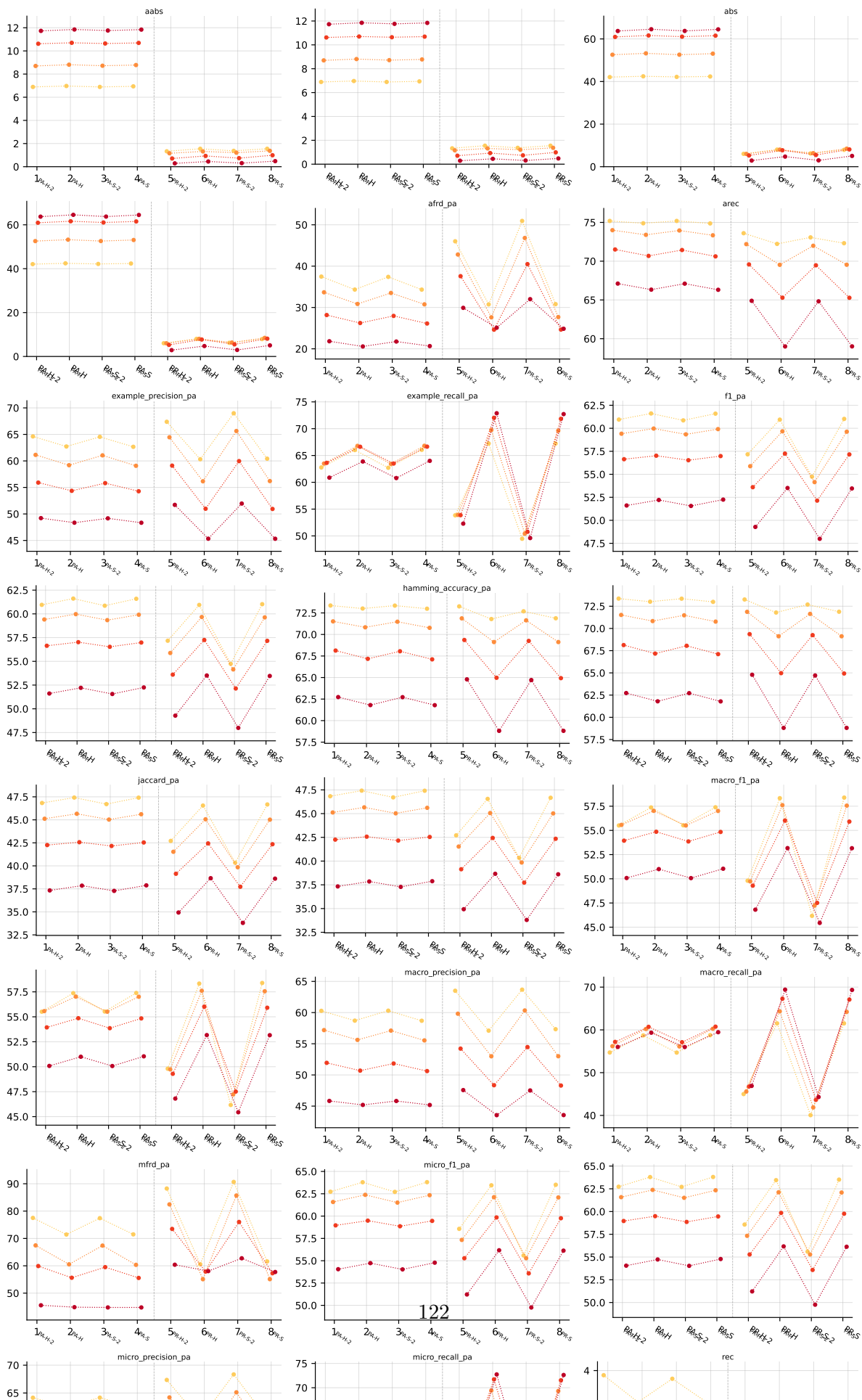
B.8 Water-quality ($K = 14$)

RF • Water-quality • BinaryVector — Standard MLC predictions

RF • Water-quality • PartialAbstention — Partial abstention

RF • Water-quality • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)





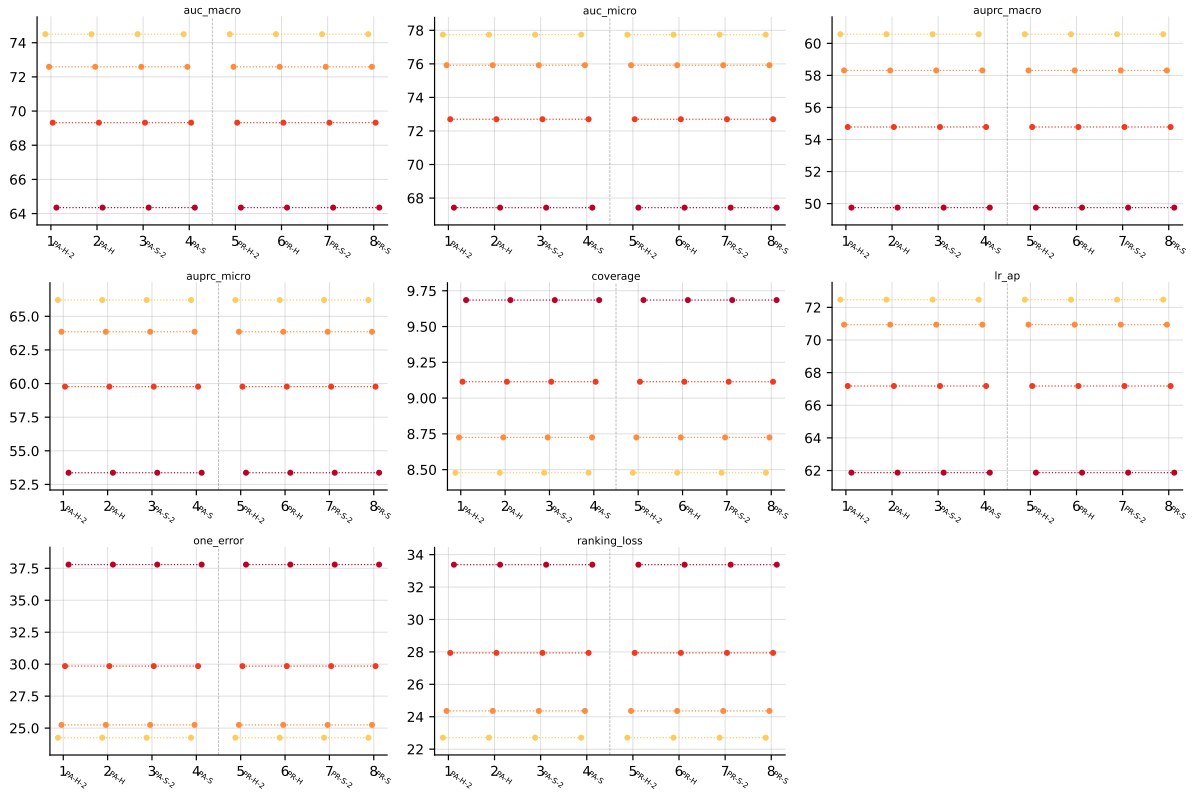


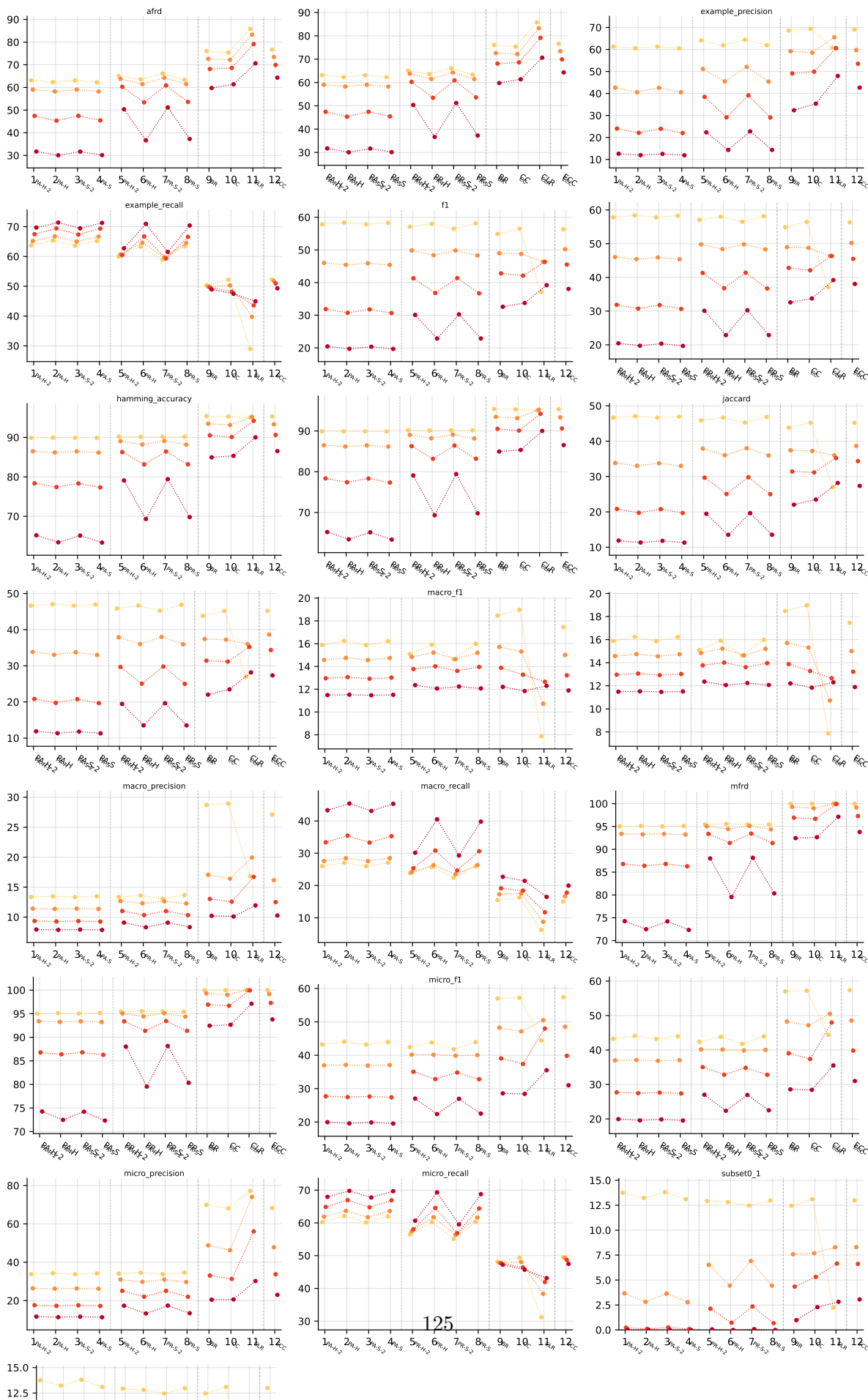
Table 43: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on Water-quality (RF base learner).

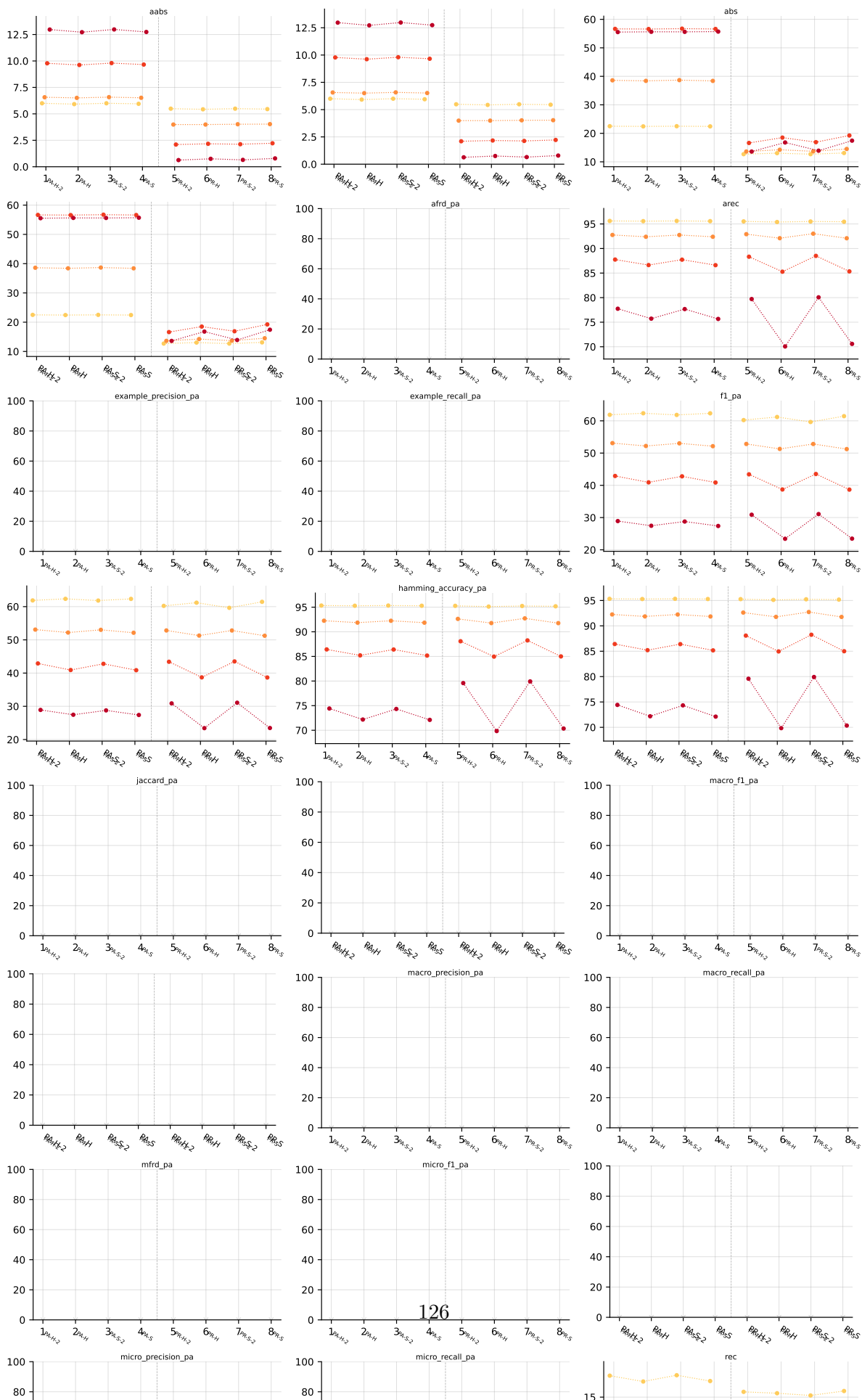
B.9 enron ($K = 53$)

RF • enron • BinaryVector — Standard MLC predictions

RF • enron • PartialAbstention — Partial abstention

RF • enron • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)





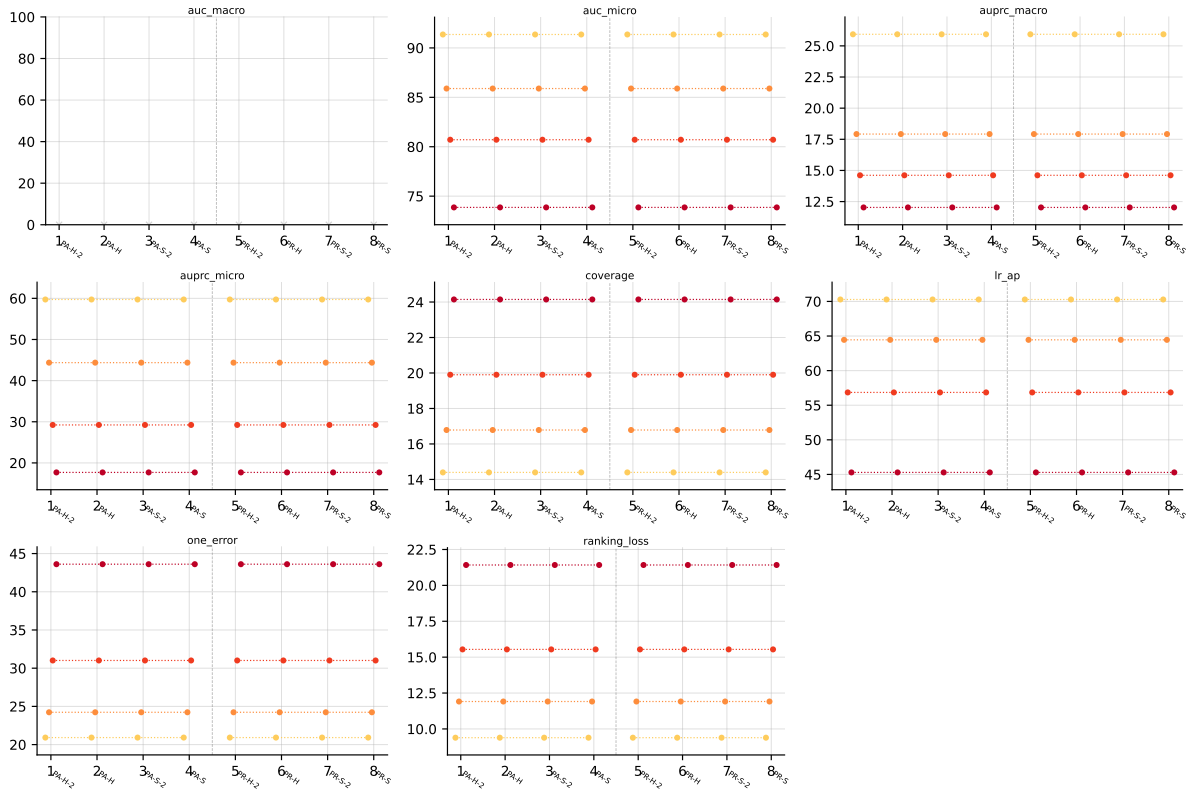


Table 46: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on **enron** (RF base learner).

C LGBM aggregated results

LGBM • Aggregate • BinaryVector — Standard MLC predictions

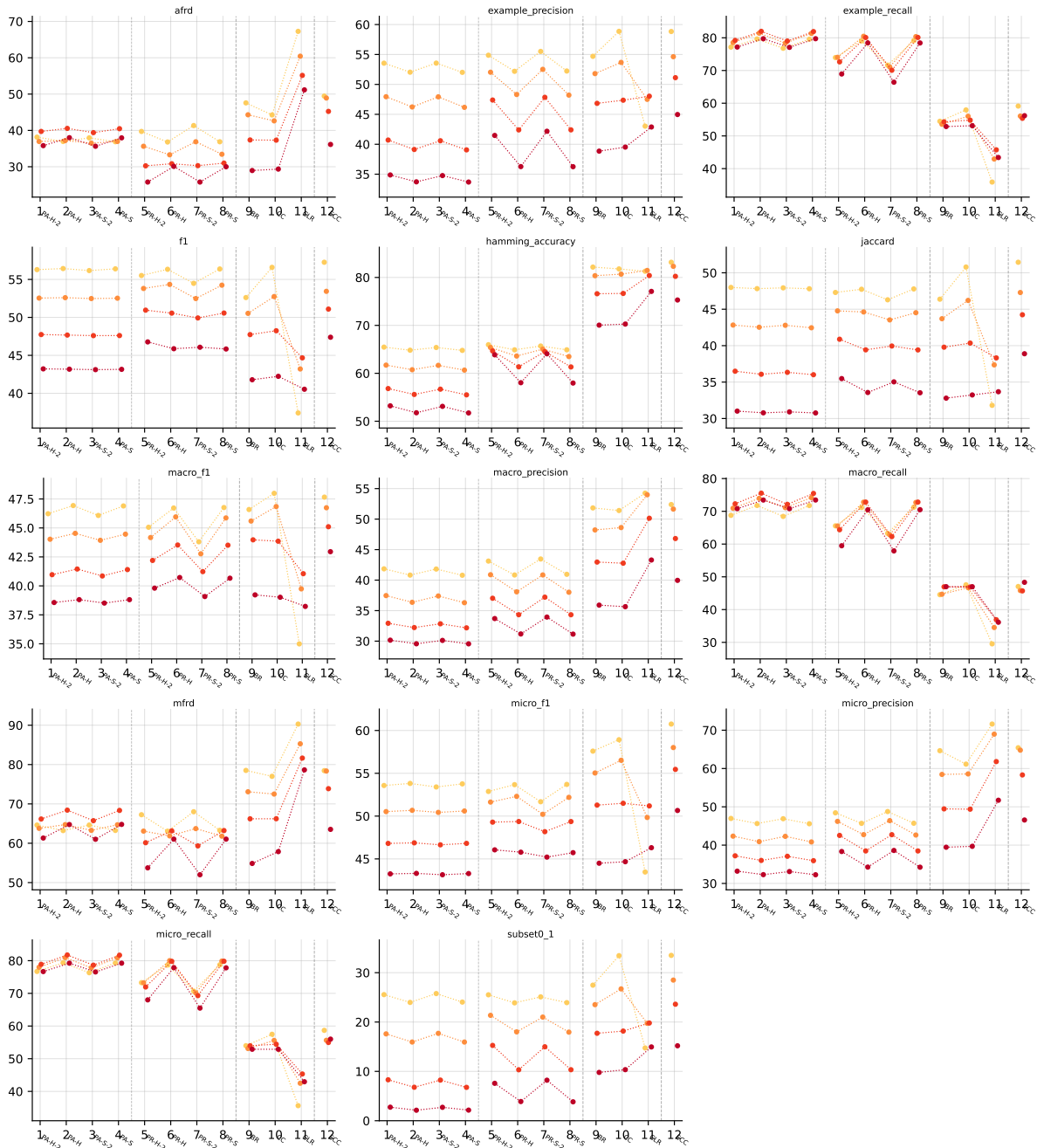


Table 47: Standard MLC predictions (BinaryVector), LGBM base learner, averaged across datasets (8 of 9: enron LGBM still training). Method indices and line colors follow the encoding paragraph above.

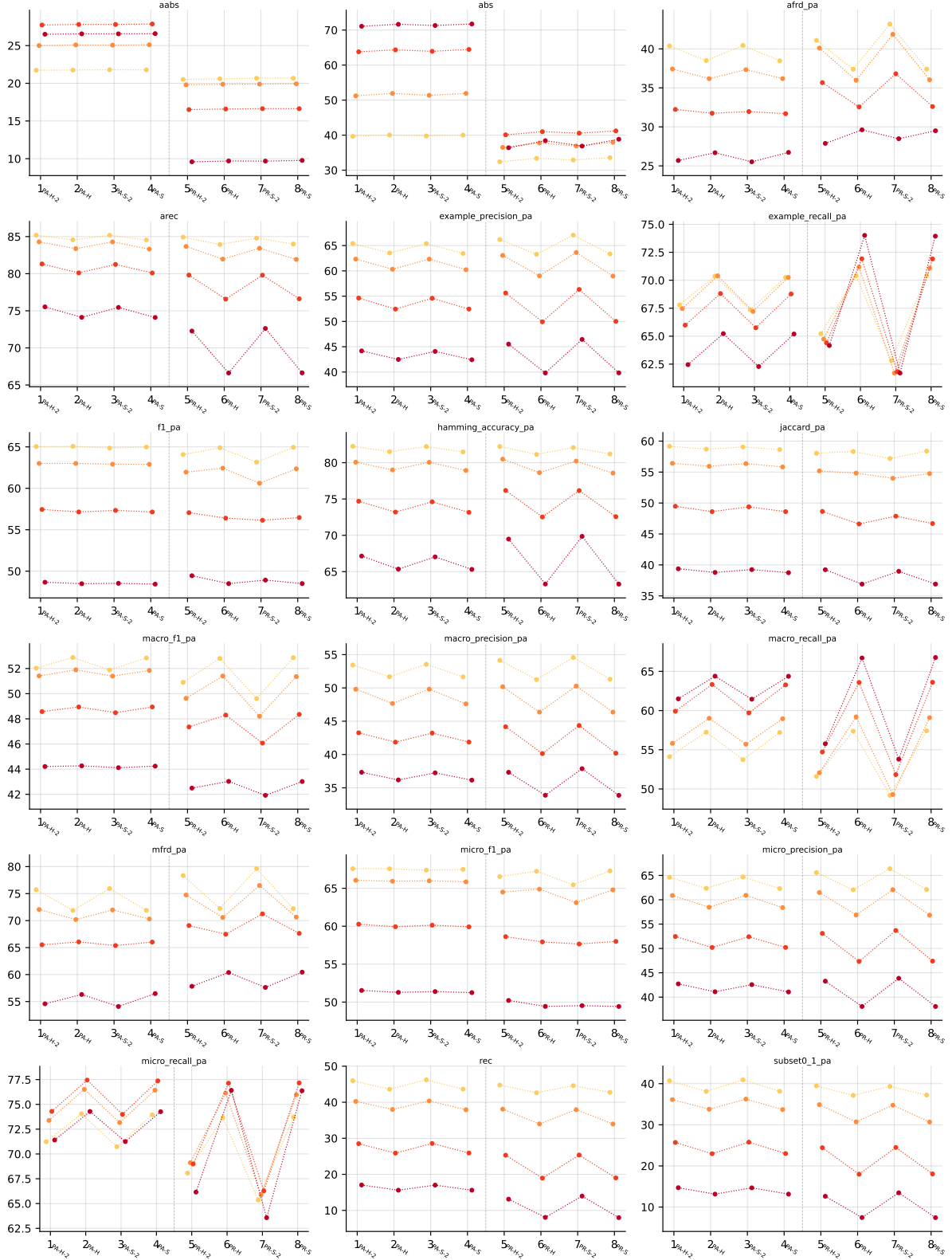


Table 48: Partial abstention (PartialAbstention), LGBM base learner, averaged across datasets (8 of 9: enron LGBM still training). Method indices and line colors follow the encoding paragraph above.

LGBM • Aggregate • PartialAbstention — Partial abstention

LGBM • Aggregate • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

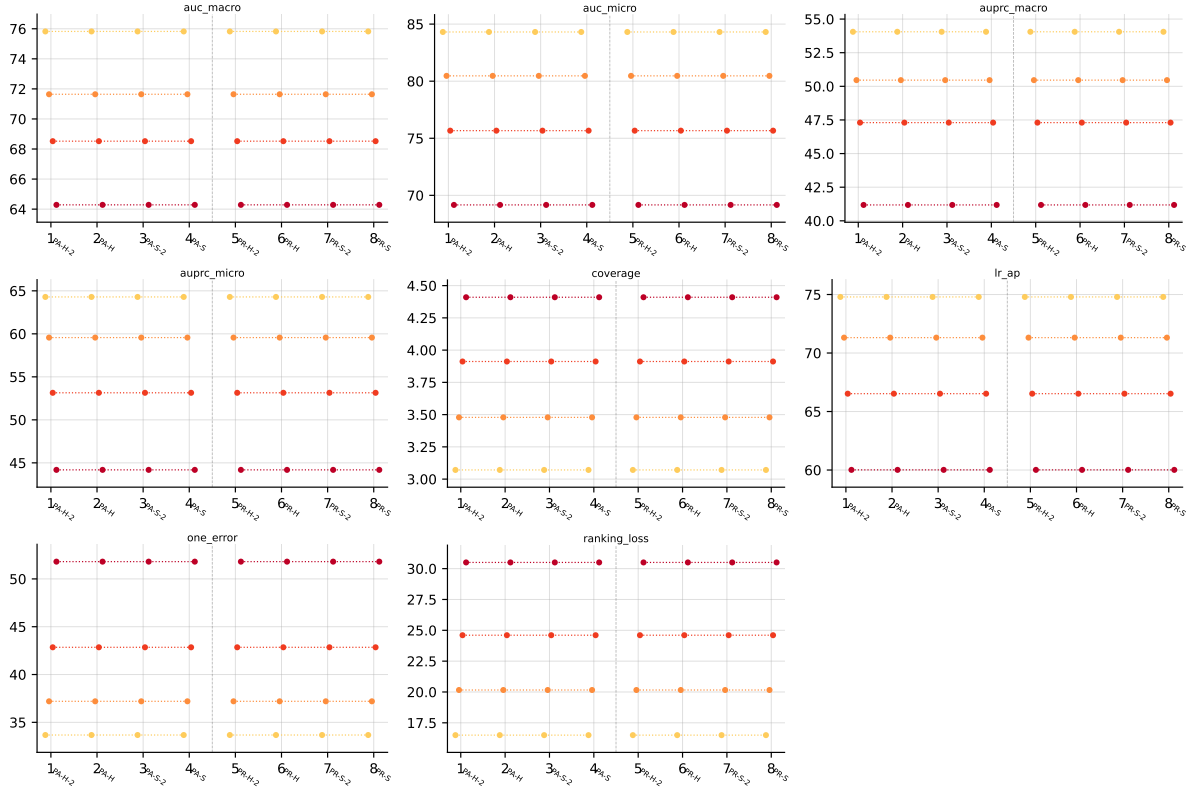


Table 49: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector), LGBM base learner, averaged across datasets (8 of 9: enron LGBM still training). Method indices and line colors follow the encoding paragraph above.